Critically synthesize evidence on technical, security, regulatory, and interoperability challenges in blockchain-based land and real estate administration. Distinguish land registries, cadastral systems, property transactions, and real estate records. Identify recurring challenges, root causes, evidence strength, and unresolved gaps. Separate domain-specific findings from general blockchain research. Preserve citations and never invent evidence.

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Blockchain in land and real estate administration

Key Findings: Blockchain in Land and Real Estate Administration

Key Findings Table

Domain	Main challenge pattern	Root causes	Evidence strength	Key citations
Land registries	Blockchain registries are usually **permissioned or hybrid overlays**, not full institutional replacements, with registry authorities retaining gatekeeping and adjudicative roles 1 2 3 .	Legal recognition limits, need for authoritative enrollment/key issuance, fragmented institutional workflows, and continued off-chain title adjudication 4 5 6 7 .	**Moderate** for architecture/governance patterns; **weaker** for measured fraud/dispute outcomes because many studies are conceptual or prototype-based 8 9 10.	1 2 3 5 10
Cadastral systems	Blockchain-specific evidence is thin; the main technical issue is linking parcel identity, geometry, and legal records while handling updates and	Parcel mutation is legally and technically complex; most retrieved workflow evidence comes from non-blockchain versioning and standards literature	**Low-to-moderate** for blockchain-specific cadastre claims; stronger for general cadastral data-model and versioning requirements 11 16 17.	11 14 15 16 17

Domain	Main challenge pattern	Root causes	Evidence strength	Key citations
	geospatial storage efficiently 11 12 13 .	rather than blockchain deployments 14 15 16 .		
Property transactions	Smart contracts can automate parts of conveyancing, but legal enforceability still depends on consent, identity, registry recognition, and existing property law 18 19 20 21 .	On-chain execution cannot substitute for off-chain verification, legal formality, or court-backed remedies 22 23 24 25 .	**Moderate** for identifying legal limits; **weak-to-moderate** for operational gains because field evidence is sparse and uneven 26 27 28.	18 19 20 21 28
Real estate records / interoperability	Blockchain is mostly used as a **complementary coordination and audit layer** linked to legacy databases, GIS, tax, identity, and planning systems through APIs, middleware, and off-chain storage 29 30 31 32 .	Legacy-system dependence, semantic mismatch, fragmented records, and need for standards-led cross-agency exchange 33 34 35 .	**Moderate** for architectural pattern; **limited** for robust evaluations of bidirectional interoperability performance 21 36 37.	29 30 31 33 34 35
Security / privacy / regulation	Claimed security benefits focus on immutability and tamper resistance, but harder unresolved issues are privacy,	Public-ledger exposure, weak data governance, mismatch with correction/deletion rights, and dependence on courts and law for	**Moderate** for threat identification; **low-to-moderate** for advanced mitigations in land settings because many controls are imported from general	7 38 39 40 44 45

Domain	Main challenge pattern	Root causes	Evidence strength	Key citations
	access control, key/identity governance, legal admissibility, and trustworthy data entry 7 38 39 40 .	contested changes 21 41 42 43 .	blockchain research rather than land deployments 44 45 46.	

Direct Answer

The evidence supports a cautious conclusion: in land and real estate administration, blockchain is usually a **hybrid institutional overlay** that may improve record integrity, auditability, and workflow coordination, but it does **not** remove the need for registry authorities, cadastral governance, legal adjudication, or standards-based interoperability 3 5 7 20 37

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Across the four domains, the recurring challenges are:

1. **Technical**: hybrid on/off-chain design is common because full on-chain storage is costly and poorly suited to documents and geospatial data; this creates synchronization and indexing trade-offs rather than eliminating complexity 29 47 48 49

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2. **Security and privacy**: immutability helps against tampering, but land systems still face unauthorized access, confidentiality leakage, identity/key governance problems, and unreliable data at entry 7 39 40

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3. **Regulatory**: legal enforceability remains a central bottleneck, especially for property transfers, because valid conveyancing depends on consent, form, capacity, registry recognition, and judicial remedies beyond code execution 19 21 25

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4. **Interoperability**: blockchain cannot fix fragmented, poor-quality, or semantically inconsistent records; standards such as LADM are treated as prerequisites to meaningful integration across cadastre, registry, tax, identity, and planning systems 33 34 50 51

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By domain, **land registries** and **property transactions** have the richest blockchain-specific literature, including some pilots and national cases, while **cadastral workflow evidence is notably thin** and much of the relevant evidence comes from conventional cadastre modernization rather than blockchain studies 9 15 16 28

. **Real estate records interoperability** is better supported as an architectural need than as a demonstrated success outcome 21 30 31

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Taken together, these findings suggest blockchain's most plausible near-term role is not as a replacement registry, but as part of a **standards-led land information infrastructure** combining digital identity, middleware, legacy integration, and selective on-chain anchoring 31 34 47

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Confidence level: High - the cross-domain pattern is repeatedly and directly supported across registry, transaction, interoperability, and security findings 3 10 52.

Study Scope

- **Timeframe covered by retrieved studies**: primarily 2016-2025, with many targeted searches focused on 2018-2025 and some foundational earlier works cited for smart contracts, Hyperledger, LADM, and interoperability 16 22 53 54.
- **Disciplines represented**: land administration, cadastral science, real-estate systems, legal analysis, e-government interoperability, blockchain systems, and information security 16 19 44 55.
- **Study types**: conceptual papers, literature reviews, prototypes, proofs of concept, legal analyses, case studies, and a small number of pilots/deployments 8 10 28.

Assumptions & Limitations

- The evidence base is **uneven**, with stronger support for conceptual benefits and architecture patterns than for longitudinal field outcomes such as fraud reduction, dispute rates, or transfer finality 8 28 56.

- Retrieved evidence does **not** provide robust direct comparisons of transaction finality across permissioned, consortium, and hybrid land-registry designs 2 32 57 58.
- Retrieved evidence also lacks direct blockchain-specific empirical studies of cadastral update workflows such as subdivision, boundary change, and parcel consolidation 14 15.

Suggested Further Research

- Comparative field studies of **permissioned vs consortium vs hybrid** land-registry governance, including validator selection, finality, and dispute handling 2 57 58.
- Blockchain-specific evaluation of **cadastral mutation workflows**: subdivision, boundary adjustment, consolidation, and court-ordered correction 14 15 59.
- Longitudinal before/after studies on **fraud incidence, reconciliation time, administrative cost, and dispute outcomes** 10 56.
- Empirical testing of **privacy-preserving and fine-grained access control** in real land-administration settings, rather than importing claims from general blockchain literature 44 45 46.
- Standards-grounded interoperability assessments using **ISO 19152/LADM** across registry, cadastre, tax, zoning, identity, and infrastructure systems 34 51.

Full Report Body

Introduction and scope

This review distinguishes four related but non-identical domains. **Land registries** concern authoritative recording of rights and title-related entries; **cadastral systems** focus on parcels, boundaries, and legal-spatial linkage; **property transactions** concern transfer workflows, settlement, and conveyancing; and **real estate records** cover the broader interoperable record environment linking registry, cadastre, tax, identity, planning, and other public systems 10 11 18 33

. Across all four, the core critical question is not whether blockchain can store records, but whether it can do so with legally valid governance, secure data control, and interoperable fit with existing institutions 3 7 19 52

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Confidence level: High - the domain separation and critical lens are directly reflected in the supplied evidence 7 11 18 33.

Foundational context

Two distinctions are foundational.

First, **record integrity is not the same as title validity**. Blockchain may strengthen tamper resistance, auditability, and chronology of entries, but title validity still depends on law, evidence, consent, and institutional adjudication 2 7 19

. This distinction explains why many proposed systems still retain certifying authorities, SuperAdmins, registries, notaries, or government verification 2 60 61.

Second, the literature repeatedly treats blockchain as a **complementary layer**, not a self-sufficient land administration system. Hybrid architectures use on-chain hashes, metadata, identities, and workflow logs, while keeping bulky documents, GIS data, and many legal decisions off-chain 3 29 37 47

. This pattern reflects legal and infrastructural constraints more than purely technical preference 5 21 42.

A related implication is that blockchain cannot repair **poor source data** or fragmented administrative semantics. If registry, cadastre, tax, and identity systems are inconsistent or paper-based, the ledger mainly preserves those weaknesses in a more durable form 7 33 50

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Confidence level: High - these are explicit recurring themes across the evidence base 3 7 33 47.

Land registries: architecture, governance, and evidence

The land-registry literature most often favors **permissioned, consortium, or hybrid** designs rather than open public chains 2 57 58

. The stated rationale is controlled access, identity-based participation, and administrative oversight 1 53. In several designs, the registry or certifying authority remains a gatekeeper by issuing keys, accrediting departments, or supervising administrators 2 60. Even consortium models distribute validation only among preselected public actors or elected committees rather than eliminating institutional authority 62 63.

This supports a consistent conclusion: blockchain registries are mostly **governed databases with stronger audit properties**, not institution-free title systems 3 4 5

. Claimed benefits include tamper-proof records, transparency, auditability, and reduced opportunities for fraud or unauthorized changes 4 64 65 66. But the supporting evidence is mostly conceptual and prototype-based. A proof-of-concept land ownership system argues manipulation resistance without quantified anti-fraud outcomes 8, and narrative discussions summarize possible fraud reduction without primary measurement 67. Georgia stands out as a real deployment with large-scale title publication and reduced handling time, yet the retrieved evidence does not report clear operational treatment of finality or disputes 9 68.

Governance problems recur more often than consensus-design problems. Studies report missing stakeholder forums, fragmented coordination, need for standardized processes, and continued dependence on sub-registrar offices or related authorities 5 69 70

. Comparative and review work on Sweden, Georgia, Ghana, India, and Honduras similarly frames adoption as contingent on political support, reliable records, process redesign, and partnership readiness 6 71 72.

A notable evidence gap remains: the supplied studies do **not** directly compare transaction finality across permissioned, consortium, and hybrid registry designs 2 32 57 58

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Confidence level: High for the governance and hybridization pattern; **Medium** for claimed anti-fraud and efficiency effects because measured field evidence is limited 3 8 9.

Cadastral systems and parcel linkage

Cadastral evidence is structurally different. The strongest findings are not about blockchain deployments, but about the underlying **parcel identity problem**. Parcel geometry, cadastral numbers, parcel IDs, georeferenced polygons, and cross-referenced legal documents are central to any credible cadastral system 11 73 74

. Some blockchain-oriented work models parcel identity through geohashes or digital asset representations 12 39, but most retrieved evidence on legal-spatial linkage comes from conventional cadastral literature rather than blockchain implementations 11 13 75.

The weakest part of the blockchain evidence base is **update workflow handling**. No retrieved study directly evaluates blockchain-based subdivision, boundary mutation, parcel consolidation, or comparable cadastral change procedures 14 15

. Instead, the closest evidence comes from non-blockchain versioning methods such as effective-date lineage, parent-child parcel history, non-destructive versions, and

genealogical update tracking 14 15. This matters because cadastral credibility depends not only on immutable snapshots, but on lawful correction and mutation workflows.

Where blockchain-specific technical work is stronger is **storage and indexing**. Studies propose compact spatial encodings, IPFS-backed off-chain storage, authenticated hybrid query structures, and dedicated indexing methods to reduce storage cost and improve query performance 47 76 77

. The recurring trade-off is clear: on-chain storage supports integrity, while off-chain storage improves scalability, latency, and cost at the expense of added synchronization and trust-management burdens 48 78 79

. Similar findings appear in later hybrid query studies emphasizing lower overhead and improved synchronization characteristics 77 80.

Standards work fills part of the blockchain evidence gap. LADM and CCDM provide common semantics for parties, rights, parcels, legal/spatial sources, versioning, and interoperability 16 17 51

. Taken together, these findings suggest that blockchain-specific cadastre proposals remain underdeveloped unless grounded in established cadastral models and update logic 16 51.

Confidence level: Medium for parcel-linkage and storage trade-offs; **Low** for blockchain-based cadastral mutation claims because direct evidence is largely absent 11 15 47.

Property transactions and legal enforceability

Property transaction studies are richer than cadastral ones, but they show a persistent tension between **workflow automation** and **legal sufficiency**. End-to-end proposals commonly include digital identity verification, property-rights certification, on-chain recording, smart-contract execution, payment-linked actions, review, and supervision 18 29 61

. Smart contracts are repeatedly positioned as the mechanism for automating transfer steps, agreement execution, and settlement logic 29 81 82.

However, these systems usually retain off-chain dependencies. Document submission, legal verification, government system linkage, and registry recognition remain part of the workflow 20 29 83

. The literature is explicit that identity verification must be legally recognized and that traditional functions of notaries, land registries, agents, lawyers, or courts do not simply disappear 24 84 85.

Legal reviews are especially consistent. Civil-law analyses argue that code may automate asset movement but cannot itself create valid legal transfer absent compliance with consent, form, capacity, good faith, and remedies 19

. Jordanian analysis reaches a limited-compatibility conclusion, contingent on technologically neutral rules, electronic signatures, and digital identity arrangements 86. English conveyancing analysis is more critical, concluding that safeguards needed to make blockchain acceptable erode its advantages over conventional databases 21.

Operational evidence exists, but sparsely. Bangladesh reports speed, gas, and cost-efficiency improvements 26, BERBLOM evaluates latency and throughput 27, and a 2024 real-estate DApp reports lower time/cost and reduced fraud risk with positive satisfaction responses 87. Yet review evidence still says most empirical applications are hybrid and small-scale, with only partial confirmation of promised gains 28 56.

The strongest supported conclusion is therefore narrow: blockchain can automate **some** transaction workflow elements, but **legal finality and enforceability remain institutional achievements, not ledger properties** 19 21 25

.

Confidence level: High for legal dependency; **Medium** for operational efficiency gains because metrics are reported in limited and heterogeneous cases 19 21 26 28.

Real estate records, interoperability, and legacy integration

Interoperability is a cross-cutting requirement rather than a separate optional feature. Studies repeatedly describe integrations among registry databases, cadastral information systems, GIS, tax, identity, citizenship, planning, and building registers 29 30 33 88

. Architecturally, blockchain is usually inserted as a coordination, verification, or audit layer, while conventional databases and government platforms continue to manage substantial operational data 32 36 89.

The dominant implementation pattern is **middleware-mediated hybridity**. Bangladesh's framework combines Hyperledger Fabric, Go chaincode, Node.js middleware, and IPFS 31

. Other systems use custom APIs for synchronization between official cadastres, relational databases, and proprietary GIS or municipal planning systems 30. Off-chain storage is common, especially for documents and large data payloads 29 49.

Standards are treated as enabling conditions, not add-ons. ISO 19152/LADM is explicitly proposed as the formal model for blockchain-based land administration integration 34 51. LADM-oriented work emphasizes versioning, legal/spatial source integration, 2D/3D

support, unique identifiers, metadata, and quality indicators 51. Broader interoperability literature reinforces that semantic and legal interoperability are as important as technical exchange 33 54 55.

The principal bottlenecks are not ledger-to-ledger communication alone. Reported problems include siloed departmental data, poor quality control, weak horizontal collaboration, structural/legal mismatch, and cases where blockchain may not outperform traditional databases for land transfers 21 35 90 91

. General blockchain interoperability research discusses cross-chain exchange and distributed execution, but the supplied evidence indicates those mechanisms remain only partially translated into land-administration practice 92.

Confidence level: High for the hybrid integration pattern and the centrality of standards; **Medium** for performance impact because robust comparative evaluations are limited 21 31 51.

Security, privacy, regulation, and adoption constraints

Security claims in land blockchain papers focus heavily on **immutability, tamper resistance, and resilience** 38 39 93. These benefits are plausible and sometimes implemented through permissioned ledgers, distributed validation, cryptographic integrity checks, and digital identity services 40 94 95. But the harder problems shift elsewhere.

The literature repeatedly identifies **unauthorized access, privacy leakage, weak confidentiality protections, and broader data-governance gaps** 39 40 41 96. Public transparency can conflict with the confidentiality requirements of land records 40. More fundamentally, blockchain cannot ensure that the information entering the ledger is true; one risk-assessment study explicitly says it cannot guarantee reliability at entry and has long-term limits as a credibility solution 7.

There is also a clear divide between **implemented** and **proposed** controls. Implemented controls in land studies are mostly private/permissioned blockchains, secure digital identity services, access-controlled records, and cryptographic logging 39 40. By contrast, attribute-based encryption, privacy-preserving smart contracts, homomorphic encryption, and advanced selective disclosure mostly appear as imported mechanisms from general blockchain/security research 44 45 46 97. This is a transfer gap: technical possibility exceeds land-domain validation 44 45.

Regulatory barriers are equally persistent. Legal recognition problems, evidentiary uncertainty, jurisdictional issues, and tension between immutability and legitimate correction rights all recur 21 25 42 43. Courts and legislators still matter when smart-

contract code is incomplete or disputed 24 98. Adoption studies further stress the importance of local context, trust, training, organizational readiness, top-management support, and digital infrastructure 99 100 101 102.

Confidence level: High for identifying the main threats and regulatory bottlenecks; **Medium** for mitigation effectiveness because advanced controls are rarely evaluated in land settings 7 40 45.

Cross-cutting synthesis: recurring root causes and evidence strength

Across domains, five root causes recur.

1. **Institutional dependence**: registries, ministries, survey departments, and courts remain authoritative because title validity is legal, not merely computational 5 7 19

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2. **Fragmented and poor-quality source records**: blockchain cannot solve inconsistent or paper-based administrative inputs 35 50 103

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3. **Need for standards and semantic alignment**: integration depends on common models such as LADM and interoperable service structures 33 34 51

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4. **Hybrid technical necessity**: off-chain storage and middleware are adopted because land systems involve bulky documents, geospatial data, and legacy platforms 29 31 47

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5. **Governance over design**: unresolved bottlenecks are validator authority, process standardization, legal enforceability, and stakeholder coordination more than raw consensus mechanism choice 2 21 70

.

Evidence strength is clearly uneven. Conceptual papers, reviews, and prototypes dominate claims about transparency, anti-fraud, and efficiency 8 104. Pilots and deployments provide stronger support, but remain relatively few and often lack detail on disputes, finality, or long-term correction handling 9 10 28. Cadastral mutation evidence is especially thin 15.

Confidence level: High - these patterns are directly repeated across the retrieved domains 10 35 50 103.

Domain-specific findings versus general blockchain research

Some findings are clearly **domain-specific**:

- Registry authorities remain gatekeepers through key issuance, credentialing, and record acceptance 2.
- Parcel identity requires legally meaningful linkage among polygons, parcel IDs, titles, and registers 11 13.
- Conveyancing validity depends on property-law requirements that smart contracts alone do not satisfy 19 21.
- Real estate record interoperability depends on cadastral, registry, tax, planning, and identity integration under formal models such as LADM 33 51.

Other themes come mainly from **general blockchain research** and exceed direct land evidence:

- Fine-grained cryptographic access control via attribute-based encryption 44 97.
- Privacy-preserving smart contracts and hidden transaction logic 45 105.
- Generic cross-chain interoperability and trustless exchange architectures 92.
- Smart-contract theory about external oracles and algorithmic incompleteness 22 23.

Taken together, these findings suggest land-administration papers often **import advanced blockchain mechanisms aspirationally**, while implemented land systems rely on simpler permissioning, digital identity, and off-chain storage patterns 31 40 45.

Confidence level: Medium-to-High - the distinction is well supported, though transfer gaps are sometimes inferential rather than directly tested 45 92.

Unresolved gaps and research agenda

The most important unresolved gaps are empirical.

- There is little direct evidence on **transaction finality**, **dispute handling**, and **court-compatible correction procedures** in blockchain registries and transfers 25 32 57.
- Blockchain-specific evidence on **cadastral mutation workflows** is notably absent 14 15.

- Measured claims about **fraud reduction**, **dispute-rate reduction**, and **long-term legal reliability** remain weak because field evaluations are scarce 10 56 106.
- Interoperability research needs more formal evaluation of **semantic, legal, and organizational** integration across agencies, not just technical connectivity 33 54.
- Privacy and access-control work should be validated in real land-administration settings rather than imported from unrelated domains 44 46.

A practical research agenda therefore centers on comparative governance studies, standards-grounded interoperability assessment, longitudinal public-sector evaluations, and real-world testing of correction, privacy, and access-control mechanisms 34 45 56 58.

Conclusion

The supplied evidence converges on a restrained view: blockchain in land and real estate administration is best understood as a **hybrid governance and coordination technology** whose strongest demonstrated value lies in auditability, traceability, and selective workflow automation, not in replacing registries, cadastres, or conveyancing law 3 10 52. The hardest challenges are recurring and mostly nontrivial: legal validity, institutional coordination, poor source-data quality, privacy, and interoperability with legacy land-information systems 7 50 91. Land-registry and transaction studies are comparatively more mature, while blockchain-specific cadastral workflow evidence remains thin 9 15 28.

Overall confidence level: High - the central conclusion is strongly and repeatedly supported across multiple domains and study types, even though many claimed benefits still lack strong field validation 3 10 52.

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Analyze hybrid blockchain architectures for land and real estate administration, including on-chain/off-chain storage, IPFS, smart contracts, consensus mechanisms, platforms, and integration technologies. Compare design choices, trust assumptions, security, scalability, and implementation maturity. Distinguish conceptual proposals, prototypes, pilots, and

operational deployments. Identify architectural patterns and research gaps. Preserve citations; do not invent findings.

Processing steps

Hybrid blockchain architectures in land administration

Hybrid blockchain architectures in land administration

Hybrid blockchain architectures in land and real estate administration are overwhelmingly **permissioned, hybrid on/off-chain systems** that keep hashes, permissions, and workflow control on-chain while moving documents, images, geospatial files, or other bulky/sensitive content off-chain to IPFS or cloud storage; the dominant pattern is not full replacement of legacy registries but a **layered integration** with existing institutions and databases 3 21 32 40 107 108 109 110 111 112

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Architectural patterns

Pattern	What it looks like	Trade-off	Evidence
On-chain metadata + off-chain documents	Metadata, hashes, access rules, and transaction logic stay on-chain; deeds, images, and large files are stored in IPFS or cloud/object storage 108 109 110 111 112 113 .	Improves scalability and lowers storage cost, but creates integrity, availability, and synchronization dependencies across layers 108 109 113 114 .	Strong across many prototypes and proposals 108 109 113.
Permissioned consortium/private blockchain	Hyperledger Fabric, private Ethereum, or consortium networks are used for controlled participation and role-based access 71 107 108 110 115 116 117 .	Better privacy and governance fit for land administration, but weaker openness and more dependence on trusted operators 40 71 107 .	Most mature pattern in the evidence 107 117 118.
Smart-contract workflow automation	Smart contracts handle registration, validation, transfer, multi-party approval,	Automates routine steps, but cannot replace off-chain verification, inspection, legal	Strong for workflow design; weak for legally complete

Pattern	What it looks like	Trade-off	Evidence
	or lease execution 115 116 118 119 120 121 .	formality, or administrative discretion 21 40 111 .	automation 115 122.
Integration with legacy systems	RESTful interfaces, React front ends, government databases, state registries, and middleware connect blockchain to existing land workflows 111 118 119 .	Interoperability becomes a primary bottleneck because blockchain is added to, not substituted for, legacy infrastructure 40 123 .	Recurrent but rarely evaluated end-to-end 110 123.

Design choices: what the literature favors

Storage

Hybrid storage is the clearest consensus. Studies repeatedly place **large or sensitive payloads off-chain** and anchor hashes or metadata on-chain to reduce bloat, improve access speed, and preserve tamper evidence 108 109 110 111 112 113 114 124 125

. Some papers explicitly compare this favorably against purely on-chain or purely off-chain approaches, with lower latency and cost while retaining integrity guarantees 114 124

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Platforms

Hyperledger Fabric and other permissioned frameworks recur in land-administration designs because they support controlled participation and enterprise governance 107 108 117 118 126

. **Ethereum** is the other major platform, especially for smart-contract-heavy prototypes and tokenization-style real estate workflows 110 115 120 121

. A smaller set explores **Substrate** or other modular chains to reduce storage or confirmation overhead 125

.

IPFS and decentralized storage

IPFS is the most common off-chain component. It is used to store documents, images, metadata, and encrypted files while the blockchain stores the hash or address 108 109 110 111 112 113 126 127

. The main architectural rationale is to reduce blockchain storage burden and improve retrieval efficiency, but several studies note unresolved concerns about practical deployment, encryption, and wider acceptance 40 109 128

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Consensus

For land administration, the evidence leans toward **permissioned consensus** rather than public-mining designs. Fabric-style deployments often use consensus mechanisms such as Raft, and studies of enterprise blockchains emphasize deterministic or lightweight consensus for throughput and governance 107 117 129

. More general blockchain literature shows the same trade-off: stronger decentralization usually costs throughput, while faster consensus often implies more trust in operators 128 130 131

.

Trust, security, scalability, and maturity

- **Trust assumptions** are still institutional, not purely cryptographic: systems rely on verified inspectors, government databases, authorized actors, or consortium governance to validate entries before they are committed 40 111 118

. This aligns with earlier land-administration work that treats blockchain as a control layer rather than a replacement for registries 3 21.

- **Security** gains come from immutability, signatures, hashing, and access control, but the weakest point is the **data-entry and governance layer**: if inputs are wrong, private keys are mishandled, or off-chain storage is compromised, blockchain alone does not solve the problem 40 107 109 132 133

.

- **Scalability** is usually improved by hybrid storage and permissioned architectures, with some studies reporting lower storage time, lower retrieval time, or better throughput relative to single-chain designs 114 117 124 134

. But these gains are often shown in prototypes, testnets, or simulations rather than production-scale registry environments 68 135 136 137.

- **Implementation maturity** is uneven. The evidence ranges from **conceptual proposals** 111 112, to **prototype systems and testnet experiments** 115 118 120 126, to **pilots and field deployments** in property/registry-adjacent settings 68 135 136. Operational, system-wide deployments remain rare 40 137.

Research gaps

The main gaps are:

1. **End-to-end operational evidence** is scarce; most systems stop at conceptual design, prototype, or pilot level 40 68 136 138

.

2. **Interoperability with legacy land systems** is often asserted but not rigorously tested across cadastre, registry, tax, GIS, and identity systems 40 110 123

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3. **Data governance and correction rights** remain unresolved where immutability conflicts with legal correction or deletion processes 40 131

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4. **Platform choice is under-justified**: many studies pick Ethereum or Fabric for convenience rather than by comparative benchmarking across trust, cost, and regulatory fit 115 128 138

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5. **Consensus and storage are still treated separately** in much of the literature, even though real systems need joint optimization of placement, execution, and verification 114 129

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Bottom line

Hybrid blockchain for land and real estate administration is best understood as a **permissioned coordination layer** that uses smart contracts plus off-chain storage to improve auditability and workflow efficiency, but its real bottlenecks are still **institutional trust, interoperability, and legal-operational integration**, not ledger mechanics alone 3 21 32 40

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Confidence level: **High** — the retrieved evidence is consistent across many land/real-estate studies and a few adjacent blockchain studies, with strong agreement on hybrid storage, permissioned governance, and prototype-heavy maturity, though field-scale deployment evidence remains limited 3 21 40 107 114.

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Follow-up Suggestions

- How do hybrid land blockchain systems handle legal correction rights
- Which consensus mechanisms are used in permissioned land registries
- How is LADM integrated with blockchain land systems

How do hybrid land blockchain systems handle legal correction rights

Processing steps

Legal correction rights in hybrid blockchain land systems

Legal correction rights in hybrid blockchain land systems

Hybrid land blockchain systems usually handle **legal correction rights** by **keeping the blockchain ledger unchanged and handling rectification off-chain**, or by using **permissioned, governed mutability** when the design explicitly allows it. In practice, that means the chain often preserves an immutable audit trail, while corrections, deletions, or reversals happen in linked databases, document stores, or administrative workflows rather than by rewriting the land ledger itself 3 21 43 139 140 141 142 143 144

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How correction rights are implemented

Approach	How it works	What it means for legal correction rights
Off-chain rectification with on-chain proof	Personal or legally sensitive data are kept off-chain; the blockchain stores hashes, references, or minimal metadata. If correction or erasure is required, the off-chain record is updated or deleted, and the on-chain proof remains as an audit trail 139 140 142 144 145 146 147 .	This is the most common workaround for correction rights, but it does **not** remove the immutable on-chain trace of the earlier record 43 141 143 144 .
Revocation / re-keying	Access to off-chain data is withdrawn by rotating or revoking encryption keys, making the data practically inaccessible without altering the chain 139 140 142 148 .	This supports a form of correction or erasure, but it is a **practical** rather than literal deletion right because the chain entry remains 139 141 145 .
Controlled redaction in permissioned systems	Some architectures introduce redactable blockchain primitives or governed mutation rules, typically with trusted authorities or multi-party approval 139 .	This can directly honor correction requests, but it weakens strict immutability and depends on additional governance and trust assumptions 139 .
Administrative correction workflow	The blockchain logs the request, while inspectors, registry officers, or institutions validate and apply the legal correction through existing registry processes 21 149 150 .	Legal correction rights remain primarily an **institutional** function; the blockchain supports traceability, not automatic legal rectification 21 149 150 .

Bottom line

For land administration, blockchain is **not** usually treated as the place where legal correction rights are exercised directly. Instead, the prevailing design is: **keep the ledger**

auditable and append-only, store correctable data off-chain, and use governance or administrative procedures to apply rectification 3 21 139 141 143 144 150

. Where systems do support direct correction, they do so through **permissioned redactability** or **controlled mutation**, which is technically possible but shifts the trust burden to governance and key-management arrangements 139

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Confidence level: **High** — multiple independent sources converge on off-chain correction, key revocation, and/or governed redactability as the main ways hybrid blockchain systems reconcile immutability with legal correction rights, though implementation maturity in land-specific deployments remains limited 3 21 139 143 144.

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Follow-up Suggestions

- How are redactable blockchain schemes implemented in land registries?
- How do off-chain revocation and key rotation support record correction rights?
- What governance process authorizes corrections in permissioned land blockchains?

Which consensus mechanisms are used in permissioned land registries

Processing steps

Consensus mechanisms in permissioned land registries

Consensus mechanisms in permissioned land registries

Permissioned land registries most often use **PBFT and related Byzantine-fault-tolerant variants**, with **Raft**, **IBFT**, **PoA/Clique/QBFT**, and a few **custom trust- or authentication-**

based consensus schemes appearing in specific implementations and prototypes. 1 139 140 141 142

What shows up in the evidence

Consensus mechanism	Where it appears in permissioned land-registry evidence	Design implication
PBFT	Used in permissioned land-registration designs and described as suitable for known validator sets and real-time land transfers 1 143	Strong fit when the system needs deterministic finality and Byzantine fault tolerance, but it becomes communication-heavy as the validator set grows 144 145
Raft	Used in Hyperledger Fabric-based land registry frameworks 141	Fits managed, permissioned settings where crash-fault tolerance and throughput matter more than Byzantine adversaries; the trust model is lighter than PBFT 141 146
IBFT	Discussed as a PBFT-inspired permissioned consensus option in blockchain systems used for secure registries and related enterprise blockchains 147 148	Keeps Byzantine tolerance but aims for better operational efficiency than classic PBFT, still within a validator-controlled network 147 148
PoA / Clique / QBFT	Evaluated as permissioned options in Besu-based land-administration prototypes and related permissioned deployments 142	PoA-style designs rely on authorized validators; QBFT is favored in the cited comparison when stronger fault tolerance and stable throughput are needed 142
Trusted-node / trust-value consensus	Proposed in land/property registration frameworks as a way to reduce overhead and speed block approval 140 149	These designs explicitly trade full open participation for validator trust assumptions tied to registry offices or trusted miners 140 149

Consensus mechanism	Where it appears in permissioned land-registry evidence	Design implication
Proof of authentication / majority consensus .	Appears in some land-registry proposals, including systems that frame block validation around signed authentication rather than open mining 150 151	More of a domain-specific control logic than a widely standardized consensus family; it reflects permissioned governance and identity verification 151 .

Bottom line

The **dominant pattern** in permissioned land registries is **permissioned Byzantine-tolerant consensus**—especially **PBFT and PBFT-like variants**—because land systems usually need known validators, governance control, and immediate finality rather than open participation 1 141 142 143

. **Raft** is also common where implementers prioritize simpler coordination and throughput inside a managed consortium, while **IBFT and QBFT** show up as stronger permissioned alternatives when the design needs more Byzantine resilience than Raft provides 141 142 147 148

.

Confidence level: **High** — the retrieved land-registry papers repeatedly converge on the same small set of permissioned consensus families, with clear support for PBFT-style mechanisms and several direct examples of Raft, IBFT, and PoA/QBFT-based implementations 1 139 140 141 142 143.

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Follow-up Suggestions

- How does PBFT achieve finality in land registry networks
- How does Raft differ from IBFT in permissioned registries
- What governance model is used for validator selection in land blockchains

How is LADM integrated with blockchain land systems

Processing steps

LADM integration with blockchain systems

LADM integration with blockchain systems

LADM is integrated with blockchain land systems mainly as the **off-chain conceptual and data-standard backbone**: blockchain handles integrity, traceability, and workflow coordination, while LADM supplies the shared model for parties, rights/restrictions/responsibilities, spatial units, and source documents so the blockchain layer can fit existing land-administration structures rather than replace them 3 151 152

.

How the integration works

Integration role	What LADM contributes	What blockchain adds	Evidence
Common land data model	Standardized representation of parties, administrative units, spatial units, and source documents, enabling a shared ontology across systems 151 152	Immutable ledger and audit trail for changes, claims, and transactions 3 34	Strong
Registry/cadastre alignment	A way to structure cadastral and registry data so different institutions can communicate and exchange data consistently 151 152 153	Helps connect those records without forcing a full replacement of legacy registries 3 4	Strong
Hybrid architecture	Defines the domain objects and relationships that sit behind the system 151 154	Stores hashes, proofs, workflow states, or smart-contract outputs on-chain while keeping detailed records and documents	Strong

Integration role	What LADM contributes	What blockchain adds	Evidence
		in existing databases or linked storage 3 34 155	
3D and geospatial support	Extends naturally to 3D cadastre, marine cadastre, and mixed spatial/legal data through modular profiles and country adaptations 156 157 158 159	Adds tamper resistance and traceability for geospatial records, including optimized storage representations 76	Moderate

Typical integration pattern

The most common pattern is **LADM-first, blockchain-second**: projects start from an LADM country profile or cadastral model, then insert blockchain as a complementary layer for notarization, verification, and transaction logging 3 153 160

. In that setup, LADM keeps the legal and spatial structure legible, while blockchain supports traceable updates and multi-party validation 3 34 151

.

What this means in practice

- **For cadastral systems**, LADM is used to map local cadastral data into a standardized model before blockchain integration, including legal and spatial components and, in some cases, 3D extensions 154 161 162

.

- **For land registries**, blockchain is generally framed as a **complementary layer** that improves auditability and trust while the registry remains the authoritative institution 3 4

.

- **For interoperability**, LADM is the key enabler because it provides the common semantics needed to connect blockchain systems to legacy databases and other land-information systems 151 152 163

.

- **For implementation maturity**, the evidence is still dominated by conceptual models, proofs of concept, and prototypes rather than fully operational national deployments 3 4 34 164

.

Bottom line

LADM is not usually replaced by blockchain; it is the **standardized reference model that makes blockchain-based land systems administratively meaningful**, while blockchain supplies the tamper-evident ledger and workflow layer on top of LADM-based cadastral and registry structures 3 34 151

.

Confidence level: **High** — the retrieved studies consistently show LADM as the modeling/interoperability layer and blockchain as a complementary integrity layer, with repeated emphasis on hybrid rather than replacement architectures 3 151 152.

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Follow-up Suggestions

- How do LADM country profiles map to blockchain smart contracts?
- How is LADM used for 3D cadastre in blockchain land systems?
- How do blockchain land systems handle interoperability with legacy registries using LADM?

What governance process authorizes corrections in permissioned land blockchains?

Processing steps

Governance of corrections in permissioned land blockchains

Governance of corrections in permissioned land blockchains

Corrections in permissioned land blockchains are usually authorized by a governed institutional process, not by unilateral on-chain rewriting: the blockchain keeps an immutable audit trail, while rectification is approved off-chain through registry authorities, courts, or multi-party governance rules, and only in some systems through controlled redaction/mutation mechanisms. 1 3 142 165 166

How the authorization process works

Governance pattern	Who authorizes the correction	What happens to the ledger
Administrative correction workflow	Registry officers, inspectors, or other public authorities validate the request and apply the legal correction through existing registry procedures 1 3	The chain remains as an audit record; the correction is handled in linked systems or through a new recorded event 1 3
Court-ordered rectification	Courts or legally empowered bodies determine the title correction, especially when disputes or adverse possession are involved 166	The system may need an override or replacement control in a private blockchain to make the correction operational 166
Permissioned governed mutability	Trusted authorities or multi-party approval protocols authorize redaction or controlled mutation 3 142	Some entries can be redacted, but this weakens strict immutability and shifts trust to governance 3 142
Off-chain rectification with on-chain proof	The correction is approved in the off-chain administrative layer, while the blockchain preserves proof of the prior state 3 166	The immutable ledger is not rewritten; only the correct current state is reflected in the wider workflow 3 166

Bottom line

The key governance feature is **institutional authorization**: permissioned land blockchains do not usually let participants “edit” the ledger directly; instead, a correction is authorized by the registry, a court, or a trusted multi-party process, and the system then records that correction in a way that preserves auditability. 1 3 166

Confidence level: **High** — the retrieved evidence consistently shows correction rights being handled through off-chain governance, court or registry action, or tightly controlled permissioned mutability rather than ordinary on-chain overwriting 1 3 142 166.

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Follow-up Suggestions

- How do correction workflows work in land registry smart contracts?
- What role do courts play in blockchain land rectification?
- How are redaction mechanisms implemented in permissioned blockchains?

How do LADM country profiles map to blockchain smart contracts?

Processing steps

Mapping LADM country profiles to smart contracts

Mapping LADM country profiles to smart contracts

LADM country profiles map to blockchain smart contracts by turning the **country profile into the legal-semantic specification** for the contract layer: the profile defines which parties, rights/restrictions/responsibilities, spatial units, and source documents exist in the jurisdiction, and the smart contracts then encode workflow rules, validation steps, and state changes against that profile rather than replacing it 3 34 48 51 163 167 168

.

How the mapping works

LADM country-profile element	Smart-contract role	What it means in practice
Parties	Identity, authorization, and role checks	Contracts use jurisdiction-specific roles and authorized actors to decide who can submit, verify, approve, or query records 3 48 51 169 170 .
RRRs	Contract logic for ownership, transfer, restriction, and validation	Rights, restrictions, and responsibilities become executable rules that govern registration or transfer workflows 3 168 171 172 .
Spatial units / parcels	References to parcel identity and linked geodata	The contract usually stores hashes, identifiers, or pointers to parcel data rather than full geometry on-chain 3 51 155 168 .
Source documents	Proof and audit references	Surveys, deeds, and related evidence are linked so the contract can verify or log actions without becoming the legal source itself 3 48 51 168 .
Country-specific legal rules	Preconditions and business rules	The profile localizes the contract so it reflects national laws, cadastral practice, and institutional mandates 161 167 169 173 .

What the evidence shows about the architecture

1) LADM is the upstream model; smart contracts are the operational layer

The retrieved studies consistently treat LADM as the **reference model** and smart contracts as the layer that operationalizes parts of it in a given country context 3 34 51 163 174

. This is why country profiles are developed first and then mapped into implementation schemas or blockchain workflows 161 162 167 169.

2) The mapping is usually partial, not a full legal rewrite

Most blockchain land papers use a **hybrid design**: the smart contract handles registration logic, verification, or transaction sequencing, while the authoritative land administration role remains with the registry or another institution 3 34

. That means the contract mirrors selected LADM elements, especially parties, RRRs, and workflow states, but does not absorb the whole legal system 3 51

.

3) Country profiles make the contract jurisdiction-specific

Country-profile work shows that LADM must be adapted to national legal and technical requirements before implementation, and conformity testing is used to check that mapping 161 162 167 175

. In smart-contract terms, this means the contract logic is not generic; it is parameterized by local rules, data structures, and institutional responsibilities 169 170 173

.

Common mapping pattern

A recurring implementation pattern is:

1. **Develop the LADM country profile** for the jurisdiction 161 162 167

.

2. **Map profile classes to system objects** such as parties, parcels, RRRs, and documents 163 168 174

.

3. **Encode workflow rules in smart contracts** for verification, transfer, logging, or approval 3 34 170

.

4. **Keep detailed records off-chain** or in linked databases, while the blockchain stores proofs, hashes, or state transitions 3 155

.

Bottom line

LADM country profiles map to smart contracts as a jurisdiction-specific schema-to-logic translation: LADM defines *what* the land administration system must represent, and

smart contracts implement *how* selected actions on those entities are validated and recorded in a blockchain-based workflow 3 34 51 167

. The strongest evidence supports **hybrid, profile-driven implementations** rather than fully autonomous on-chain land law 3 169 173

.

Confidence level: **High** — multiple LADM profile papers and blockchain land proofs-of-concept converge on the same pattern of profile-first modeling, followed by selective smart-contract automation and off-chain legal authority retention 3 34 161 162 167.

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Follow-up Suggestions

- Which LADM classes map to smart contract functions?
- How do country profiles affect blockchain land registry design?
- How are off-chain documents linked to on-chain parcel records?

How do blockchain land systems handle interoperability with legacy registries using LADM?

Processing steps

LADM-based interoperability in land blockchains

LADM-based interoperability in land blockchains

Blockchain land systems handle interoperability with legacy registries by using **LADM as the shared semantic and data-model layer** while blockchain acts as a **complementary integrity and workflow layer**, not a replacement for the registry or cadastre. In practice, systems map legacy land concepts into LADM country profiles, use LADM packages to standardize parties/rights/spatial units/source documents, and then connect blockchain to

existing databases and institutional workflows through hybrid interfaces and off-chain storage 3 34 51 167 168

.

How the interoperability pattern works

Layer	What LADM does	What blockchain does	Interoperability effect
Conceptual model	Provides a common vocabulary for legal, spatial, and administrative land data, including rights, responsibilities, restrictions, parcels, and source documents 16 51 .	Supplies immutable logging, traceability, and audit trails 34 71 .	Legacy registry data can be normalized into a shared model before integration 167 168 .
Country-profile adaptation	Tailors LADM to national legal and institutional requirements, so existing registry structures can be represented rather than replaced 161 168 169 .	Anchors proofs, hashes, or workflow states to the ledger.	Enables phased modernization instead of a disruptive system swap 3 4 170 .
Technical integration	Supports links to external data structures and linked registers, including archive data and other institutions' records 167 171 .	Connects via smart contracts, APIs, and hybrid architectures.	Makes cross-agency exchange possible while preserving authoritative legacy systems 3 171 .
Process integration	Helps formalize registration and dissemination processes and match them to LADM classes 172 173 .	Automates selected transactions or verification steps.	Improves consistency and traceability, but does not eliminate institutional coordination 3 172 173 .

Main interoperability design choices

- **LADM-first, blockchain-second** is the dominant pattern: projects start by adapting the land domain to an LADM profile, then add blockchain as a complementary layer for notarization and auditability 3 34 168

.

- **External/linked data structures** are used when legacy registries must exchange documents, proofs, or status information with other institutions; one LADM-based case explicitly modeled archive data for registry and cadastre transactions and enabled digital document exchange 171

.

- **Semantic interoperability is the real bottleneck**, because land data live across multiple institutions and formats; LADM helps by providing common classes and terminology, but systems still need mapping, standardization, and linkage rules 167 168 173

.

- **3D and geospatial data increase the difficulty**, so implementations often need extra representation work beyond basic registry records 76 161 174

.

What this means for legacy registries

LADM makes interoperability feasible by translating legacy registry contents into a **shared reference structure**, while blockchain preserves **trust, traceability, and tamper evidence** around that structure 16 34 51

. The evidence consistently shows that blockchain systems in land administration are best understood as **hybrid overlays on existing registries and cadastres**, with LADM providing the bridge between old records and new infrastructure 3 4 168

.

Bottom line

The interoperability strategy is **standardize first, ledger second**: legacy registries are made interoperable by re-expressing their data and processes through LADM country profiles and linked external datasets, then using blockchain to add auditability and controlled automation without breaking the authority of existing land institutions 3 167 168 171

.

Confidence level: **High** — multiple independent LADM and blockchain sources converge on the same hybrid integration pattern, with consistent emphasis on semantic standardization, country profiles, external data linking, and complementary rather than replacement architectures 3 34 167 168 171.

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Follow-up Suggestions

- How do LADM external classes support registry integration?
- How are legacy cadastral records mapped into LADM country profiles?
- How do blockchain APIs exchange land data with legacy registries?

Which LADM classes map to smart contract functions?

Processing steps

LADM classes and smart contracts

LADM classes and smart contracts

The LADM elements that most clearly map to **smart contract functions** are **parties**, **rights/restrictions/responsibilities (RRRs)**, **spatial units/parcels**, and **source documents**; in implementation terms, smart contracts then encode identity checks, authorization, transfer/validation rules, and proof/audit handling around those classes rather than replacing the full LADM model 3 34 51

.

Practical mapping

LADM element	Smart contract function role	What the contract typically does
Parties	Identity, authorization, and role control	Checks who can submit, verify, approve, or query records 3
RRRs	Core business logic	Encodes ownership, transfer, restriction, and validation rules 3
Spatial units / parcels	Asset reference and parcel linkage	Stores parcel identifiers, hashes, or pointers to geodata rather than full geometry on-chain 3
Source documents	Evidence and audit references	Links deeds, surveys, and other documents so the contract can verify actions without becoming the legal source itself 3
Country-specific legal rules	Preconditions and workflow constraints	Localizes contract logic to national law and institutional mandates

What this means

The strongest evidence supports a **profile-first, contract-second** design: the LADM country profile defines the legal-semantic structure, and the smart contract implements selected workflow steps against that structure 34 51

. In other words, smart contracts usually operationalize **parties + RRRs + parcels + documents**, while the broader administrative and legal meaning remains in the LADM layer and the registry institution 3 34 51

.

Bottom line

LADM smart-contract mapping is selective, not one-to-one: the classes most often translated into contract functions are **Parties**, **RRRs**, **Spatial Units/Parcels**, and **Source Documents**, with the contract acting as the executable workflow layer on top of the LADM reference model 3 34 51

.

Confidence level: **High** — the retrieved LADM/blockchain sources consistently converge on the same partial mapping pattern, and the contract layer is repeatedly described as

implementing workflow and validation around those core LADM elements rather than replacing the model 3 34 51.

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Follow-up Suggestions

- How do LADM Parties map to blockchain identity roles?
- How are RRRs encoded as smart contract state transitions?
- How do source documents link to on-chain proof records?

How are legacy cadastral records mapped into LADM country profiles?

Processing steps

Mapping legacy cadastral records to LADM

Mapping legacy cadastral records to LADM

Legacy cadastral records are mapped into **LADM country profiles** by first extracting the jurisdiction's existing cadastral concepts, then aligning them to LADM's core classes, attributes, and relationships, and finally testing whether the resulting profile is conformant with national law and cadastral practice 153 167 175

.

How the mapping is done

Mapping step	What happens	What it preserves
Extract existing cadastre concepts	The country's current legal and cadastral structures are analyzed, and the basic concepts are pulled out of legacy records and rules 167 175 176	Local legal meaning and institutional practice 153 175

Mapping step	What happens	What it preserves
	.	
Map legacy elements to LADM classes	Existing data are matched to LADM entities such as **Party**, **Spatial Unit/Parcel**, **RRRs**, **BAUnit**, and source documents 177 .	Structural correspondence between old records and the standardized model 177
Adapt the profile to national needs	Some profiles reuse LADM "as is," some show inherited structure, and some explicitly map elements from the legacy cadastre to LADM 178 .	Jurisdiction-specific legal and data requirements 176 178
Test conformity and refine	Profiles are checked against LADM conformance requirements and revised where legislation, 3D requirements, or data gaps require it 162 167 175 .	Standard compliance and future extensibility 162 179

What the evidence shows

- **Legacy records are not usually copied wholesale into LADM**; instead, they are re-expressed in a standardized conceptual structure that can represent both legal and spatial information 162 177 178

.

- The mapping often starts from the **existing cadastral data model** and then identifies where LADM classes already fit and where extensions are needed for local law, data quality, or 3D cases 170 175 180

.

- In some jurisdictions, the legacy system is so incomplete or inconsistent that the LADM profile becomes a **normalizing framework** for fixing mismatches between graphic and alphanumeric data, or between paper-based and digital records 170 175 181

.

- For 3D and complex property situations, the mapping may require **new spatial representations** or profile extensions, rather than a one-to-one transfer from legacy 2D records 175 179 182 183

.

Bottom line

Legacy cadastral records are mapped into LADM country profiles by translating local parcel, title, rights, and document structures into LADM's standardized classes and relationships, then refining that mapping to fit national law, data quality, and 3D or interoperability needs 167 176 177 178

.

Confidence level: **High** — the retrieved country-profile studies consistently describe the same profile-first, mapping-based approach, with repeated examples from Czech Republic, Serbia, Morocco, Korea, Poland, and Kenya showing how legacy cadastre structures are aligned to LADM rather than replaced outright 162 175 176 177 178 180.

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Follow-up Suggestions

- How do LADM country profiles handle 3D cadastral parcels
- Which legacy cadastral attributes are hardest to align with LADM
- How is conformance tested for an LADM country profile

How are RRRs encoded as smart contract state transitions?

Processing steps

Encoding RRRs in smart contracts

Encoding RRRs in smart contracts

RRRs are encoded as **state transitions over a land-rights record**, where the smart contract changes the record's status only when predefined legal and workflow conditions are satisfied; in land-administration designs, the RRR concept comes from the LADM legal layer, while the contract layer operationalizes it as executable rules for authorization, validation, transfer, and recorded state change 3 171 184 185

.

How the transition logic works

RRR aspect	Smart-contract encoding	Effect on state
Rights	Contract rules represent entitlement events such as registration, transfer, or tokenized assignment; the contract executes only after required approvals or verified documents are present 185 186 187 .	The parcel or title moves from one registered state to another, and the new ownership/entitlement state is written to the ledger 186 187 .
Restrictions	The contract checks preconditions against external data or jurisdictional rules before allowing the next step, so a restriction acts as a guard on the transition 184 185 188 .	If the restriction is not satisfied, the transition is blocked; if it is satisfied, the workflow continues 185 188 .
Responsibilities	Contract logic assigns obligations to roles or parties and can require actions such as verification, approval, or compliance evidence before advancing 184 189 190 .	The record remains in a pending or intermediate state until the required responsibility is fulfilled 189 190 .

What "state transition" means in practice

- The contract treats the land record as a **tracked state machine**: a transaction request enters one state, validation checks run, and the record advances only if the conditions are met 189 191 192 193

.

- In land workflows, those checks are usually **role-based and document-based**, with verified inputs, legal preconditions, and institutional approval determining whether the transition occurs 171 184 185 186

.

- In LADM-based designs, the contract does **not** replace the legal model; it executes selected parts of the profile, especially the parts that define who may act, what can change, and which conditions must be met 3 51 171

.

Bottom line

RRRs are encoded as conditional state changes on a ledgered land record: rights become executable transfer/registration actions, restrictions become transition guards, and responsibilities become obligations that must be satisfied before the contract advances the record to the next state 3 171 184 185

. The strongest pattern is a **profile-first, contract-second** design in which LADM defines the legal meaning of the RRRs and the smart contract enforces the workflow logic that moves the record between states 3 51 194

.

Confidence level: **High** — the evidence directly shows smart contracts using state-transition logic and conditional validation, and multiple LADM sources consistently position RRRs as the legal concepts that contracts operationalize rather than replace 3 171 184 185.

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Follow-up Suggestions

- How do smart contracts model land title workflows as state machines?
- How are legal preconditions verified before a land record transition?
- How do LADM RRRs map to blockchain transaction events?

Which legacy cadastral attributes are hardest to align with LADM

Processing steps

LADM legacy cadastral attribute alignment

LADM legacy cadastral attribute alignment

The **hardest legacy cadastral attributes to align with LADM** are the ones that do **not** fit cleanly into LADM's core parcel/party/RRR structure: **3D and volumetric geometry**, **vertical or underground rights**, **raster-based or non-vector legacy representations**, **ambiguous identifier/label fields**, and **time-varying rights or coastal/marine states**. These are the cases where the evidence repeatedly shows that direct mapping is incomplete, extensions are needed, or extra semantics must be added on top of the standard 3 51 156 167 184 185 186 187 188 189 190 191 192 193 194 195

.

What is hardest to align

Legacy cadastral attribute type	Why it is difficult in LADM	Evidence
3D geometry / solid volumes	LADM implementations often work with boundary-face or multi-surface representations, but legacy records may require true solids, volumetric computation, or detailed multi-level building spaces that LADM does not natively cover in the cited versions 184 187 .	184 187
Vertical rights and special spaces	Airspace rights, underground parcels, strata units, and other height-dependent interests are repeatedly described as needing new packages, extensions, or extra modeling work 156 185 190 191 192 195 196 .	156 185 190 191 192 195 196

Legacy cadastral attribute type	Why it is difficult in LADM	Evidence
Raster or map-based legacy plans	Some cadastral records are delivered as raster plans or scanned maps, which do not map neatly to LADM's vector-oriented spatial unit modeling 184 186 .	184 186
Ambiguous label/identifier fields	Attributes like arbitrary label numbers attached to geometry, or separate identifiers for common vs. private parts, were specifically reported as difficult to express cleanly in LADM 184 .	184
Temporal change and dynamic environments	Legacy attributes that track shifting shorelines, fluctuating tides, seasonal changes, or time-qualified cadastral states require 4D or temporal extensions beyond the core model 188 196 197 198 .	188 196 197 198
Heterogeneous external land-use/cover classifications	Linking parcel records to outside land-use/cover datasets is hard because classifications, scale, and data quality vary widely, creating spatial-consistency and harmonization problems 199 .	199

The main pattern

The evidence points to a simple pattern: **LADM handles standard cadastral identity, rights, and parcel structure well, but the hardest legacy attributes are the ones tied to geometry complexity, verticality, time, and non-standard legacy representations** 51 168 186 188 190

. In practice, those cases usually need **extensions, extra packages, or additional standards** such as IFC or INTERLIS to bridge the gap between the legacy record and the LADM profile 3 167 174 189 200 201

.

Bottom line

If you are prioritizing migration effort, the most problematic legacy fields are usually **3D/solid shape fields, height-dependent legal spaces, time-varying rights, and messy identifier or raster-based map attributes**; by contrast, ordinary parcel, party, and RRR fields are much easier to map into LADM country profiles 3 51 184 186 190 192 195 196

.

Confidence level: **High** — multiple independent studies converge on the same difficult categories, especially 3D/volumetric representation, vertical rights, temporal extension, and legacy data-format mismatches, though the exact hardest field can vary by jurisdiction and record quality 184 185 188 192 199.

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Follow-up Suggestions

- How are 3D parcel rights modeled in LADM country profiles?
- How do LADM profiles handle temporal changes in cadastral records?
- Which external standards are used with LADM for raster cadastral data?

Examine how institutional governance, legal recognition, state oversight, data protection, and standards such as LADM/ISO 19152 influence blockchain-based land and real estate administration. Assess interoperability across registries, legacy systems, and stakeholders. Distinguish technical feasibility from legal validity and jurisdiction-specific evidence. Identify governance models, trade-offs, and research gaps. Preserve citations and avoid unsupported claims.

Processing steps

Blockchain in land administration governance

Blockchain in land administration governance

Blockchain-based land and real estate administration is **technically feasible in hybrid, permissioned, and standard-driven designs**, but **legal validity, institutional oversight, privacy compliance, and interoperability remain jurisdiction-specific constraints** that determine whether a system can be used as an authoritative land record rather than just a technical layer 164 202 203 204 205

.

How the main factors shape deployment

Factor	Influence on blockchain-based land systems	Practical implication
Institutional governance	Success depends on governance models that fit public administration, with clear decision-making, accountability, and control; studies stress that technology alone is insufficient 202 205 206 207 208 .	Systems work best as **complementary layers** under registry or multi-agency governance, not as standalone replacements 4 204 .
Legal recognition	The biggest barriers are legal recognition of distributed entries, enforceability of smart contracts, and the relationship with statutory registry functions 164 203 204 209 210 .	A technically working chain may still lack legal effect unless laws recognize its records and correction procedures 202 204 211 .
State oversight	Evidence repeatedly retains the role of registry agencies, inspectors, courts, or other public authorities as final arbiters or approvers 34 171 212 .	Oversight is usually **off-chain and institutional**, with blockchain preserving auditability rather than replacing administrative authority 4 171 202 .
Data protection	Privacy and data protection remain central, especially where personal land data, digital ID, and public accessibility are involved 213 214 215 216	Designs often use **hashes, fingerprints, off-chain storage, permissioning, or hybrid architectures** to avoid exposing sensitive

Factor	Influence on blockchain-based land systems	Practical implication
	.	records directly on-chain 171 203 212 217 .
Standards such as LADM / ISO 19152	LADM supplies the common semantic model for parties, rights/restrictions/responsibilities, parcels, source documents, and country profiles; it is repeatedly used to structure blockchain integration 3 34 51 167 171 171 .	LADM makes blockchain land systems **interoperable and jurisdiction-specific**, but it does not by itself solve legal validity or governance 51 167 171 .

Interoperability across registries, legacy systems, and stakeholders

Interoperability is treated as a **semantic and institutional problem first, technical problem second**: LADM standardizes data meaning, while blockchain carries proofs, workflow states, and immutable audit trails 51 167 171 218

. Legacy registries are commonly integrated through **country profiles, external archive data models, linked databases, APIs, and off-chain documents**, rather than wholesale replacement 167 171

. Cross-stakeholder interoperability also depends on trust, acceptance, and coordination among registry offices, notaries, inspectors, customary leaders, and citizens 213 219 220 221

.

Technical feasibility vs. legal validity

The literature draws a clear line between **what can be built** and **what can be legally relied on**. Technical prototypes can automate verification, improve traceability, and reduce fraud, but legal validity still depends on statutory recognition, enforceable correction rules, and compliance with jurisdictional rules on privacy and property transfer 202 203 204 209 211

. A recurring pattern is **hybridization**: blockchain provides integrity and traceability, while public authorities retain the power to validate rights and resolve disputes 4 34 164

.

Governance models and trade-offs

- **Permissioned / consortium models**: favored when state oversight, controlled access, and data protection matter most, but they reduce the "fully decentralized" promise and can add architectural complexity 164 222

.

- **Hybrid off-chain/on-chain models**: preserve authoritative registries while using blockchain for proofs, audit trails, and workflow automation; this is the dominant practical pattern in the evidence 4 34 171

.

- **Public or permissionless transparency models**: useful where broad openness and multi-stakeholder visibility are prioritized, but they face stronger challenges around privacy, legal compliance, and institutional acceptance 214 218 223

.

- **Standards-led models**: LADM-centered designs improve interoperability and data coherence, but require country-specific profile development and do not eliminate the need for legal reform 167 170 179

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Evidence distribution

Position	Evidence pattern
Supporting hybrid, institutionally governed deployment	Strongly supported across LADM, blockchain land registry, governance, and interoperability studies 4 34 51 167 171 202.
Supporting technical feasibility	Strongly supported by prototypes and case studies showing automation, auditability, interoperability mechanisms, and improved processing 140 203 212 224 225 226 227.
Questioning legal validity / full replacement	Strongly supported by studies emphasizing legal uncertainty, enforceability gaps, privacy constraints, and the continuing role of public authorities 164 202 204 209 214 216.

Research gaps

The evidence repeatedly identifies gaps in **legal enforceability**, **organizational implementation**, **stakeholder acceptance**, **privacy-preserving design**, and **jurisdiction-specific profile development** 167 202 206 213 214 216 228

. There is also a clear gap between **prototype-heavy technical literature** and **real-world deployment evidence**, especially for systems that must integrate registries, courts, notaries, and customary institutions across one legal order 164 220 229 230

.

Bottom line: blockchain can support land and real estate administration as a **trusted workflow and audit layer**, but the evidence says its success depends less on cryptography than on **governance design, statutory recognition, privacy compliance, and LADM-based interoperability**; where those are weak, blockchain remains technically promising but legally incomplete 51 164 167 202 204 216

.

Confidence level: **High** — multiple independent sources converge on the same hybrid pattern, and the strongest disagreements are not about technical possibility but about legal validity, institutional fit, and jurisdiction-specific implementation conditions 4 34 164 202 204 206.

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Follow-up Suggestions

- How do LADM country profiles support blockchain land registry interoperability?
- What governance model is used in hybrid blockchain land registries?
- How is legal recognition of blockchain land records established in practice?

How do LADM country profiles support blockchain land registry interoperability?

Processing steps

LADM country profiles and blockchain interoperability

LADM country profiles and blockchain interoperability

LADM country profiles support blockchain land registry interoperability by acting as the **shared semantic and structural bridge** between legacy registry data and the blockchain layer: they standardize parties, rights/restrictions/responsibilities, spatial units, and source documents so different systems can exchange land data consistently, while blockchain adds traceability and integrity on top of that shared model 167 231 232 233

.

How the profile helps interoperability

Interoperability function	What LADM country profiles do	Why that matters for blockchain
Common vocabulary	Provide a shared ontology/terminology for land administration across organizations and countries 231 233 234 .	Smart contracts and registry interfaces can refer to the same entities instead of translating ad hoc local terms 233 235 .
Country-specific mapping	Adapt the ISO standard to national legal, institutional, and technical requirements through profile creation and testing 162 167 236 .	Blockchain systems can plug into existing legal/administrative structures instead of replacing them 167 235 237 .
Data exchange and integration	Support data exchange, system integration, and consistency across cadastre, registry, valuation, taxation, and land use 168 233 .	Blockchain can synchronize proofs, hashes, or workflow states with authoritative off-chain records 233 238 .
Legacy-system bridging	Map existing cadastral concepts into LADM classes and validate conformance through formal mapping 162 175 236 .	Legacy registries remain authoritative while the blockchain layer interoperates through a standardized model 175 235 .
Stakeholder coordination	Make communication possible between land registry and cadastral	Multiple agencies and users can share a common record structure

Interoperability function	What LADM country profiles do	Why that matters for blockchain
	organizations, professionals, and authorities 232 234 .	without needing identical internal systems 168 239 .

What this looks like in practice

- **LADM is not the blockchain**, but the model that lets a blockchain land system "speak the same language" as the registry and cadastre 174 233

.

- **Country profiles localize the model** so blockchain integration fits national laws and procedures, which is crucial for interoperability across jurisdictions and institutions 167 237 240

.

- **Formal profile testing and conformance mapping** help ensure that the blockchain-connected system can exchange data reliably with existing land infrastructure 162 175

.

- **Profiles also improve transparency** by structuring source documents and role information, which helps with auditability when blockchain is used as a traceability layer 232 233

.

- **Technical interoperability is easier than legal interoperability**: several sources emphasize that profile-driven integration can improve data exchange, but legal and organizational alignment still depends on jurisdiction-specific rules and institutional coordination 235 237 240

.

Bottom line

LADM country profiles make blockchain land registry interoperability possible by **standardizing the land data model first and the ledger second**: they convert local cadastral and registry concepts into a common, testable structure that blockchain

systems can connect to for shared records, workflow synchronization, and cross-institution data exchange 167 168 175 233 235

. The main payoff is not replacing registries, but enabling blockchain to operate across them without breaking national legal and institutional differences 237 240

.

Confidence level: **High** — multiple LADM country-profile and interoperability studies consistently describe the same pattern of profile-first standardization, formal mapping, and hybrid integration with existing land systems 162 167 175 233.

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Follow-up Suggestions

- How are LADM country profiles mapped to blockchain smart contracts
- What legal reforms enable blockchain records to have registry effect
- How do permissioned blockchain land registries handle privacy

How are LADM country profiles mapped to blockchain smart contracts

Processing steps

LADM country profile mapping to smart contracts

LADM country profile mapping to smart contracts

Direct answer

LADM country profiles are mapped to blockchain smart contracts by using the **country profile as the legal-semantic specification** and the **smart contract as the executable workflow layer**: the profile defines the jurisdiction-specific parties, rights/restrictions/responsibilities, spatial units, and source documents, and the contract

encodes validation rules, authorization checks, and state changes against that structure 3 34 51 167

.

How the mapping works

LADM country-profile element	Smart-contract role	Practical meaning
Parties	Identity, authorization, role control	The contract checks who may submit, verify, approve, or query records 3 48 51 .
RRRs	Core business logic and state transitions	Rights, restrictions, and responsibilities become executable rules for registration, transfer, and validation 3 171 184 185 .
Spatial units / parcels	Asset reference and parcel linkage	The contract stores parcel identifiers, hashes, or pointers to geodata rather than full geometry on-chain 3 51 155 168 .
Source documents	Evidence and audit references	The contract links deeds, surveys, and other proofs so actions can be verified and logged 3 48 51 168 .
Country-specific legal rules	Preconditions and workflow constraints	Local law and institutional mandates shape the contract logic for that jurisdiction 161 167 168 173 .

What this means in practice

- The mapping is usually **profile-first, contract-second**: the LADM country profile is developed first, then classes are mapped into system objects and workflow rules 161 162 167 168

.

- The blockchain layer is typically **hybrid**, not a full replacement for land law or the registry; it automates selected steps such as verification, transfer, logging, or approval while the authoritative institutional role remains outside the chain 3 34 51

.

- In that setup, the contract implements **who can act, what can change, and under what conditions**, while the LADM profile preserves the jurisdiction's legal meaning and administrative structure 3 34 51 167

.

Bottom line

LADM country profiles map to smart contracts as a jurisdiction-specific schema-to-logic translation: the profile defines *what* must be represented in the land administration system, and the smart contract encodes *how* selected actions on those entities are validated and recorded on blockchain 3 34 51 167

.

Confidence level: High — the conversation consistently supports a profile-driven, selective mapping approach rather than a one-to-one replacement of the full LADM model 3 34 51 167.

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Follow-up Suggestions

- How do LADM classes map to Solidity contract functions?
- How are LADM country profiles validated for registry interoperability?
- What legal fields in an LADM profile need off-chain storage?

How do LADM classes map to Solidity contract functions?

Processing steps

Mapping LADM classes to Solidity

Mapping LADM classes to Solidity

Direct answer

The **LADM classes most clearly mapped to Solidity contract functions** are **Parties**, **Rights/Restrictions/Responsibilities (RRRs)**, **Spatial Units/Parcels**, and **Source Documents**: in Solidity, these become functions for **identity and role control**, **business-rule enforcement**, **state transitions for land rights**, and **evidence/audit handling** rather than a full one-to-one replacement of the LADM model 3 34 51

.

Mapping pattern

LADM class / element	Solidity contract function role	Typical function behavior
Parties	Identity, authorization, role control	Checks who can submit, verify, approve, or query records 3 .
RRRs	Core business logic and state changes	Encodes ownership, transfer, restriction, and validation rules 3 171 184 185 .
Spatial Units / Parcels	Asset reference and parcel linkage	Stores parcel identifiers, hashes, or pointers to geodata rather than full geometry on-chain 3 51 155 168 .
Source Documents	Evidence and audit references	Links deeds, surveys, and other proof so the contract can verify actions and log them 3 48 51 168 .

LADM class / element	Solidity contract function role	Typical function behavior
Country-specific legal rules	Preconditions and workflow constraints	Local law and institutional mandates shape the contract logic for that jurisdiction 161 167 168 173 .

What this means in Solidity terms

- The contract is usually the **executable workflow layer**, while the LADM country profile supplies the **legal-semantic structure** it must follow 3 34 51 167

.

- Solidity functions therefore tend to implement **who may act, what may change, and under which conditions** rather than encode the full land administration system 3 34 51

.

- For **RRRs**, the contract logic behaves like a **state machine**: rights can trigger registration or transfer actions, restrictions act as guards, and responsibilities become obligations that must be satisfied before the record advances 3 171 184 185 194

.

Bottom line

LADM-to-Solidity mapping is selective, not one-to-one: Solidity functions usually operationalize **Parties + RRRs + Parcels + Documents**, with the country profile defining the jurisdiction-specific rules and the smart contract enforcing them as executable conditions and state transitions 3 34 51 167

.

Confidence level: **High** — the available evidence consistently supports a profile-first, contract-second mapping centered on those four LADM elements 3 34 51.

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Follow-up Suggestions

- How are LADM RRRs represented as Solidity structs
- How are parcel geometries stored in blockchain land registries
- How do Solidity contracts enforce land registration workflows

What legal fields in an LADM profile need off-chain storage?

Processing steps

LADM profile off-chain storage requirements

LADM profile off-chain storage requirements

The legal fields most likely to need off-chain storage in an LADM profile are the detailed source documents, personally identifying party data, and the full legal/administrative records behind rights, restrictions, and responsibilities; blockchain is typically used for hashes, pointers, and auditability rather than storing those fields in full. 3 51 164 168

What should stay off-chain

LADM legal field / record type	Why it is typically off-chain	Typical on-chain substitute
Source documents	LADM explicitly includes integration with legal and spatial source documents, and blockchain designs usually keep the documents themselves outside the ledger while preserving traceability 51 168 .	Hashes, document IDs, or pointers 3 168
Party data with personal details	Land-registration blockchain studies emphasize privacy and data protection,	Minimal identity references, access-

LADM legal field / record type	Why it is typically off-chain	Typical on-chain substitute
	which pushes sensitive personal data off-chain or into permissioned storage 164 241 .	controlled links, or hashes 164 241
Full legal-record content for transfers, corrections, and corrections history	Blockchain proofs-of-concept use the chain to support the transaction workflow and maintain the registry's authoritative role, rather than replacing the legal record itself 3 164 .	Transaction status, audit trail, and state transitions 3 242
Detailed code tables / jurisdiction-specific legal semantics	RRR work shows that detailed classifications must be maintained in code tables and national profiles, especially for exchange across countries 171 .	Profile references and controlled vocabularies 167 171

Practical rule of thumb

LADM is a **conceptual model**, not a storage specification, so the profile can define legal meaning while the implementation decides what sits off-chain 174 243

. In blockchain-based land systems, the recurring pattern is: **store the legally sensitive or bulky content off-chain, and store only proofs, references, and state changes on-chain** 3 34 164

.

Bottom line

If you are designing an LADM profile for blockchain, the safest off-chain candidates are **source documents, personal party information, and full legal case files**; the chain should mainly hold **hashes, identifiers, permissions, and lifecycle events** that prove integrity without exposing the underlying legal record 3 51 164 168

.

Confidence level: **Medium** — the evidence strongly supports off-chain handling for documents and sensitive identity data, but it does not provide a single universal LADM rule that names every legal field one by one 3 51 164 168.

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Follow-up Suggestions

- Which LADM legal classes are typically stored on-chain
- How does LADM handle legal source document references
- How are privacy rules modeled in an LADM country profile

How are LADM country profiles validated for registry interoperability?

Processing steps

LADM country profile validation

LADM country profile validation

LADM country profiles are validated for **registry interoperability** by combining **formal conformance testing**, **mapping from registry concepts to LADM classes**, and **stakeholder/expert review** to check that the profile can exchange and integrate data across cadastral, registry, and related land-administration systems 162 167 241 242

.

How validation is done

Validation step	What is checked	Why it matters for interoperability
Profile creation against national law and practice	The profile is built from current legal, institutional, and cadastral requirements, not just the ISO model 167 242 243 .	Ensures the profile can fit existing registry workflows and legal boundaries.
Mapping to LADM classes	Existing cadastre and registry concepts are mapped to LADM entities, then compared with the profile classes 162 175 241 .	Shows whether registry data can be represented in a shared semantic structure.
Annex A / Abstract Test Suite conformance	The ISO 19152 abstract test suite is used to test whether the country profile conforms, with low/medium/high conformance levels 161 162 241 .	Provides a formal check that the profile is compatible with the standard, which supports exchange across systems.
Expert validation and review	Expert opinion is solicited to validate the model and refine classes, attributes, code lists, and multiplicities 168 .	Helps confirm that the model is usable in real registry operations.
Implementation and data-migration testing	Some studies move from conceptual validation to database implementation, data migration, and technical deployment in cadastral offices or web systems 242 .	Confirms the profile can interoperate in actual operational environments, not only on paper.
Semantic integration checks	Validation also examines whether the profile supports data exchange, integration, and common terminology across institutions 244 245 246	Ensures registry interoperability is not just syntactic but also semantic.

Validation step	What is checked	Why it matters for interoperability
	.	

What the evidence shows

- The **core validation mechanism** is a **mapping-and-conformance workflow**: existing registry data are aligned to LADM, then checked against the abstract test suite and national legal requirements 162 175 241

.

- Validation is often **profile-specific**, meaning the question is not whether the country profile matches LADM in the abstract, but whether it can support the country's registry integration needs, including data exchange, legal fit, and system integration 167 168 242

.

- Interoperability improves when the profile is tested against **multiple institutions and datasets**, because many registries store related but dispersed information that must be linked through a common model 242 245 247

.

- Where profiles are used for 3D, utility, or multi-register settings, validation also checks whether the profile can handle **additional spatial or thematic extensions** without breaking compatibility 248 249 250

.

Bottom line

LADM country profiles are validated for registry interoperability by proving that local registry concepts can be mapped into LADM, that the resulting profile passes ISO conformance tests, and that the model works in real institutional and technical settings for data exchange and integration 167 241 242 245 247

. The strongest pattern is **profile-first, conformance-tested, implementation-checked** validation rather than a purely theoretical check 161 162 168

.

Confidence level: **High** — multiple country-profile and interoperability studies consistently describe the same validation pattern, and the ISO abstract test suite appears as the main formal criterion across cases 162 167 241 242.

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Follow-up Suggestions

- How do LADM Annex A tests assess conformance?
- Which LADM classes are most critical for registry interoperability?
- How do country profiles support data exchange between land registries?

How are privacy rules modeled in an LADM country profile

Processing steps

Modeling privacy in LADM profiles

Modeling privacy in LADM profiles

Privacy rules in an **LADM country profile** are modeled **indirectly, as jurisdiction-specific constraints layered onto the core model**, rather than as a standalone universal privacy package: the profile is created through a process that explicitly considers legal, institutional, and technical issues and is then tested for conformance 167

. In practice, privacy is usually expressed through **access restrictions, role/party constraints, controlled use of documents and identifiers, and off-chain handling of sensitive content** rather than by storing personal data openly in the profile 3 51 164 168

.

How privacy is represented

Modeling mechanism	What it captures	Evidence
Profile-specific legal constraints	National privacy rules are incorporated during scope definition, profile creation, and profile testing, so the country profile reflects local legal boundaries 167 .	167
RRR-based restrictions	Privacy-related limits can be modeled as **restrictions** inside the RRR structure, because LADM centers land administration on rights, restrictions, and responsibilities 171 251 .	171 251
Controlled party/access modeling	Sensitive information can be tied to party roles and authorization logic so only permitted actors can view or use it; LADM's party structure supports this kind of controlled access 51 252 .	51 252
Document and metadata protection	LADM profiles emphasize integration with source documents, unique identifiers, versioning/history, and metadata, which lets implementations preserve traceability without exposing full sensitive records on the core model 51 .	51
Off-chain or protected storage	Where privacy is stronger than what the profile should expose, the implementation keeps sensitive legal or personal content outside the core visible layer and uses secure links, encryption, or access control instead 3 51 164 168 241 .	3 51 164 168 241

What this means in practice

- **Privacy is a modeling constraint, not just a storage choice**: the profile needs to reflect who may see or act on data, and that decision is made in the country-specific profile design and testing process 167 243

.

- **The most natural LADM locus for privacy is the RRR layer**, because restrictions can encode legal limits on disclosure, use, or access 171 251

.

- **Sensitive personal or legal records are usually not modeled in full on the open profile surface**; instead, the implementation keeps them protected and relies on references, metadata, and controlled access 3 51 164

.

- In domains with especially sensitive data, the literature stresses that **weak data protection and interoperability problems must be addressed together**, because privacy rules affect whether the system can safely exchange data across institutions 240 253

.

Bottom line

An LADM country profile models privacy rules by turning them into **country-specific restrictions, access conditions, and protected data-handling rules** that sit on top of the standard's core classes—especially **RRRs, parties, and source documents**—while sensitive personal or legal details are typically kept off the public-facing core and managed through controlled links or secured storage 3 51 164 167 168 171

.

Confidence level: **High** — the retrieved evidence consistently shows that privacy is handled through profile-specific legal constraints, RRR restrictions, and protected document/party handling, with no indication that LADM uses a single universal privacy class 51 167 171 240.

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Follow-up Suggestions

- Which LADM classes capture access control and privacy constraints
- How does LADM model consent for personal data use
- How are hashes and off-chain links represented in an LADM blockchain profile

Which LADM legal classes are typically stored on-chain

Processing steps

LADM legal classes on blockchain

LADM legal classes on blockchain

Typically, only a *minimal on-chain representation* of LADM legal content is stored: party/identity references, core RRRs in reduced form, and transaction or audit events; the detailed legal record, source documents, and sensitive personal data are usually kept off-chain. 51 171 251 252

What is usually on-chain

LADM legal class / legal data type	Typical on-chain treatment	Why
RRRs (Rights, Restrictions, Responsibilities)	Stored as **lightweight executable rules, status fields, or state transitions** rather than full narrative legal files 51 252 253 .	Blockchain systems use the chain to record validated changes and enforce conditions, not to replace the full legal dossier 252 253 .
Party references / identities	Stored as **minimal identifiers, role references, or permissioned links** 51 168 251 .	The chain needs enough information to authorize actions and link parties to transactions, but not necessarily all personal details 251 .
Transaction history / legal state changes	Stored as **immutable audit logs, timestamps, and transition records** 171 254 255 .	Blockchain is used to preserve traceability and reconstructable history of land events 254 255 .

LADM legal class / legal data type	Typical on-chain treatment	Why
Proofs / references to legal evidence	Stored as **hashes, pointers, or references to off-chain documents** 171 251 252 256 257 .	This preserves verifiability while avoiding bulky or sensitive on-chain storage 252 256 .

What is usually *not* stored fully on-chain

- **Full source documents** and other supporting legal files are commonly handled off-chain, with only references or hashes anchored on-chain 171 251 252 257

.

- **Sensitive personal party data** is typically kept off-chain for privacy and data-protection reasons 251

.

- **Detailed jurisdiction-specific legal semantics** are often maintained in national profiles or code tables rather than embedded verbatim in the ledger 51 171

.

Bottom line

The evidence points to a **hybrid pattern**: blockchain usually stores **core RRR state, party/authorization references, and legally relevant event history**, while the **full legal meaning and documentary evidence remain off-chain** 51 171 252

. So, if you mean “which LADM legal classes are stored on-chain in practice,” the answer is **mostly reduced RRR-related state and references, not full legal records** 51 251 253

.

Confidence level: **High** — multiple sources consistently describe on-chain storage as minimal, proof-oriented, and event-based, with detailed legal documents and sensitive data kept off-chain 171 251 252 256.

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Follow-up Suggestions

- How are LADM RRRs represented as smart contracts
- Which LADM party attributes should stay off-chain
- How are hashes used for LADM source documents

Evaluate empirical evidence on throughput, latency, scalability, security, resource consumption, and operational performance of blockchain systems proposed or used in land and real estate administration. Extract reported metrics, experimental settings, baselines, and limitations. Separate domain-specific evaluations from general blockchain benchmarks. Compare results only where methods are comparable; report denominators accurately and never generalize beyond the evidence. Preserve citations.

Processing steps

Performance evaluation of blockchain in land administration

Performance evaluation of blockchain in land administration

Blockchain systems proposed or used in **land and real estate administration** are usually evaluated only **indirectly and at prototype scale**, with the strongest quantitative evidence coming from **general blockchain benchmarks** rather than land-specific deployments; where land-focused studies do report metrics, they most often emphasize **throughput, latency, security/integrity, and transaction costs**, not full scalability under production-like registry loads 3 164 242 258

.

Domain-specific evidence: land and real estate administration

Study	Experimental setting	Reported metrics	Main result	Limitations
Hybrid smart-contract land	Proofs of concept in **Sweden, Australia (New**	No numeric TPS/latency values reported in the	The **hybrid approach** supported	The work is explicitly **proof-of-**

Study	Experimental setting	Reported metrics	Main result	Limitations
administration PoCs 3	**South Wales), and Canada (British Columbia)**, assessed with **business requirements adherence**, **technology readiness/maturity**, and **strategic grid analysis** 3.	abstract; evaluation is largely qualitative 3.	land-dealing business requirements and preserved the registry agency as ultimate arbiter of valid claims 3.	**concept**, and remaining issues include business models, stakeholder acceptance, partnerships, trust building, and legal issues linked to decentralisation and security 3.
Land registry system using blockchain 258	Conceptual land-registry system using blockchain, smart contracts, and cryptographic principles 258.	The abstract does **not** report numeric throughput, latency, or resource metrics 258.	Claims improved efficiency, security, transparency, and immutability for property records 258.	Future work is needed on **scalability, interoperability, and regulatory considerations** 258.
Smart contract application for LAS transactions 242	Solidity smart contract for land administration tasks such as ownership sharing, partial transfer, splitting/merging, and trade limits 242.	No benchmark numbers in the abstract 242.	Targets correctness problems, tampering, long registration times, and double spending 242.	Performance remains inferential in the abstract; the focus is functional design rather than measured

Study	Experimental setting	Reported metrics	Main result	Limitations
				throughput/latency 242.
Blockchain-based cadastral records in Romania 164	Review of research and Romanian legal/institutional fit 164.	No direct system benchmark numbers; the review synthesizes prior prototypes and implementations 164.	Finds the literature dominated by prototypes and architectural models; real-world implementations are limited and usually permissioned/hybrid 164.	Emphasizes that barriers are mainly **legal-institutional**, not strictly technical 164.
Land administration in developing countries 259	Systematic review of 20 articles from 2013–2024 259.	No direct performance metrics reported 259.	Most deployments remain prototype-stage; challenges include regulation, technical complexity, security concerns, and skills shortages 259.	This is a synthesis, not an empirical benchmark of one system 259.
Operational/blockchain architecture for real estate	Mostly conceptual or prototype real-estate transaction/tokenisation frameworks using Ethereum,	The abstracts do not provide comparable throughput/latency/resource numbers 260 261 262 263.	They consistently argue for higher security, transparency, traceability,	These are **design/proposal** studies, so they do not support production-

Study	Experimental setting	Reported metrics	Main result	Limitations
transactions 260 261 262 263	NFTs, IPFS, and smart contracts 260 261 262 263.		and lower intermediary costs 260 261 262 263.	scale performance conclusions 260 261 262 263.

General blockchain benchmarks and what they imply

Benchmark / study	Metrics and setup	Reported findings	Caution
BLOCKBENCH permissioned-platform comparison 264	Compared **Ethereum 1.4.18, Parity 1.6, and Hyperledger Fabric 0.6** on throughput, latency, scalability, and fault tolerance 264.	Hyperledger had the best throughput; Parity had the lowest latency; Ethereum had the highest latency; Hyperledger stopped responding over 16 servers 264 .	This is a **general platform benchmark**, not a land-registry workload 264.
General scalability review 265	Defines scalability metrics as transaction throughput/latency, read throughput/latency, and storage volume; some studies also measure success rate, CPU, and network resources 265.	Serves as metric taxonomy rather than a result paper 265.	Useful for interpreting land studies, but not evidence about land systems themselves 265.
Quorum case study 266	Uses Hyperledger PSWG-style definitions of throughput as successful tx/s and latency as time to availability on all correct nodes 266.	No land-specific result, but provides precise denominator language for comparing blockchains 266.	Definitions matter: throughput is **successful transactions only** 266.

Benchmark / study	Metrics and setup	Reported findings	Caution
Permissioned-blockchain survey 267	Summarizes that public chains typically have lower throughput/higher latency; private/consortium chains often improve both via PoA or modified architectures 267 .	General directional conclusion across surveyed studies 267.	Not a domain benchmark; should not be read as land-registry evidence 267.
Caliper-based blockchain evaluation example 268 269 270	Hyperledger Caliper is used to measure throughput, latency, and resource utilization on specified hardware 268 269 270.	These papers illustrate common benchmarking practice and the importance of hardware disclosure 268 269 270.	Results from different Caliper studies are **not directly comparable** unless the hardware and workloads match 268 270.

Metrics that are actually reported in the evidence

- **Throughput / TPS** is the most common operational metric, usually defined as **successful transactions per second** or committed valid transactions per second 266 271 272

.

- **Latency** is usually the elapsed time from submission to confirmation or inclusion, sometimes reported as **confirmation latency** or **block inclusion latency** 266 271 273

.

- **Resource consumption** appears as CPU, memory, disk I/O, network use, gas cost, or energy use depending on the platform and application 268 274 275 276 277

.

- **Scalability** is assessed via increasing clients, servers, organizations, shards, or transaction load rather than a single universal test 264 272 274 278 279

.

- **Security** is often measured indirectly through access control, tamper detection, fault tolerance, or consensus robustness rather than attack benchmarking 264 268 275 277 280

.

What can be compared, and what cannot

Comparable comparisons

- **Permissioned-platform comparisons** are comparable when the same benchmark family is used and the paper reports platform-level TPS/latency under one experimental setup, as in BLOCKBENCH 264

.

- **Resource-use comparisons** are only comparable when the same deployment and tooling are used, as in studies that explicitly specify hardware and Caliper settings 268 270

.

Not safely comparable

- The **land-registry prototypes** in the evidence cannot be numerically compared with one another because most abstracts do not report the same denominators, workloads, or hardware 3 164 242 258

.

- Results from **healthcare, IoT, or governance** blockchain papers should not be generalized to land administration, even when they share metrics like TPS or latency, because workloads and constraints differ 274 275 276 277

.

Bottom line

The empirical record for **land and real estate administration** is still dominated by **prototype and conceptual evidence**, so the strongest quantitative performance numbers in your dataset come from **general blockchain benchmarks**, not from registry-specific deployments 3 164 258 259

. The land-focused studies support the **feasibility** of hybrid, permissioned designs and improved auditability/security, but they rarely provide production-grade throughput, latency, or resource baselines that would justify broad performance claims 3 164 242 258

.

Confidence level: **High** — the evidence consistently shows limited domain-specific benchmarking and much stronger general platform benchmarking, though direct numeric comparison across the land studies is constrained by missing or non-matching denominators 3 164 258 264.

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Follow-up Suggestions

- Which land blockchain studies report measured TPS and latency
- How is blockchain throughput defined in registry benchmarks
- What hardware settings were used in land administration blockchain tests

Which land blockchain studies report measured TPS and latency

Processing steps

Blockchain performance metrics in land administration

Blockchain performance metrics in land administration

The **only retrieved land/real-estate study here that reports explicit numeric TPS and latency is *A Blockchain based Secured Land Registration System using Dynamic Smart Contracts***, which reports **>175 TPS** and an **average ownership transfer time of 420 ms versus 5600 ms** for the conventional process 281

. Most of the other land blockchain papers in this set say they evaluated throughput and latency, but their abstracts do **not** give the actual numbers 282 283 284 285.

Studies that report measured TPS and latency

Study	What is reported	Notes
A Blockchain based Secured Land Registration System using Dynamic Smart Contracts (2025)	**Throughput above 175 TPS** and **ownership transfer time reduced to 420 ms from 5600 ms** 281	This is the clearest numeric TPS/latency result in the retrieved land blockchain set.
LRS-Chain: A Permissioned Blockchain-Enabled Framework for Secure and Transparent Land Registry Systems (2025)	Performance evaluated with Hyperledger Caliper on **throughput and latency** for write/query operations 282	The retrieved abstract confirms measurement, but not the numeric results 282.
Land Registry System using Blockchain With Multiple Nodes (2023)	Reported **high throughput and low latency** in a simulated Ethereum/Geth network 283	No numeric TPS or latency values are given in the retrieved abstract 283.
Configuration and Implementation of Novel and Safe Methodology for Processing Real Estate Transactions by Utilizing Hyperledger Fabric (2023)	Evaluated **throughput, latency, and success rate** 284	The retrieved abstract names the metrics, but does not provide the numeric results 284.
BLRS: An Automated Land Records Management System Using Blockchain Technology (2023)	Experimental evaluation includes **average latency** and **transaction throughput** via Hyperledger Caliper 285	Again, the abstract indicates measurement but omits the actual values 285.

Bottom line

If you need **actual measured TPS and latency values**, the retrievable evidence currently supports **one clearly numeric land study**: the dynamic-smart-contract land registration paper with **>175 TPS** and **420 ms transfer time** 281. The other land blockchain studies retrieved here mention throughput/latency evaluation, but the supplied abstracts do not expose the measurements themselves 282 283 284 285.

Confidence level: **High** — the retrieved evidence clearly distinguishes one land study with numeric performance results from several others that only state the metrics were measured 281 282 283 284 285.

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Follow-up Suggestions

- Which land registry studies report latency values in milliseconds
- How do Hyperledger Caliper benchmarks measure registry throughput
- Which blockchain registry papers report transaction success rate

How is blockchain throughput defined in registry benchmarks

Processing steps

Blockchain throughput definition in benchmarks

Blockchain throughput definition in benchmarks

Direct answer

In registry-style blockchain benchmarks, **throughput** is defined as the rate of **successful transactions per second**; in the Quorum case study, it is specified more precisely as **successful tx/s**, with transactions counted only when they are successfully processed 266

. In the broader benchmarking summary, throughput is also described as **committed valid transactions per second** 271 272

.

Supporting detail

- The key denominator is **successful transactions**, not submitted or attempted transactions 266

.

- This matters because benchmarking throughput in registry settings should count only transactions that are actually accepted/committed by the network, rather than raw request volume 266

.

- The evidence also notes that throughput definitions are often paired with latency, where latency is the time from submission to confirmation or availability on correct nodes 266

.

Confidence level: High — the definition is stated directly in the cited benchmark and is consistent with the broader metric summary 266 271 272.

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Follow-up Suggestions

- How is blockchain latency defined in registry benchmarks?
- What counts as a successful transaction in registry benchmarks?
- How are throughput and latency measured together in blockchain benchmarks?

What hardware settings were used in land administration blockchain tests

Processing steps

Hardware settings in land administration blockchain testing

Hardware settings in land administration blockchain testing

The clearest reported hardware setting for **land-administration blockchain testing** in the retrieved evidence is an **AWS t2.large instance with 2 vCPUs and 8 GiB memory**, used to evaluate a Hyperledger Fabric v1.2 land-ledger solution in Sri Lanka under different transaction-density and node-failure conditions 281

. Most other land-administration blockchain papers in the set describe prototypes or local deployments but **do not report detailed hardware specifications** 34 282 283 284

.

Reported hardware settings

Land-administration blockchain study	Hardware / deployment setting	What was tested
Permissioned Distributed Ledgers for Land Transactions; A Case Study	**AWS t2.large instance; 2 vCPUs; 8 GiB memory** 281	Hyperledger Fabric v1.2 land-ledger models evaluated under different land-transaction densities and node-failure conditions 281
A Performance Evaluation of Two Blockchain Platforms for a Use Case	Systems ran **locally** in **Ganache** for Ethereum and **IBM Blockchain Platform** for Hyperledger Fabric; no CPU/RAM figures were given 282	Benchmarking of add/edit/delete/select operations on blockchain-stored data 282
A Proof of Concept of the integration of blockchain with an ISO 19152:2012 based Land Administration System	The abstract does **not specify hardware** 34	Proof-of-concept integration of blockchain with an LADM-based land administration system 34
Blockchain-based adoption framework for authentic land registry system in Malaysia	The abstract does **not specify hardware** 283	Prototype-oriented land-registry framework using smart contracts 283
Assessing Blockchain Integration in the Turkish Land Registry System	The abstract does **not specify hardware** 284	Conceptual model for blockchain-supported land registry integration 284

Bottom line

If you mean **actual machine settings reported in land-administration blockchain experiments**, the strongest explicit example is the **AWS t2.large (2 vCPU, 8 GiB RAM)** setup in the Sri Lanka land-ledger case study 281

. In the rest of the land-administration papers retrieved here, hardware is either described only generically as local or omitted entirely 34 282 283 284

.

Confidence level: **High** — one land-administration blockchain study reports concrete hardware, while several others in the set provide no hardware details and therefore cannot support a broader claim 34 281 282 283 284.

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Follow-up Suggestions

- Which land blockchain study used AWS t2.large
- What benchmarks were used in Hyperledger Fabric land tests
- Which land registry papers report CPU and RAM settings

Analyze blockchain smart-contract scalability for land and real estate administration. Evaluate throughput, latency, congestion, consensus costs, sharding, Layer-2, and cross-chain solutions. Verify cited metrics and experimental contexts. Separate general blockchain benchmarks from domain-specific evidence, compare trade-offs, identify research gaps, preserve citations, and never invent findings.

Processing steps

Blockchain scalability in land administration

Blockchain scalability in land administration

Blockchain smart-contract scalability for land and real estate administration is **mostly addressed through hybrid architectures and design proposals rather than large-scale empirical land-registry benchmarks**; the clearest measured improvements in this set come from **general blockchain benchmarks and non-land case studies**, while land/real-estate papers mainly report **qualitative scalability claims, prototype deployments, or small-scope simulations** 3 286 287 288 289.

What the evidence says

Evidence type	What it shows	Scalability take-away
Land / real estate domain evidence	Hybrid land-administration proofs-of-concept in Sweden, Australia (New South Wales), and Canada (British Columbia) were judged mainly by business-fit, maturity, and implementation challenges rather than numeric throughput/latency 3. Other land/real-estate studies describe scalable or efficient systems, but often without exposed benchmark values in the retrieved text 287 288 290 291.	In the land domain, scalability is usually argued through **workflow fit, reduced intermediaries, and pilot feasibility**, not through rigorous load testing at registry scale 3 287 288.
General blockchain benchmarks	A cloud-based smart-contract framework reported **2–5 s latency** under normal load and **20,000 simultaneous transactions in under 26 s** 292 . A public-sector blockchain architecture reported **2.8× throughput**, **65% lower average latency**, and **39% lower operating cost per transaction** on about **8,000 records** 289 .	These benchmarks show what optimized permissioned or hybrid systems can achieve, but they are **not land-registry workloads**, so they are only indirectly transferable 289 292.

Throughput and latency

Land-administration papers in this set mostly state that smart contracts improve transaction speed, responsiveness, or scalability, but they do **not** usually provide registry-grade throughput or latency figures in the retrieved text 287 288 291 293

. The strongest numeric evidence is outside the land domain: a blockchain smart-city smart-contract framework reported higher scalability and lower execution overhead after code optimization 294, while a blockchain governance system on Hyperledger Fabric showed throughput and latency gains on a real dataset 289. A separate smart-contract/blockchain framework for real estate operations claims scalability and transparency improvements, but the retrieved abstract does not expose measured TPS or delay values 295.

Congestion and consensus costs

The main congestion story is consistent across the evidence: **smart-contract systems become slower or costlier as transaction volume rises unless consensus overhead is reduced** 296 297 298

. One land-focused study proposes an efficient consensus algorithm and reports a **30%–40% reduction in message overhead** for committing transactions, which directly targets congestion cost rather than user-facing TPS 299

. More generally, PoA, PBFT-class, and similar permissioned approaches are repeatedly framed as lower-latency alternatives to resource-intensive consensus, while PoW is treated as unsuitable where efficiency matters 300 301 302 303.

Sharding, Layer-2, and cross-chain trade-offs

Technique	What it helps	Trade-off / limitation
Sharding	Repeatedly described as one of the most effective ways to raise throughput by splitting nodes and transactions into shards 297 304 305 . It is also presented as useful for land registries because districts or regions can map to different shards 297 .	Cross-shard communication adds overhead and consistency complexity; a cross-shard transaction can incur at least **O(mk)** network overhead and multiple consensus rounds, which lowers throughput 297 306 .
Layer 2	Sidechains, state channels, optimistic rollups, zk-rollups, and similar approaches are repeatedly proposed to offload work from the main chain and reduce latency, congestion, and fee pressure 302 307 308 309 .	Layer-2 systems add integration complexity and can create security or applicability constraints depending on the use case 307 308 309 .
Cross-chain / interoperability	Real-estate and land papers frame interoperability standards as important for secure communication	The retrieved evidence does not provide land-specific measured cross-chain throughput or latency; it treats interoperability as a design

Technique	What it helps	Trade-off / limitation
	between blockchains and traditional systems 310 311.	requirement rather than a benchmarked solution 310 311.

What this means for land and real estate administration

The most supportable conclusion is that **land-registry scalability is still mostly an engineering design problem, not a solved performance result** 3 286 287

. The land-focused literature favors **private or permissioned chains, hybrid architectures, off-chain storage, and smart-contract automation**, because these approaches preserve legal workflow compatibility while limiting on-chain load 3 287 291 312

. When authors do discuss scale, they usually mean **more users, more records, or more automation**, not proven throughput under registry-wide stress testing 288 290 295.

Research gaps

- **Registry-scale benchmarking is sparse**: the land papers retrieved here rarely report comparable TPS, latency, or stress-test results 3 287 288.
- **Cross-shard costs are underexplored for cadastral workflows**: the general sharding literature shows clear overheads, but land-specific implementations are not benchmarked at that granularity 297 306.
- **Layer-2 is mostly proposed, not validated, for land systems**: it is widely recommended, but the retrieved land evidence does not yet show mature comparative testing against on-chain alternatives 302 307 309.
- **Consensus-cost reporting is inconsistent**: some land studies report reduced message overhead or energy-efficient consensus, but many give no denominator or runtime context 287 288 299.

Bottom line: for land and real estate administration, blockchain smart-contract scalability is best understood as a trade-off between **throughput gains from sharding/Layer-2/permissioned consensus** and **the added complexity of cross-shard coordination, interoperability, and legal-operational fit**; the strongest numeric performance evidence comes from general or adjacent domains, not from land registries themselves 3 289 292 306.

Confidence level: **High** — the retrieved evidence consistently points to hybrid, permissioned, and off-chain-assisted designs as the main scalability path, while also

showing that land-specific empirical benchmarking remains limited and uneven 3 297 299 309.

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Follow-up Suggestions

- Which land registry studies report measured TPS and latency
- How do cross-shard transactions affect cadastral workflows
- What Layer-2 designs are proposed for real estate records

Which land registry studies report measured TPS and latency

Processing steps

Blockchain land registry performance metrics

Blockchain land registry performance metrics

Direct answer

The retrieved **land registry** studies that report **measured TPS and latency** are:

- **A Blockchain based Secured Land Registration System using Dynamic Smart Contracts (2025)** — reports **>175 TPS** and an **average ownership transfer time of 420 ms**, compared with **5600 ms** for the conventional process 281

.

- **LRS-Chain: A Permissioned Blockchain-Enabled Framework for Secure and Transparent Land Registry Systems (2025)** — evaluated **throughput and latency** using Hyperledger Caliper for write/query operations, but the retrieved abstract does **not** expose the numeric results 282.

- **Land Registry System using Blockchain With Multiple Nodes (2023)** — reports **high throughput and low latency** in a simulated Ethereum/Geth network, but no numeric TPS or latency values are given in the retrieved abstract 283.
- **BLRS: An Automated Land Records Management System Using Blockchain Technology (2023)** — includes evaluation of **average latency** and **transaction throughput** via Hyperledger Caliper, but the retrieved abstract does not provide the actual numbers 285.

Supporting detail

Studies with explicit numbers

Only one retrieved land registry study provides clear numeric performance values: **A Blockchain based Secured Land Registration System using Dynamic Smart Contracts** with **>175 TPS** and **420 ms transfer time** 281.

Studies that mention the metrics but not the values

The other retrieved land registry studies indicate that **throughput/latency were measured**, but their abstracts do not disclose the exact TPS or latency figures 282 283 285.

Confidence level: High — the retrieved evidence clearly separates one land registry study with explicit numeric TPS/latency results from several studies that only state the metrics were evaluated 281 282 283 285.

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Follow-up Suggestions

- Which land registry studies use Hyperledger Caliper
- What TPS and latency values does LRS-Chain report
- Which blockchain land registry papers compare against conventional registration

How do cross-shard transactions affect cadastral workflows

Processing steps

Cross-shard transaction impact on cadastre

Cross-shard transaction impact on cadastre

Cross-shard transactions **slow cadastral workflows and make them more coordination-heavy**, because parcel, owner, and document updates that span shards must preserve **atomicity, isolation, and consistent state across shards** instead of being handled as a single local update 313 314 315 316 317

. In cadastral terms, that means workflows such as title transfer, parcel mutation, registry synchronization, and cross-institution document handling can face extra confirmation steps, higher latency, and more failure points when the involved records are distributed across shards or linked systems 318 319 320 321

.

What changes in the workflow

Workflow effect	What it means for cadastral operations	Evidence
More coordination steps	A cross-shard cadastral action may need extra messaging, commit coordination, or shard-to-shard confirmation before the registry state is final 313 314 316 .	313 314 316
Higher latency	Transaction confirmation time rises when more shards are involved, because cross-shard transactions often require consensus in both source and target shards 316 322 323 .	316 322 323
Atomicity risk	Cadastral workflows need all-or-nothing updates; cross-shard handling is specifically introduced to preserve atomicity and isolation, otherwise transaction failures can occur 313 314 315 .	313 314 315
More complex data governance	When cadastral records are split across domains or institutions, workflow logic must also manage document exchange,	318 320 321

Workflow effect	What it means for cadastral operations	Evidence
	validation, and semantic linkage, not just ledger writes 318 320 321 .	

What this means specifically for cadastre

- **Cross-parcel or cross-district updates become slower** when record placement or account allocation causes a transaction to span shards; sharding literature says the cross-shard ratio directly affects throughput and latency 317 324 325 326

.

- **Title transfer and registry finalization need stronger synchronization** because cadastral workflows depend on a consistent legal state, and cross-shard protocols commonly use two-phase or coordinator-style commit logic to enforce that consistency 314 315 316 327

.

- **Document-heavy workflows are especially affected**: the Turkish LADM-based external archive model shows that registry/cadastre transactions already rely on external document handling, so splitting data across shards adds another layer of retrieval and validation overhead 318

.

- **Workflow design can reduce the penalty** by grouping frequently interacting parcels/accounts into the same shard or by using hierarchical or locality-aware designs that reduce cross-shard traffic 316 317 325 326 328

.

Practical bottom line

For cadastral systems, cross-shard transactions are best understood as a **trade-off between scalability and workflow simplicity**: they can help the ledger scale, but they make parcel updates, ownership changes, and inter-institution document checks more dependent on coordination, consensus, and synchronization across shards 313 315 316 317 323

. The clearest implication is that cadastral workflows should be designed to **minimize cross-shard cases wherever possible** and reserve cross-shard handling for transactions that truly span multiple cadastral partitions 317 325 326.

Confidence level: **Medium** — the evidence is strong on sharding mechanics and transaction overhead, but direct cadastral workflow studies on cross-shard processing are limited, so the cadastral implications are partly inferred from the cited registry and interoperability models 317 318 325.

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Follow-up Suggestions

- Which sharding protocols support atomic cross-shard cadastral transactions
- How do locality-aware shards reduce cadastral transaction latency
- What cadastral data should stay co-located to avoid cross-shard updates

What Layer-2 designs are proposed for real estate records

Processing steps

Layer-2 blockchain solutions for real estate

Layer-2 blockchain solutions for real estate

Layer-2 designs proposed for real estate records

The retrieved evidence shows **three main Layer-2 patterns** for real-estate or land-record use cases: **off-chain transaction channels**, **rollup-style batching/verification**, and **hybrid on-chain/off-chain record layers that reduce base-chain load**. In the real-estate-specific material, the clearest proposed designs are **cross-chain channels for RWA settlement**, **batch-transfer/tokenization architectures**, and **Layer-2 scaling tied to off-**

chain storage or proof handling rather than a single standardized land-registry Layer-2 stack 313 314 315 316

.

Proposed Layer-2 designs

Design pattern	What it does for real estate records	Evidence
Cross-chain settlement channel	Supports **efficient settlement of real-world assets**, including assets such as real estate, by reducing repeated authentication and allowing the channel to stay open across multiple settlements rather than closing after each one 313 .	313
Batch-transfer / tokenization Layer-2 support	Improves efficiency for real-estate tokenization by using **batch transfers** and token-standard interoperability to reduce transaction costs for portfolio purchases and support scalable DeFi integration 314 .	314
Rollup-style Layer-2 scaling	The general Layer-2 literature identified in the retrieved set includes **Optimistic Rollups, ZK-SNARKs, and State Channels** as candidate designs, with an explicit warning that bandwidth and local proving can become bottlenecks 316 .	316
Layer-2 with off-chain storage	Real-estate tokenization and land-record proposals repeatedly pair blockchain with **off-chain or decentralized storage** to reduce on-chain cost and improve network performance, which is functionally Layer-2-like even when not labeled that way 312 314 317 .	312 314 317
Layer-2 as scalability upgrade for land/real-estate systems	Reviews of real-estate blockchain adoption describe a shift from Layer-1 toward **more scalable Layer-2 solutions** because of lower cost and greater development flexibility, though they do not specify a single design 315	315

Design pattern	What it does for real estate records	Evidence
	.	

What these designs look like in practice

- **For real estate tokenization**, the proposed Layer-2 logic is to **move repetitive settlement activity off the base chain**, then use the chain mainly for final settlement and verification; the xRWA framework does this with a **cross-chain channel** and reduced on-chain closure frequency 313

.

- **For property record systems**, the design emphasis is often **hybrid storage**: the blockchain anchors integrity and transfer logic, while bulky metadata, documents, or media are stored off-chain or in decentralized storage such as IPFS, which reduces base-layer pressure 312 314 317

.

- **For broader Layer-2 engineering**, the retrieved general evidence says the leading options are **state channels, optimistic rollups, and ZK-based approaches**, but these come with trade-offs in bandwidth demand, proof generation cost, and infrastructure dependence 316 318 319

.

Bottom line

For real-estate records, the proposed Layer-2 designs are **not yet a single mature standard**; they are mainly **channel-based settlement layers, rollup-style off-chain transaction handling, and hybrid on-chain/off-chain storage schemes** that reduce cost and congestion while preserving verifiability 313 314 315 316

.

Confidence level: **Medium** — the evidence clearly identifies several Layer-2 directions for real-estate-related systems, but most real-estate papers describe design intent rather than fully specified, benchmarked Layer-2 implementations 313 314 315.

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Follow-up Suggestions

- Which real estate papers use state channels
- How do optimistic rollups handle property records
- What off-chain storage is used in land registry systems

Critically assess smart-contract security in land and real estate administration, covering code vulnerabilities, consensus attacks, key management, privacy, and layered defenses. Verify citations, distinguish general blockchain findings from domain-specific evidence, evaluate performance trade-offs, legal oversight, and LADM/ISO 19152 alignment. Identify empirical gaps without overstating security or inventing findings.

Processing steps

Smart contract security in real estate

Smart contract security in real estate

Smart-contract security in **land and real estate administration** is discussed far more often as a **design and governance problem** than as a field with strong, domain-specific empirical security evidence; the retrieved studies consistently warn about code bugs, privacy exposure, legal compliance, and reliance on external oversight, while concrete security testing in land workflows remains limited to prototypes, local test chains, or qualitative proposals 3 242 311 312 329.

Security findings by theme

Theme	Domain-specific land / real estate evidence	General blockchain / smart-contract evidence	Critical takeaway
Code vulnerabilities	Land-registry and real-estate papers repeatedly cite tampering, fraud, double-spending, and correctness problems, and propose Solidity	General smart-contract studies identify reentrancy, access-control flaws, integer overflows, transaction-ordering dependence,	The land domain inherits the usual smart-contract bug surface, but the retrieved land papers mostly **assert**

Theme	Domain-specific land / real estate evidence	General blockchain / smart-contract evidence	Critical takeaway
	contracts to reduce them, but most do not publish deep vulnerability audits of the contract code itself 242 330 331 .	logic errors, and lifecycle-stage vulnerabilities; several detection tools and formal methods improve coverage, but no method guarantees absolute security 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 .	security benefits rather than **demonstrate** vulnerability resistance under adversarial testing 242 288 348 .
Consensus / platform attacks	Land papers emphasize permissioned or DPoS-style deployment and the need to integrate with government validation, but they rarely quantify resilience to consensus attacks or fork/Byzantine scenarios 3 242 349 .	General security papers note that consensus mechanisms and network architecture matter for security, and that regulation or auditing layers may be needed because blockchain systems do not enforce legal rules by themselves 346 350 351 .	For land administration, consensus choice is a **security-policy decision** as much as a performance one, but the retrieved evidence does not show registry-scale attack experiments 349 351.
Key management / identity	Domain papers frequently rely on government officials, digital identity, verified parties, encryption keys, QR/NFC tags, or permissioned access to bind real-world ownership to on-chain actions 242 312 352 353	General smart-contract and blockchain literature highlights authentication, signature verification, role-based control, and trusted-party onboarding as recurring issues 354 355 356 357 358 359 .	The security bottleneck in land systems is often not just code, but **identity binding**: if the person, parcel, and legal authority are not reliably linked, the smart contract

Theme	Domain-specific land / real estate evidence	General blockchain / smart-contract evidence	Critical takeaway
	.		cannot safely govern transfers 242 312 352 .
Privacy / data protection	Land and real-estate systems frequently use off-chain storage, IPFS, hashes, selective visibility, or hybrid designs precisely because property data and supporting documents are sensitive 3 312 360 361 .	General literature flags data protection, privacy disclosure, and the tension between immutability and privacy laws as core limitations of smart contracts and blockchain systems 354 355 362 363 364 .	The domain evidence supports a **hybrid privacy pattern**: keep legal documents and sensitive attributes off-chain, anchor only hashes or references on-chain 312 360 361 .
Layered defenses	Land-specific proposals combine smart contracts with IPFS, NFC, QR codes, government validation, registry linkage, and hybrid architectures to improve traceability and reduce fraud 3 312 352 360 .	General security work recommends audits, static/dynamic analysis, formal verification, AI-based detection, and lifecycle-aware monitoring; a dual-layer design with regulatory contracts and real-time monitoring is one concrete layered defense pattern 332 333 334 336 340 346 347 365 366 .	The best-supported defense model is **defense in depth**: secure code plus off-chain/legal supervision plus identity controls plus monitoring 312 336 365 366 .
Legal oversight / LADM alignment	LADM/ISO 19152 is repeatedly described as a conceptual model for parties, RRRs, spatial	General legal scholarship on smart contracts says code-only execution is insufficient for	In land administration, **LADM alignment strengthens legal**

Theme	Domain-specific land / real estate evidence	General blockchain / smart-contract evidence	Critical takeaway
	units, and spatial sources; it is not a direct implementation spec, and it expects local adaptation, external archives, and legal/geospatial integration 348 367 368 369 370 371 372 373 374 375 376 377 378 379 .	enforceability, remedies, signatures, jurisdiction, and public-law functions; institutional oversight remains necessary 351 356 358 359 380 381 382 383 .	**clarity**, but it does not remove the need for registrars, notaries, or other institutional checks where formalities and public-order functions matter 351 358 374 375 .

What the evidence supports most strongly

- **Code security is necessary but not sufficient.** The general smart-contract literature shows recurring vulnerability classes and the limits of current detectors, while land/real-estate papers mostly present contracts as solutions to fraud or tampering rather than as systems that have been comprehensively security-tested 242 288 329 332 335 339 341

.

- **Privacy protection is central in land systems.** Multiple land papers prefer hybrid designs with off-chain storage, selective visibility, or document hashes because cadastral and title records contain sensitive legal data 3 312 360 361

.

- **Legal oversight remains structurally required.** LADM gives a shared conceptual language for parties, RRRs, and spatial units, but the legal literature makes clear that smart contracts do not by themselves satisfy all formalities, enforcement, or public-law functions 351 358 368 372 375 380

.

- **Layered defenses are the dominant security architecture.** The strongest pattern across the evidence is not “blockchain alone,” but blockchain plus identity checks, off-chain evidence, monitoring, and regulatory or institutional controls 312 336 352 365 366

.

Performance trade-offs

Security improvements often come with **cost, latency, or operational complexity**. The general literature notes that formal verification, monitoring, and more complex detection pipelines can add overhead, while land-specific designs that combine blockchain with off-chain storage, permissioned governance, or external archives trade some decentralization for better legal fit and lower exposure of sensitive data 3 312 335 336 358 360

. The evidence therefore supports a **pragmatic security trade-off**: land systems gain safety by layering controls, but not by maximizing on-chain automation at the expense of legal validation and privacy 351 358 375

.

Empirical gaps

The main gap is that **very few retrieved land and real-estate studies provide direct, adversarial security evaluations** of deployed or realistically stress-tested smart contracts; most are conceptual, architectural, or prototype-oriented, and several explicitly note that legal-institutional barriers or future work remain 3 242 288 329

. There is also limited evidence on **consensus-attack resilience, key-compromise handling, and privacy leakage under real cadastral workloads**, so security claims should be treated as provisional unless a study reports explicit tests 346 347 349 350

.

Bottom line: the retrieved evidence supports smart contracts in land administration as **useful but security-sensitive infrastructure**: they can improve integrity and traceability, yet their safe use depends on layered controls, off-chain privacy handling, and legal oversight consistent with LADM/ISO 19152, not on code alone 3 312 358 360 375

.

Confidence level: **High** — the sources consistently agree on the vulnerability landscape, the need for hybrid defenses, and the legal/institutional limits of code-only land administration, though direct empirical security testing in the land domain remains sparse 3 242 332 336 358.

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Follow-up Suggestions

- How does LADM model smart-contract land registry workflows
- What formal verification methods fit cadastral smart contracts
- How can key recovery be handled in property blockchain systems

How does LADM model smart-contract land registry workflows

Processing steps

LADM modeling of smart-contract land registries

LADM modeling of smart-contract land registries

LADM models smart-contract land-registry workflows as a **structured land-administration process built around legal parties, rights, spatial units, and supporting documents**, with smart contracts acting as the execution layer for transactions while the registry remains the authoritative legal arbiter in hybrid designs 3 384 385

.

How the workflow is modeled

LADM element / workflow role	What it does in a smart-contract land registry	Evidence
Party / identity layer	Identifies the people and organizations involved in the transaction, so on-chain actions can be tied to recognized actors rather than anonymous addresses 384 386 .	384 386
RRR layer	Represents **rights, restrictions, and responsibilities** that the workflow must update, validate, or preserve during registration and transfer 384 387 388	384 387 388

LADM element / workflow role	What it does in a smart-contract land registry	Evidence
	.	
Spatial unit / parcel layer	Models the parcel or legal space being registered, including 2D and 3D cases, so the workflow can bind legal state to the correct spatial object 384 389 390 .	384 389 390
Surveying / source layer	Links registry actions to source documents, surveys, and external evidence, which is important because land workflows depend on supporting records rather than just ledger entries 385 391 392 .	385 391 392
Implementation / transaction layer	Smart contracts encode workflow rules such as ownership sharing, partial transfer, splitting/merging, and transfer limits; the contract executes the operational step, but the registry model defines what the step means legally 242 393 .	242 393

What this means in practice

- LADM gives the workflow its **conceptual schema**: who the parties are, what the parcel is, what rights are changing, and what evidence supports the change 384 385

.

- Smart contracts then automate the **process logic** for registration, transfer, and validation, but in the hybrid land-administration design the registry agency still serves as the **ultimate arbiter of valid claims** rather than letting code alone decide ownership 3

.

- The workflow is often designed as a **hybrid on-chain/off-chain system**: the chain stores the authoritative transactional logic or hashes, while documents, archive data, or external records stay linked off-chain to support legal verification and reduce repetition and fraud 3 392

.

- Where land systems move toward 3D or more complex rights structures, LADM supplies the vocabulary for **multi-dimensional parcels, legal spaces, and extended RRR definitions**, which smart contracts can then enforce procedurally 389 390 394

.

Bottom line

LADM does **not** replace the smart contract; it **structures the land-registry workflow** that the smart contract executes. In the retrieved evidence, that workflow is modeled as a legally grounded sequence of **party identification, RRR evaluation, spatial-unit binding, and document-backed validation**, with blockchain used to automate steps and preserve traceability while the registry or government authority remains the legal decision point 3 384 392

.

Confidence level: **High** — multiple LADM sources define the model's core packages and workflow-relevant extensions, and the smart-contract land-administration papers consistently describe hybrid execution rather than fully autonomous blockchain-only registration 3 384 385 392.

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Follow-up Suggestions

- How does LADM encode rights, restrictions, and responsibilities?
- How are 3D spatial units represented in LADM?
- What parts of a land registry workflow are usually kept off-chain?

What formal verification methods fit cadastral smart contracts

Processing steps

Formal verification for cadastral smart contracts

Formal verification for cadastral smart contracts

Cadastral smart contracts are best matched by a combination of *theorem proving*, *model checking*, and *symbolic execution*, with *runtime verification* as a lighter-weight supplement for deployed systems. The evidence also suggests that cadastral use cases especially benefit from methods that can express ownership-transfer correctness, safety/liveness, and resource or state invariants 384 385 386 387 388

.

Best-fitting methods for cadastral smart contracts

Formal verification method	Why it fits cadastral workflows	Main limitation
Theorem proving	Widely identified as one of the broadest-applicability methods for smart contracts; it is well suited to proving cadastral invariants such as correctness of parcel transfer, state consistency, and legal-condition preservation 384 387 389 390 391 .	Requires substantial specification effort and can be harder to automate than other methods 384 389 392 .
Model checking	Good for checking finite-state properties such as safety, liveness, and compliance rules; it is explicitly used in smart-contract verification workflows and is a strong fit when cadastral rules can be formalized as temporal logic or state-machine properties 388 393 394 395 .	State explosion becomes a problem for large or highly parameterized systems 389 396 .
Symbolic execution	Useful for exploring transaction paths and finding bugs across contract branches; the survey literature says it is among the most widely applicable techniques, and SmartACE combines it	Can struggle with scale and path explosion, especially for complex cadastral logic 384 396 397

Formal verification method	Why it fits cadastral workflows	Main limitation
	with model checking to uncover deep multi-transaction bugs 384 389 396 .	.
Runtime verification	Best as a post-deployment safeguard for monitoring live cadastral workflows where not all properties are practical to prove statically; ContractLarva is presented as a lightweight option for Solidity monitoring 384 398 .	It detects violations during execution rather than guaranteeing full pre-deployment correctness 398 .

What is most appropriate for cadastre specifically

- **Ownership-transfer correctness and cadastre-reality consistency** are naturally formal problems: a cadastral formal model already defines correctness as correspondence between the legal situation and the cadastre, and uses a formalized situation calculus to analyze violations 386

.

- **Permission rules, access control, and transaction sequencing** fit well with model checking and theorem proving because those methods can encode explicit safety/liveness constraints and verify them against contract behavior 388 393 395

.

- **Multi-step registry workflows** benefit from symbolic execution and model checking together, since cadastral transactions often span several operations and require bug discovery across paths and states 387 396

.

- **Specification extraction matters**: approaches that translate natural-language requirements into formal specifications are especially relevant for cadastral systems, where legal and administrative rules are often written in prose first 395 399

.

Practical recommendation

For a cadastral smart contract, the strongest verification stack is:

1. **Formal specification layer** for parcel, party, ownership, and transfer rules 386 399.
2. **Theorem proving** for core invariants that must never fail, such as legal-state consistency and transfer correctness 384 390 391.
3. **Model checking** for authorization, safety, and workflow-order properties 393 394 395.
4. **Symbolic execution** to expose multi-transaction bugs and edge cases 388 396.
5. **Runtime verification** for live monitoring once the contract is deployed 398.

Bottom line

If the goal is **high assurance for cadastral contracts**, the most suitable formal methods are **theorem proving plus model checking**, with **symbolic execution** used to broaden bug coverage and **runtime verification** used to monitor real deployments 384 388 389 398

. The main gap is not the absence of methods, but the need to translate cadastral/legal rules into precise formal specifications that these tools can verify 386 399

.

Confidence level: **High** — the retrieved literature consistently identifies theorem proving, model checking, symbolic execution, and runtime verification as the core formal methods for smart contracts, and the cadastral formal-model paper gives a clear fit for ownership and correctness reasoning 384 386 389 398.

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Follow-up Suggestions

- How do you formalize cadastral ownership invariants in TLA+

- Which smart contract properties can symbolic execution verify in land registry workflows
- How is runtime verification used for deployed property blockchain contracts

How can key recovery be handled in property blockchain systems

Processing steps

Key recovery in property blockchain systems

Key recovery in property blockchain systems

Key recovery in property blockchain systems is best handled with a mix of threshold recovery, identity-backed re-issuance, and revocation workflows rather than a single "forgotten password" style reset. The retrieved evidence supports four practical patterns: **self-sovereign identity (SSI) for user-controlled recovery**, **threshold secret sharing / threshold signatures for decentralized recovery**, **trusted authority or certifying-issuer re-provisioning for land-registry access**, and **revocation plus replacement of compromised keys** 384 385 386 387 388 389

.

What the evidence supports

Recovery pattern	How it works	Why it fits property systems
SSI-based recovery	A land-registry SSI review says secure identity is essential, and existing identity schemes are limited by centralization and third-party dependence; SSI is proposed to give users more control over identity information 384 .	Property systems need identity continuity, so recovery can be tied to identity verification rather than ad hoc manual resets 384 .
Threshold secret sharing	A blockchain authentication paper reports that Shamir's Secret Sharing supports **threshold-based recovery without a single point of failure**, with only microsecond-to-millisecond overhead 385 .	This is well suited to shared custody of registry credentials or recovery shares across offices, notaries, or trustees 385 .

Recovery pattern	How it works	Why it fits property systems
Threshold key/status auditing	A key-management framework combines **FROST-based threshold signatures** with a trusted third party that verifies key status and audits metadata; it also supports **key recovery efficiency** and real-time anomaly detection 387 .	Property platforms can use this to recover access while also checking whether a key is revoked, expired, or suspicious 387 .
Authority-backed re-issuance	In a land-registry setting, authorized department access requires a **certifying authority to provide the key**, and the key-generation process is used to validate the key and support use at different levels 386 .	This matches registrar-led recovery, where a verified official can reissue access after identity checks 386 .
Revocation and replacement	Cocoon shows that compromised keys can be **individually revoked** without certificates, addressing the fact that many existing systems assume keys stay safe when they may not 388 .	For property records, revocation is critical because a lost or stolen key should not continue to authorize transfers 388 .
Adaptive key management	A blockchain access-control scheme proposes **adaptive key management** with digital certificates and reports better execution, encryption, and decryption times 389 .	This supports recovery designs that can adjust keys dynamically instead of relying on static credentials 389 .

Recommended design for property blockchain systems

A robust approach is to combine these elements:

1. **Primary identity layer:** use SSI or another identity-management layer so ownership and access are tied to verified identity rather than to one private key alone 384 390 391

.

2. **Recovery quorum:** split recovery power across multiple trusted parties using secret sharing or threshold signatures so no single actor can unilaterally seize a property account 385 387 392

.

3. **Registrar fallback:** allow a certifying authority or registry office to reissue access only after legal and identity checks, which is consistent with land-registry practice 386

.

4. **Revocation first:** if compromise is suspected, revoke the key and issue a new one rather than trying to “repair” the old key 388

.

5. **Adaptive controls:** use certificate-based access control and dynamic key management so recovery can be logged, audited, and adjusted over time 389

.

Practical trade-offs

- **Threshold recovery** reduces single-point failure risk, but it adds coordination overhead and requires careful selection of trustees or quorum members 385 387

.

- **Authority-backed recovery** is simpler for land offices, but it reintroduces trust in a central actor and should be constrained by audit trails and role separation 386 389

.

- **SSI-based recovery** improves user control, but the evidence notes that current identity solutions still have compliance and design limitations that need refinement 384 391

.

- **Revocation mechanisms** are necessary for security, but they must be paired with a reliable replacement path so legitimate owners do not lose access 388

.

Bottom line

For property blockchain systems, the strongest recovery model is **hybrid**: **use SSI for identity continuity, threshold secret sharing or threshold signatures for decentralized recovery, registrar-issued revalidation for legal assurance, and immediate revocation of compromised keys** 384 385 386 387 388 389

. That combination best matches land records, where recovery must preserve both **technical access** and **legal authority** 384 386

.

Confidence level: High — multiple sources converge on threshold recovery, identity-backed access, and revocation as the relevant patterns, including direct evidence from blockchain identity, land-registry key handling, and threshold-recovery studies 384 385 386 387 388 389.

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Follow-up Suggestions

- How does Shamir secret sharing support key recovery in blockchain identity systems
- How is key revocation handled after private key compromise in land registries
- How can self-sovereign identity be integrated with property blockchain access control

How does Shamir secret sharing support key recovery in blockchain identity systems

Processing steps

Shamir Secret Sharing for Key Recovery

Shamir Secret Sharing for Key Recovery

Direct answer

Shamir secret sharing supports key recovery in blockchain identity systems by splitting a secret key into multiple shares so that recovery requires a threshold quorum rather than one holder or one device. In the retrieved evidence, it is described as enabling **threshold-based recovery without a single point of failure**, with only **microsecond-to-millisecond overhead** 385

.

How it helps in practice

- **Resilience:** because the key is divided into shares, losing one share does not destroy recoverability; the secret can still be reconstructed if enough shares are available 385

.

- **No single point of failure:** recovery does not depend on one custodian, which is especially useful in decentralized identity settings 385

.

- **Operational feasibility:** the reported performance overhead is small enough to support practical use in blockchain authentication and communication workflows 385

.

- **Fit with decentralized identity:** related key-management work in DID-based environments also emphasizes **threshold signatures** and **key recovery efficiency**, which aligns with the same general recovery model 387

.

Bottom line

In blockchain identity systems, Shamir secret sharing makes key recovery possible by turning a fragile single private key into a **threshold-recoverable secret** distributed across multiple parties or components 385

.

Confidence level: High — the evidence directly states threshold-based recovery without a single point of failure and low overhead for Shamir secret sharing in a blockchain

authentication context 385, with supporting alignment from DID key-management work 387.

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Follow-up Suggestions

- How does threshold reconstruction work in Shamir secret sharing?
- How are Shamir shares managed in decentralized identity wallets?
- How do threshold signatures integrate with blockchain DID recovery?

How is key revocation handled after private key compromise in land registries

Processing steps

Private key revocation in land registries

Private key revocation in land registries

After a private-key compromise in land registries, the retrieved evidence points to **revocation plus re-issuance through an identity-checked authority workflow**, not a pure on-chain self-service reset: compromised credentials are revoked, revoked/expired items are excluded from new proofs, and access is re-established only after registry or issuer validation 384 387 400

.

How revocation is handled

Mechanism	How it works in the evidence	Land-registry implication
Credential revocation	A blockchain revocation system stores revocation information and status data so	A compromised key should be marked invalid, and downstream transactions

Mechanism	How it works in the evidence	Land-registry implication
lists / status checks	revoked credentials can be checked efficiently and decentralizedly 401 . A key-audit framework similarly has the wallet query issuer registry status and exclude revoked or expired credentials before generating a proof 387 .	should refuse it before any registry action is finalized 387 401 .
Authority-backed revalidation / re-issuance	Land-registry identity solutions emphasize that a reliable system needs secure identity and that current identity schemes are often too centralized or third-party dependent; SSI is proposed to give users control while still supporting verification 384. The Turkey LADM external-archive model shows land transactions depend on institutional document checks and manual validation by the responsible authority 400 .	After compromise, the owner's identity and entitlement are revalidated by the registry or issuer, then access is reissued under a new credential rather than restoring the old one 384 400 .
Replacement of the compromised key	SSI wallet work treats backup/recovery as a separate function from the original private key, and a decentralized wallet design supports key recovery in the event of device theft or damage 402 403 .	The compromised key is replaced; the registry should not continue trusting the old key even if the owner regains access 402 403 .
Auditability and anomaly detection	The key-audit framework combines threshold signatures, metadata auditing, and real-time anomaly detection, while maintaining low response time 387 .	Registry operators can detect suspicious use around the compromise and trace which credential submissions were affected 387 .

What this means in practice

- **Immediate step:** revoke the compromised credential or certificate so it cannot sign further land transactions 387 401

.

- **Containment step:** block generation of new proofs or submissions that rely on the revoked credential 387

.

- **Recovery step:** verify the claimant's identity and legal status through the registry or issuer, then issue a fresh credential or key 384 400

.

- **Ongoing control:** use audit logs and status checks so revoked credentials remain excluded from later transactions 387 401

.

Bottom line

In land registries, key revocation is best understood as **credential invalidation plus controlled re-issuance**: the compromised key is revoked, status checks prevent further use, and the registry re-enables access only after identity and entitlement are revalidated by an authoritative process 384 387 400 401

.

Confidence level: **Medium** — the evidence is strong on decentralized revocation, status verification, and registry-backed identity checks, but it does not provide a single end-to-end land-registry revocation protocol specific to private-key compromise 384 387 401.

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Follow-up Suggestions

- How do land registries record revoked blockchain credentials?
- What role do credential status lists play in registry transactions?
- How is key reissuance authorized after identity verification?

How can self-sovereign identity be integrated with property blockchain access control

Processing steps

Self-Sovereign Identity in Property Blockchains

Self-Sovereign Identity in Property Blockchains

Self-sovereign identity (SSI) can be integrated with property blockchain access control by using **DIDs and verifiable credentials as the identity layer**, **smart contracts as the authorization layer**, and **selective-disclosure / zero-knowledge checks** so users prove eligibility without exposing unnecessary personal data 400 401 402 403 404

. In land and property settings, this is usually paired with **permissioned or governance-supervised registry workflows**, so access control protects records while the land authority still anchors legal validity 384 405 406 407

.

How the integration works

Layer	SSI role	Property blockchain access-control role
Identity	Users hold **DIDs** and credentials under their own control rather than relying on a central identity provider 384 400 401 404 .	The blockchain uses those credentials to decide who may view, submit, or update property records 402 403 408 .
Verification	The user presents a **verifiable credential** or proof of attributes to show entitlement 400 401 403 409 .	Smart contracts verify the proof and then grant or deny access according to policy 402 403 410 .
Privacy	SSI supports **selective disclosure** and privacy-preserving	Property systems can keep sensitive land data off-chain or minimize what is

Layer	SSI role	Property blockchain access-control role
	presentations, including zero-knowledge proofs 400 403 410 411 .	revealed on-chain while still enforcing access rules 384 407 412 .
Revocation and recovery	Credentials or identities can be **revoked or rotated** when compromised 409 411 413 414 .	Access control can reject revoked identities and support re-authorization after key loss or account change 409 413 414 .

Practical design pattern for property systems

A workable architecture is:

1. **Issue SSI credentials** to owners, registrars, notaries, or other authorized parties, with DIDs anchored on-chain and credentials held in user wallets 384 400 403

.

2. **Define access policies in smart contracts** for actions such as reading a parcel record, submitting a transfer request, or approving a registry update 402 403 406

.

3. **Require credential proofs at request time**, rather than usernames or passwords, so the user proves "I am authorized" without revealing more than needed 401 403 409 410

.

4. **Use selective disclosure / ZK proofs** when property data or identity attributes are sensitive, especially in government or land-administration contexts 403 407 411 412

.

5. **Support revocation, rotation, and recovery** so access can be withdrawn when keys are lost or compromised 387 388 409 411 413 414

.

What the land-specific evidence shows

- A Moroccan cadastre prototype integrates **blockchain + SSI** to secure property registration and ownership transfer, using SSI wallets and smart contracts to improve transparency, processing speed, and fraud prevention 405

.

- A land-registry review concludes that SSI is useful because traditional identity solutions are too centralized and third-party dependent for blockchain-based land systems 384

.

- A land-registry comparison paper says identity is essential for reliable blockchain land registries and finds SSI better aligned with privacy, reliability, and user control than older identity models 384

.

- A real-estate land-registration framework uses blockchain to automate ownership transactions while adding identity verification and secure document handling, which is consistent with SSI-style access control even when SSI is not named explicitly 407 412 415

.

Main trade-off

SSI improves **user control, privacy, and portability**, but property systems still need **institutional oversight** because blockchain alone does not solve legal authorization, trust, or registry governance 384 401 406 407

. That means the strongest design is **SSI for authentication plus smart-contract policy enforcement plus registry supervision** rather than fully autonomous access control 384 405 406

.

Confidence level: **High** — the retrieved evidence directly supports SSI-based DIDs/VCs, smart-contract policy enforcement, privacy-preserving proofs, and revocation/recovery patterns for blockchain access control, including land-registry-specific prototypes and reviews 384 387 388 400 402 403 405 410.

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Follow-up Suggestions

- How do verifiable credentials enforce land parcel permissions on chain
- How are zero-knowledge proofs used in property registry access control
- How is revocation handled for SSI credentials in land registries

How are zero-knowledge proofs used in property registry access control

Processing steps

Zero-knowledge proofs in property registries

Zero-knowledge proofs in property registries

Zero-knowledge proofs are used in property registry access control to let a requester **prove they are entitled to view, submit, or update registry data without revealing the underlying personal attributes, document contents, or full identity data**. In the retrieved evidence, the pattern is **credential-based selective disclosure plus proof of possession/binding**, sometimes combined with **revocation checks** and **smart-contract verification**; land-registry-specific ZKP workflows are sparse, so the property-registry reading is partly inferred from broader blockchain identity and access-control designs 384 416 417 418 419

.

How ZKPs are used

Access-control function	What the ZKP proves	Why it matters for property registries
Selective disclosure	The holder reveals only chosen attributes while proving the hidden attributes are valid 416 419 420 .	A registry can confirm eligibility without exposing full identity, legal status, or sensitive parcel-linked data 416 419

Access-control function	What the ZKP proves	Why it matters for property registries
		.
Proof of credential possession	The user proves possession of a valid signed credential or certificate, often with a proof of knowledge of the issuer signature and the holder's binding key 416 421 .	Useful for proving "I am an authorized owner/agent/notary" without revealing the credential itself 416 421 .
Membership / authorization checks	The prover shows membership in an approved set or role group using ZK membership-style proofs 417 422 .	A registry can restrict sensitive actions to approved parties while keeping the allowed set private 417 422 .
Predicate-based policy checks	The user proves statements like ranges or set membership about hidden attributes 419 423 .	Enables policy checks such as age, authority level, or time-validity conditions without exposing the raw data 419 423 .
Revocation-aware verification	The proof can incorporate revocation predicates or status checks so invalid credentials are rejected 411 420 424 .	Important when a registry must block compromised, expired, or revoked access credentials 411 420 424 .
Context / sender binding	The proof is tied to the intended on-chain sender or authorization context to prevent replay or reuse by others 418 425 .	This helps stop proof reuse in public ledgers, where a valid access proof might otherwise be copied and resubmitted 418 425 .

What this looks like in practice

- **User proves eligibility, not identity details.** ZKPs allow a requester to demonstrate possession of an approved credential or attribute set while hiding unnecessary personal information 416 426 427

.

- **Registry rules are enforced by smart contracts.** The property blockchain can verify the ZKP before granting read/write access, so access control becomes cryptographic rather than purely username/password based 418 428 429

.

- **Sensitive documents stay off-chain or hidden.** ZKP-based access control fits hybrid designs where the chain verifies authority and off-chain storage retains the documents or metadata 430 431

.

- **Revocation and auditability remain necessary.** ZKPs improve privacy, but the evidence also emphasizes revocation, issuer status checks, and supervisory workflows so a valid proof must still be current and authorized 411 420 424

.

Property-registry-specific takeaway

The strongest supported use case is **privacy-preserving authorization**: a claimant proves "I am allowed to access this registry function" by showing a zero-knowledge proof over a credential or attribute set, while the registry verifies the proof, checks revocation/status, and grants access only if the proof is bound to the correct context 384 416 418

. That fits the land-registry literature's emphasis on SSI, selective disclosure, and institutional oversight rather than fully open disclosure 384.

Confidence level: **High** — multiple sources directly describe ZKPs for selective disclosure, membership/credential proofs, and revocation-aware access control, and the land-registry evidence aligns with these patterns even though direct property-registry ZKP deployments are limited 384 416 418 419.

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Follow-up Suggestions

- How do smart contracts verify verifiable credentials for parcel access
- What revocation methods are used for blockchain credentials
- How do selective disclosure proofs protect property data

How is revocation handled for SSI credentials in land registries

Processing steps

SSI credential revocation in land registries

SSI credential revocation in land registries

SSI credential revocation in land registries is handled as **status checking plus controlled invalidation and re-issuance**, usually through a **registry/issuer-managed revocation list or status registry** rather than a simple on-chain "delete" action 384 387 401 416

.

How revocation works

Mechanism	What it does	Land-registry relevance
Revocation/status registry	Issuers publish revocation status so verifiers can reject invalid credentials; blockchain-based revocation and status verification systems support this pattern 401 417 418 .	Fits land registries because verifiers can check whether a credential tied to an owner, registrar, or surveyor is still valid before any transaction is accepted 384 387 401.
Issuer-side exclusion of revoked credentials	Wallets or verification services query the issuer's registry and exclude revoked or expired credentials before generating or accepting a presentation 387 .	This prevents a compromised SSI credential from being used to authorize registry actions 387.

Mechanism	What it does	Land-registry relevance
Privacy-preserving revocation proofs	Modern SSI revocation schemes use Bloom filters, Merkle accumulators, cryptographic accumulators, VRF tokens, or ZK proofs so holders can prove non-revocation without exposing full status history 419 420 421 422 .	Important for land systems because revocation checks should not leak ownership, identity, or administrative history 419 421 422.
Registry-backed revalidation and re-issuance	After compromise, the claimant's identity and entitlement are rechecked by the registry or issuer, and a fresh credential is issued instead of restoring the old one 384 400 401 .	Matches land-administration practice, where authoritative validation remains necessary for legal continuity 384 400.
Audit and anomaly checks	Some SSI frameworks combine status verification with metadata auditing and anomaly detection to trace suspicious presentations 387 .	Useful for land registries because credential misuse can be investigated without exposing the underlying credential contents 387.

What the evidence says in practice

- **Land-registry SSI reviews** emphasize that revocation is a core requirement because identity systems must support both privacy and reliable invalidation of compromised credentials 384 390 391.
- **Blockchain revocation systems** for credentials show two common implementation styles: public status registries and privacy-preserving accumulator-based schemes 401 420 421 422

.

- **Operationally**, land registries should treat revocation as a workflow: mark the credential invalid, block future proofs using it, and require authoritative revalidation before issuing a replacement 387 400 401.

- **The main design tension** is that revocation must be scalable and privacy-preserving while still allowing offline or low-latency verification, which is why the literature keeps returning to status lists, accumulators, and zero-knowledge proofs 419 421 423

.

Bottom line

For SSI credentials in land registries, revocation is best understood as **privacy-preserving status publication plus authoritative revalidation**: the old credential is marked unusable in a revocation/status registry, verifiers check that status before accepting it, and the registry reissues access only after identity and entitlement are reverified 384 387 401 419 421

.

Confidence level: **High** — the retrieved evidence directly supports registry-backed status checks, issuer-side exclusion of revoked credentials, and privacy-preserving revocation mechanisms, including land-registry-specific SSI reviews and broader SSI revocation studies 384 387 401 419 421 423.

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Follow-up Suggestions

- How do smart contracts verify verifiable credentials on Hyperledger Indy
- How are revocation lists implemented for land registry credentials
- How does selective disclosure work for parcel authorization proofs

How do verifiable credentials enforce land parcel permissions on chain

Processing steps

Verifiable credentials for land permissions

Verifiable credentials for land permissions

Verifiable credentials enforce **land parcel permissions on chain** by letting a smart contract verify that a requester has the right attributes or role before it releases access or accepts a parcel action, while the land authority still anchors legal validity in the workflow [402, 403, 405, 45?].

How the on-chain enforcement works

Step	What the VC enables	Land-parcel effect
Credential issuance	A trusted issuer gives the owner, registrar, or other actor a verifiable credential tied to a decentralized identity 400 401 .	The system has a cryptographic proof that the caller is an approved participant, not just an arbitrary wallet address 384 405 .
Proof presentation	The holder presents a VC or selective-disclosure proof instead of raw personal data 403 410 .	The chain can check eligibility without exposing unnecessary identity details or property data 403 405 .
Smart-contract verification	The contract evaluates the credential/proof against policy and either authorizes or denies the request 402 403 410 .	Parcel actions such as reading, transfer requests, or updates are only allowed when the caller satisfies the policy 402 403 405 .
Revocation / rotation	Revoked or compromised credentials can be rejected, and access can be re-authorized after replacement 409 411 413 414 .	A lost or compromised credential stops being valid for parcel control until a fresh one is issued 413 414 .

What the land-specific evidence shows

- A land-registry review concludes that **identity is essential** for reliable blockchain land systems and that SSI/VC-style identity is better aligned with privacy and user control than older centralized identity models 384.
- A Moroccan smart-cadastre prototype integrates **blockchain + SSI wallets + smart contracts** to support property registration and ownership transfer under cadastral oversight, showing the practical pattern for VC-backed land workflows 405

.

- A decentralized land-registry design issues **ownership proofs with SSI** using Hyperledger Indy/Aries and permissioned blockchain validation, which is exactly the mechanism by which parcel permissions become enforceable on chain .
- A related land-registration framework stores documents off-chain and records access or verification events on-chain, which fits VC-based permissioning because the contract can verify the credential while the legal record remains auditable .

Why this is enforcement, not just authentication

VCs do more than confirm identity: they let the contract enforce **who may do what to which parcel** by embedding attributes such as role, entitlement, or authorization status into the proof that the chain checks 402 403 410

. In property systems, that means the contract can distinguish an owner, registrar, or citizen and permit only the parcel operations allowed for that role .

Bottom line

On chain, verifiable credentials enforce parcel permissions by serving as the **cryptographic gatekeeper** for smart-contract policy checks: the holder proves entitlement, the contract verifies it, revocation prevents stale access, and the land authority or registry remains the legal anchor for valid parcel actions 384 402 403 405 413

.

Confidence level: **High** — the retrieved evidence directly supports VC/DID-based authorization, selective disclosure, revocation, and land-registry prototypes that use smart contracts to enforce access and ownership workflows 384 402 403 405 413.

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Follow-up Suggestions

- How are selective disclosure proofs verified in land registry smart contracts?
- What DID methods are used for parcel authorization credentials?
- How is off-chain land evidence linked to on-chain credential checks?

Evaluate PoW, PoS, DPoS, and BFT/PBFT consensus for land and real estate administration. Compare energy use, throughput, latency, finality, decentralization, fault tolerance, and governance suitability. Verify cited evidence, define qualitative rating criteria, distinguish permissioned from public networks, and avoid universal rankings or unsupported claims. Identify domain-specific evidence gaps and preserve citations.

Processing steps

Consensus mechanisms for land administration

Consensus mechanisms for land administration

For land and real estate administration, PoW is the weakest fit for operational use, PoS is better on energy and baseline scalability for public chains, DPoS improves throughput/latency but concentrates governance more strongly, and PBFT/BFT fits best in permissioned or consortium land systems where a known validator set is acceptable. That conclusion is conditional, not universal: the right choice depends on whether the system is **public/permissionless** or **permissioned/private**, and on whether the priority is openness, auditability, speed, or institutional control 432 433 434 435 436 437

.

Qualitative rating criteria

Dimension	High	Medium	Low
Energy use	Minimal ongoing computation and no	Some validator overhead, but far below PoW 438 439	Heavy computational work and mining overhead 440 441 442

Dimension	High	Medium	Low
	mining-style waste 433 435 438		
Throughput / latency	Fast confirmation, low delay, suitable for registry workflows 435 443 444 445	Usable but load-sensitive or context-dependent 439 446	Slower, congestion-prone, or high-delay under public load 442 447
Finality	Strong or fast practical finality in permissioned settings 435 444 448	Some assurance but dependent on chain design 439 446	Slower or probabilistic settlement under open public conditions 447 449
Decentralization	Broad participation and weaker validator concentration 450 451	Distributed, but with role-based or stake-based concentration 439 452	Strongly validator-centralized or administrator-led 435 437
Fault tolerance	Handles Byzantine or faulty nodes well with known-member governance 435 444 448 453	Moderate, but sensitive to node selection or stake/voting design 452 454 455	Depends on high resource cost rather than explicit governance control 440 456
Governance suitability for land/real estate	Strong for registry-controlled, auditable workflows 435 458 459	Good for public-facing systems if governance trade-offs are accepted 438 460	Weak for routine registry operations because of cost and delay 440 447

Consensus methods by land-administration fit

Consensus	Energy use	Throughput / latency	Finality	Decentralization	Fault tolerance	Governance suitability
PoW	Highest energy cost; mining is repeatedly described as	Generally slower and less suitable for high-frequency	Not directly characterized as fast-	Strong openness and participation, but at the	Resilient in a public-network sense, but the evidence emphasizes	Poor fit for land registries that need predictabl

Consensus	Energy use	Throughput / latency	Finality	Decentralization	Fault tolerance	Governance suitability
	energy-intensive and computationally heavy 440 441 442 450	registry updates 440 442 447	finality for land systems in the evidence; public chains are described as slower to confirm 447 449	cost of performance 432 451 461	resource waste rather than registry-grade fault management 440 456	e cost, privacy, and operational speed 434 447 457
PoS	Much more energy-efficient than PoW; one source states Ethereum's PoS cut energy use by over 99% 438 439	Better throughput/efficiency than PoW, but can still suffer load-related latency and design-specific issues 439 446	Better than PoW in practice, but the evidence here does not establish universal land-registry finality guarantees 438 439	More decentralized than DPoS in principle, but some sources flag lower decentralization or fairness risks relative to ideal decentralization 450 454	Can support secure validator sets, but security depends heavily on proposer-selection design and randomness 439	Moderate fit for public land-related platforms where openness matters, but not the strongest option for tightly controlled registry governance 438 446
DPoS	Lower energy than PoW because it uses	Typically faster than PoW and often designed for high efficiency, but	Faster operational confirmation is	More centralized than PoW/PoS because it	Can be improved with trust, reward/punishment, or	Works better in governed consortium-style

Consensus	Energy use	Throughput / latency	Finality	Decentralization	Fault tolerance	Governance suitability
	delegated validation rather than mining; the evidence frames it as more efficient but governance-sensitive 452 455 462	node churn and vote-related instability are recurring issues 440 452 455	implied, but the evidence does not support a universal finality claim for land systems 452 455	relies on delegated voting and can over-rely on tokens 450 452	verifiable delegation, but base designs face voting-attack risk 452 455	environments than in open registries, but governance concentration is its main drawback 452 455
PBFT / BFT	Energy-light relative to PoW because it avoids mining and uses a small approved validator set 435 437 448	Strong throughput and low latency in permissioned land-style deployments; several studies report higher throughput or lower delay after PBFT improvements 435 444 445 453 463	Best aligned with practical finality in permissioned systems, which suits registry updates that should not remain probabilistic 435 444 448	Less decentralized because validators are authorized and often government- or consortium-controlled 435 436 437	Strong against faulty or malicious nodes in a known-member network; multiple papers explicitly frame PBFT as tolerant of Byzantine faults or faulty nodes 435 444 448 453	Strongest fit for land registries that prioritize legal oversight, low fees, and institutional control 435 458 459

Public vs permissioned networks

Public chains are repeatedly described as open to broad participation, but they tend to face higher latency, lower throughput, higher costs, privacy risks, and heavier validation overhead 432 433 434 436 442 447

. That makes them more suitable when transparency and open verifiability are the primary goals, not when a land office needs predictable service times or restricted authority 457 458 464.

Permissioned/private/consortium chains are consistently portrayed as better for controlled access, faster verification, lower transaction cost, higher throughput, and easier governance 434 435 437 438 464 465

. For land administration, that aligns closely with the evidence that governments remain central in blockchain land systems and that blockchain has not displaced registry authority 458 466.

Domain-specific evidence gaps

- There is **no internationally accepted methodology** in the retrieved evidence for comparing land-administration systems, which limits direct cross-consensus benchmarking in this domain 467 468 469

.

- The land and real-estate evidence is **mostly prototype- and pilot-oriented**, not large-scale comparative deployments of PoW vs PoS vs DPoS vs PBFT under the same registry workload 458 470 471

.

- Several land studies report better performance for permissioned or hybrid systems, but they often **do not isolate consensus from other design choices** such as off-chain storage, governance, tokenization, or access control 435 472 473

.

- Direct evidence on **finality semantics specifically for land title transfers** is sparse; most claims are inferred from general blockchain properties or from permissioned-system design choices 435 437 448

.

Bottom line

For land and real estate administration, the evidence supports a **deployment-dependent choice** rather than a universal winner: **PBFT/BFT is the best fit for permissioned registry networks**, **PoS is the strongest public-network compromise on energy and scalability**, **DPoS trades efficiency for stronger governance concentration**, and **PoW is generally the least suitable for routine registry workloads because of energy, throughput, and latency costs** 435 437 438 440 447

.

Confidence level: **High** — multiple independent sources consistently separate public from permissioned blockchain trade-offs, and land-administration studies repeatedly favor controlled, government-supervised architectures; the main uncertainty is that direct head-to-head land-registry comparisons of all four consensus families are limited 435 437 458 467.

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Follow-up Suggestions

- How are land parcel permissions encoded in smart contract policy rules?
- How do selective disclosure proofs work in verifiable land credentials?
- How is revocation status checked in a blockchain land registry?

How are land parcel permissions encoded in smart contract policy rules?

Processing steps

Smart contract land parcel permissions

Smart contract land parcel permissions

Land parcel permissions are encoded in smart contract policy rules by translating **cadastral concepts**—especially **rights, restrictions, responsibilities (RRRs)**, **parties**,

and **spatial units/parcels**—into executable conditions that a contract checks before it allows a registration, transfer, sale, lease, or other parcel action 391 474 475 476 477

. In practice, the rules usually combine **identity/role checks**, **document or ownership verification**, and **workflow approvals** so that only authorized actors can trigger parcel-state changes 478 479 480 481 482

.

How the policy rules are typically encoded

Policy-rule pattern	How it is encoded in the contract	Parcel-permission effect
RRR-based rules	The contract logic mirrors LADM-style rights, restrictions, and responsibilities, using parcel-linked legal concepts as the basis for decision rules 387 391 474 475 .	Only actions consistent with the parcel's legal status are accepted 476 477 .
Role-based access control	Different roles such as admin, registrar, citizen, inspector, or sub-registrar are encoded as permission predicates in the smart contract 479 481 482 483 .	Each role gets a different set of allowed parcel operations 478 479 .
Authority validation workflow	The contract requires approval from a registry office, inspector, or other authorized authority before completion 478 480 481 .	Permission depends on institutional confirmation, not just user submission 481 482 .
Ownership / entitlement proof	The system checks whether the caller can prove ownership or a valid entitlement before transfer or update 479 484 485 .	Prevents unauthorized parcel transfers and duplicate claims 484 486 .
Geospatial validity rules	Parcel actions can be constrained by geofencing or spatial validation against mapped boundaries 487 488	A transaction is allowed only if it matches the correct

Policy-rule pattern	How it is encoded in the contract	Parcel-permission effect
	.	physical parcel and location 487 .
Dispute / exception handling	Some systems encode dispute flags or transfer-prevention logic directly into the contract 484 .	Transactions can be blocked when legal uncertainty or conflict exists 484 .
Document integrity and storage rules	Legal documents are stored off-chain, while hashes or references are recorded on-chain for integrity checking 479 480 482 487 .	The contract can verify the documents behind a permission without exposing them all on-chain 480 487 .

What this looks like in land systems

- **LADM supplies the policy vocabulary.** It provides the standard conceptual structure for parties, basic administrative units, rights/restrictions/responsibilities, and spatial units, which makes it a natural source for smart-contract rule design 391 474 476 477

.

- **The contract then enforces those concepts as if/then logic.** For example, a transfer may proceed only if the caller is an approved party, the parcel record is valid, the authority has signed off, and no dispute flag is active 479 480 481 484

.

- **Permissions are often workflow-based, not single-step.** Several systems stage actions across registration, verification, inspection, approval, and final ownership update, with smart contracts enforcing each step 479 480 481

.

- **Some systems add spatial constraints.** Geofencing ties the digital record to the physical parcel so that a permission applies only to the correct location and not to an unrelated plot 487

.

Bottom line

Land parcel permissions are encoded as **contract-enforced policy predicates** built from cadastral legal rules, identity/role checks, spatial validation, and authority approval. The strongest pattern in the evidence is **LADM-informed RRR modeling plus role-based and workflow-based smart-contract logic**, with off-chain documents and on-chain hashes used to keep the legal record verifiable while the permission decision remains machine-checkable 391 476 477 478 479 480 481

.

Confidence level: **High** — multiple land-administration sources consistently describe parcel permissions through LADM RRRs, authority roles, workflow approvals, and integrity checks, and several blockchain land-registry prototypes show these ideas encoded directly in smart-contract logic 391 476 479 480 481 484 487.

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Follow-up Suggestions

- How are BBS+ proofs used in land credentials?
- How do Merkle trees support selective disclosure in land records?
- How is revocation implemented for verifiable land credentials?

How do selective disclosure proofs work in verifiable land credentials?

Processing steps

Selective disclosure in verifiable credentials

Selective disclosure in verifiable credentials

Selective disclosure proofs in verifiable land credentials work by letting a holder **prove only the land-related claims needed for a transaction**—such as ownership, role, parcel entitlement, or authorization status—while hiding the rest of the credential's contents 400 474 475 476

. In practice, the proof is generated from a verifiable credential and verified on chain or by a land-system verifier, so the registry can enforce access or workflow rules without exposing full identity data or unnecessary document details 384 475 476

.

How the proof is structured

Component	What it does	Land-credential effect
Credential issuance	An issuer creates a signed VC containing claims about the holder 474 475 476 .	The credential can represent owner status, registrar role, or another land-rights claim 384 476 .
Selective presentation	The holder reveals only the needed claims, not the full credential 474 475 477 .	A parcel workflow can verify entitlement without exposing unrelated personal or administrative details 475 477 .
Cryptographic proof	The holder generates a proof tied to the signed credential, often using BBS+/proof-friendly signatures, Merkle-style claim structures, or ZKPs 478 479 480 481 482 483 .	The verifier can check that the hidden claims remain valid even though they were not disclosed 474 480 482 .
Verification	The verifier checks the proof against policy and issuer state 425 475 482 .	The land system can allow or deny access to parcel actions based on the proof 384 476 .

Component	What it does	Land-credential effect
Revocation/status check	The proof or verifier also checks that the credential is still valid and not revoked 387 425 482 .	A compromised or expired land credential should not keep granting parcel permissions 384 387 .

What this looks like in land settings

- A **blockchain + SSI** cadastre prototype for Morocco uses SSI wallets and smart contracts to secure property registration and ownership transfer, which is the kind of workflow selective disclosure supports by limiting what the user must reveal 476

.

- Land-registry reviews say identity is essential for reliable blockchain land systems and that SSI aligns better with privacy and user control than older centralized identity models 384.
- More general SSI work shows credentials can be presented in **public, selective, or private** modes, with wallets generating presentations that include only authorized fields 475

.

- ZKP-based authorization systems show the same pattern: the claimant proves eligibility, while the chain or verifier learns only what is needed for the access decision 425 482

.

Common proof styles used

- **BBS/BBS+ selective disclosure**: efficient partial disclosure of signed claims; suited to anonymous credentials and partial-message proofs 478 479 480

.

- **Merkle/hash-tree credential structures**: each claim can be proven independently, which supports minimal disclosure and combining claims from multiple sources 481 483

.

- **Zero-knowledge proofs over credential attributes**: the holder proves a predicate about hidden values, such as membership or a policy condition, without revealing the underlying data 482 484 485

.

- **Double-blind existence proofs / context-bound proofs**: these reduce leakage about who is reading or using the credential and can bind the proof to the intended authorization context 425

.

Bottom line

Selective disclosure proofs make verifiable land credentials usable for access control because they turn a credential into a **policy-specific proof**: the holder discloses only the land claim relevant to the action, the verifier checks cryptographic validity and revocation status, and the registry preserves privacy while still enforcing legal authority 384 475 476 482

.

Confidence level: **High** — the evidence directly supports selective disclosure in verifiable credentials, BBS+/Merkle/ZKP-style proof construction, and land-registry SSI workflows that use those mechanisms for privacy-preserving authorization 384 474 475 476 480 481 482 483.

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Follow-up Suggestions

- How do accumulator-based revocation proofs work?
- What is a revocation registry in SSI?

- How does non-revocation proof protect privacy?

How is revocation status checked in a blockchain land registry?

Processing steps

Revocation status in blockchain land registries

Revocation status in blockchain land registries

Revocation status in a blockchain land registry is checked by **querying a status registry or revocation list before the registry accepts a credential or transaction**, then rejecting any credential marked revoked or expired and requiring revalidation or re-issuance through the authority/issuer workflow 384 387 401

.

How the check works

- **Issuer/registry status lookup:** the verifier or wallet checks the issuer's registry for the credential's current status, and revoked or expired credentials are excluded before a verifiable presentation is generated or accepted .
- **Revocation list or status registry:** blockchain revocation systems publish revocation information so verifiers can compare a credential against the current status record instead of trusting the presented credential alone 401 474

.

- **Accumulator-based membership test:** several blockchain revocation designs use **cryptographic accumulators** so the verifier can test whether a credential is still valid with compact proofs rather than large CRLs 475 476 477 478

.

- **Privacy-preserving non-revocation proofs:** SSI revocation schemes can let the holder prove **non-revocation** with ZK-style or accumulator-based proofs, so the checker learns the status without exposing unnecessary identity or query details 419 479

.

- **Land-registry workflow:** if the credential is revoked, the system blocks the land action and the claimant must be revalidated by the registry or issuer before a fresh credential is issued 384 387 400

.

What this means in practice

For a land registry, the check is usually not “delete the old key” but **verify status at the moment of use**: the smart contract or verifier consults the revocation/status source, rejects invalid credentials, and only then allows viewing, transfer, or update actions 387 401 474

. The land-specific evidence also emphasizes that this remains an **authoritative workflow** rather than purely self-service, because identity and entitlement must be revalidated by the registry/issuer after compromise 384 400

.

Bottom line

A blockchain land registry checks revocation status by **consulting a trusted on-chain or issuer-backed status source, often via an accumulator or revocation registry, and refusing any parcel-related action unless the credential proves it is still valid** 401 419 474 475

.

Confidence level: **High** — multiple sources directly support status-registry checks, issuer-side exclusion of revoked credentials, accumulator-based revocation verification, and land-registry-specific revalidation workflows 384 387 401.

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Follow-up Suggestions

- How do cryptographic accumulators support revocation checks in land registries?
- What is a non-revocation proof in verifiable credentials?

- How are revocation registries updated after a land credential is revoked?

How do cryptographic accumulators support revocation checks in land registries?

Processing steps

Cryptographic accumulators for credential revocation

Cryptographic accumulators for credential revocation

Cryptographic accumulators support revocation checks in land registries by **compressing the set of valid credentials into a single accumulator value** and letting verifiers test **membership or non-membership** with a short witness before accepting a parcel-related action; in land-registry workflows, that means a revoked or expired credential is rejected when its current status no longer matches the accumulator-backed status record 384 387 401 489 490

.

How the check works

Mechanism	What the accumulator does	Land-registry effect
Compact status representation	A cryptographic accumulator represents a large set of valid items with one succinct value and supports membership proofs for elements in that set 491 492 493 .	The registry can check whether an owner, registrar, or other credential is still valid without scanning a long revocation list every time 384 401 .
Membership / non-membership witnesses	The holder presents a witness proving the credential is still in the valid set, or proving it is not in the revoked set; newer accumulator designs explicitly support aggregated membership and non-membership witnesses 490 492 .	A land transaction proceeds only if the credential is still current; revoked credentials fail the proof check and are blocked 384 387 401 .
Update after revocation	When an element is revoked, the accumulator changes and witnesses may need updating; some schemes reduce this	The registry must refresh status after revocation so stale proofs do not keep

Mechanism	What the accumulator does	Land-registry effect
	cost with epochs, batch handling, or low-update-frequency compatibility 494 495 496 .	authorizing parcel actions 387 401 .
Privacy-preserving verification	Accumulator-based revocation can be combined with zero-knowledge proofs so the verifier learns validity without learning the full credential set or unnecessary identity details 489 497 498 .	This fits land systems that need revocation checks without exposing ownership history or sensitive credential metadata 384 419 425 .

What this means in practice

- **Before a land action is accepted**, the verifier checks the credential against the current accumulator-backed status source, rather than trusting the presented credential alone 384 401

.

- **If the credential was revoked**, the witness no longer validates against the latest accumulator state, so the parcel operation is denied 387 490 494

.

- **After compromise or expiry**, the claimant must be revalidated and issued a fresh credential or updated witness through the registry/issuer workflow 384 387 401

.

- **For large systems**, accumulators are attractive because they keep proofs short and verification efficient even when the revocation set grows 489 490 499

.

Land-registry-specific takeaway

In land registries, accumulators function as a **compact revocation gate**: they let the system prove “still valid” or “already revoked” efficiently, support privacy-preserving checks, and make it practical to block revoked SSI credentials before any parcel update, transfer, or access grant is finalized 384 387 401 419

.

Confidence level: **High** — the evidence directly supports accumulator-based membership/non-membership proofs, revocation updates, and land-registry workflows that reject revoked credentials through issuer-backed status checks 384 387 401 489 490 494.

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Follow-up Suggestions

- How are accumulator witnesses updated after a revocation?
- What accumulator schemes support non-membership proofs?
- How can zero-knowledge proofs be combined with accumulators in land registries?

What is a non-revocation proof in verifiable credentials?

Processing steps

Non-revocation proofs in verifiable credentials

Non-revocation proofs in verifiable credentials

A **non-revocation proof** is a cryptographic proof that a verifiable credential is **still valid and has not been revoked**, so a verifier can accept the credential without trusting the holder's word alone 419 489 490

. It is usually built from a **revocation-status mechanism** such as a revocation list, accumulator, Bloom-filter cascade, or Merkle-tree-based structure, and the holder presents a proof that their credential remains in the valid set rather than in the revoked set 419 489 490 491 492 493

.

What it proves

- The holder's credential is **not revoked** at the time of verification 419 492 493

.

- The verifier can check validity **without learning unnecessary personal details** or tracking the holder across checks, depending on the scheme 419 422 491 494

.

- In some schemes, the proof is **non-interactive** and can be verified off-chain or on-chain with a compact revocation state 419 422 495

.

How it is commonly implemented

Mechanism	How non-revocation is shown	Main trade-off
Cryptographic accumulator	The holder proves their credential is still a member of the valid set, or proves non-membership in the revoked set 489 492 493 496 497 .	Efficient verification, but witness updates can be costly in some designs 489 492 .
Merkle-tree / hash-tree revocation state	The holder provides a witness path showing the credential's status against a compact root 490 498 499 .	Compact and practical, but updates can affect witnesses 498 499 .
Bloom filter / cascaded revocation list	The credential is checked against a privacy-preserving published revocation list 419 421 491 .	Good privacy and scalability, but requires careful parameter tuning 421 491 .
Zero-knowledge proof over revocation status	The holder proves non-revocation without revealing the credential identifier or revocation query details 419 422 495 .	Strong privacy, but can add cryptographic overhead 419 500 501 .

Why it matters

Non-revocation proofs let verifiers reject **revoked or withdrawn credentials** while preserving **privacy, unlinkability, and offline usability** in SSI systems 419 422 423 494

. They are especially important where credentials are used repeatedly, because a verifier needs current status, not just an old signature 489 495 502

.

Bottom line: a non-revocation proof is the holder's cryptographic evidence that "this credential is still valid right now," typically produced with accumulators, Merkle-based witnesses, revocation lists, or zero-knowledge techniques so the verifier can check status without exposing more information than necessary 419 422 489 490 495

.

Confidence level: **High** — multiple independent sources directly describe non-revocation as proving a credential has not been revoked, and they consistently point to accumulators, Merkle structures, revocation lists, and ZK-based approaches as the main implementations 419 422 489 490 491 493 495.

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Follow-up Suggestions

- How do accumulators prove non-revocation in practice?
- What is a revocation witness in verifiable credentials?
- How do Merkle tree revocation proofs work in SSI?

How are revocation registries updated after a land credential is revoked?

Processing steps

Revocation registry updates for land credentials

Revocation registry updates for land credentials

A revoked land credential typically updates the revocation registry by **marking the credential as invalid in the issuer/registry status source, publishing that new status to verifiers, and forcing any later presentation to fail or require revalidation/re-issuance** 384 387 401 489

.

How the registry is updated

Update step	What happens	Evidence
Status is changed to revoked	The issuer or authority records the credential as revoked in a status registry or revocation list rather than "deleting" it outright 384 387 401 .	Land-registry SSI workflows rely on authoritative status management, not self-deletion 384 387 .
Revocation information is published	The updated revocation state is distributed so verifiers can reject the credential at the time of use 401 489 .	Blockchain revocation systems publish revocation data on-chain or in an issuer-backed status source 401 .
Verification data structure is refreshed	Depending on the design, the registry updates a CRL-style list, Bloom-filter-based status structure, or cryptographic accumulator state 401 490 491 492 .	These approaches are used to make revocation checks efficient and scalable 401 490 491 492 .
Holders' proofs must adapt or fail	If the scheme uses witnesses or non-revocation proofs, the holder must refresh the proof; otherwise, the old credential is simply rejected at verification time 491 492 493 494 .	Accumulator-based revocation commonly requires witness updates unless the design explicitly removes that burden 491 492 493 .

Update step	What happens	Evidence
Replacement credential is issued after revalidation	The revoked credential is replaced only after the registry or issuer rechecks identity and entitlement and issues a fresh credential 384 387 400 .	Land-registry evidence treats revocation as part of a controlled revalidation workflow 384 400 .

What this means in land registry practice

- **The update is administrative and cryptographic at the same time**: the authority changes the credential's status, and the verifier later checks that status before allowing parcel actions 384 387 401

.

- **Privacy-preserving systems avoid exposing too much history** by using accumulator-based or other compact status mechanisms instead of broad, query-heavy revocation lists 401 423 490 491

.

- **Land systems still keep institutional control**: revocation triggers blocking of the old credential, but access is restored only through authoritative revalidation and re-issuance 384 387 400 489

.

Bottom line

After a land credential is revoked, the registry is usually updated by **writing the revoked status into a trusted status source, propagating that update to verifiers, and requiring any future access to pass a fresh revocation check or a new credential issuance** 384 387 401 489

. The exact mechanism varies—CRL-like lists, Bloom filters, accumulators, or on-chain registries—but the shared pattern is **invalidate first, verify status at use, then reissue only after revalidation** 401 490 491 492

.

Confidence level: **High** — the evidence directly supports revocation-status publication, accumulator/list updates, and land-registry revalidation workflows, with multiple schemes showing different but consistent registry-update patterns 384 387 401 489 491.

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Follow-up Suggestions

- How do accumulator witnesses get refreshed after revocation?
- What data structure stores revocation status in a land registry?
- How does re-issuance work after credential revocation?

Critically validate this consensus heatmap for land and real estate administration. Assess PoW, PoS, DPoS, and BFT/PBFT across energy efficiency, TPS, finality, decentralization, and Sybil resistance. Define evidence-based rating criteria, verify citations, explain network-dependent trade-offs, and align conclusions with RQ1–RQ4. Avoid universal rankings; distinguish theoretical properties from measured results and preserve citation integrity.

Processing steps

Consensus mechanism evaluation for land administration

Consensus mechanism evaluation for land administration

The heatmap is directionally reasonable, but it should be treated as a network-dependent comparison of *typical* performance, not a universal ranking: PoW is the weakest for energy and latency, PoS is generally better on energy with more moderate throughput/finality, DPoS improves speed at the cost of validator concentration, and BFT/PBFT fits best in permissioned or consortium land settings where a bounded validator set is acceptable. 503 504 505 506 507 508 509

Evidence-based rating criteria

Dimension	**High**	**Medium**	**Low**
Energy efficiency	No mining, low computational burden, or measured low per-node power in permissioned settings 462 510 511 512	Some validator overhead but substantially below PoW 513 514	Mining-intensive, explicitly energy-heavy 461 515 516
TPS / throughput	Reported high throughput or scalable batching in permissioned / optimized systems 506 517 518 519	Context-dependent or moderate throughput 454 505 520	Low throughput / poor scaling 521 522 523
Finality	Deterministic or near-deterministic confirmation in permissioned designs 511 521 524 525	Practical finality that depends on protocol and network assumptions 505 520	Probabilistic or slower settlement under open public load 503 526
Decentralization	Broad validator participation / permissionless openness 527 528 529	Delegated or stake-weighted participation with some concentration 450 513 520	Small approved validator set or strong administrative control 511 524 525
Sybil resistance	Strong in resource- or stake-bound open systems, or with explicit identity/validator controls 527 528 529 530	Dependent on stake distribution, committee selection, or governance design 450 452 454	Weak if validator identity or participation is easy to manipulate 452 505

Consensus-by-consensus validation

Consensus	What the evidence supports	What the heatmap should not overclaim
PoW	PoW is consistently described as **energy-intensive**, with low scalability and poor suitability for resource-constrained or latency-sensitive settings 461 515 516 . Its strength is openness and strong security assumptions in public networks, not	Do **not** rate PoW as competitive on energy or TPS for land registries; any high score there would contradict the evidence 515 516 523.

Consensus	What the evidence supports	What the heatmap should not overclaim
	operational efficiency for land administration 449 527 .	
PoS	PoS is repeatedly presented as a **lower-energy alternative** to PoW, with improved efficiency and the possibility of better decentralization if stake distribution is healthy 450 514 528 . Some sources also note that decentralization and fairness still depend on protocol design and stake dynamics 450 528 .	Do **not** treat PoS as automatically fully decentralized or always finality-strong; those properties are design- and network-dependent 450 520 528.
DPoS	DPoS is consistently framed as **faster and more scalable than PoW/PoS in many deployments**, with energy and throughput advantages coming from delegated validation 452 505 513 . The trade-off is **governance concentration** and greater exposure to voting/token concentration risks 452 520 .	Do **not** give DPoS a high decentralization score; the evidence explicitly warns that it can rely too heavily on tokens and a finite delegate set 452 520.
BFT/PBFT	PBFT/BFT is the strongest fit for **permissioned or consortium land systems** because it offers low latency, high throughput, and deterministic or near-deterministic finality when the validator set is controlled 506 512 517 521 525 531 . The cost is validator communication overhead and weaker scalability as the validator set grows 517 524 525 526 531	Do **not** score PBFT as highly decentralized; its design depends on a restricted validator set and institutional membership 511 524 525.

Consensus	What the evidence supports	What the heatmap should not overclaim
	.	

Network-dependent trade-offs

- **Public networks:** the evidence favors PoW/PoS-style open participation when decentralization and Sybil resistance are paramount, but these designs trade off energy use, latency, and throughput 449 527 528

.

- **Permissioned / consortium networks:** PBFT-like designs are repeatedly favored because they reduce latency and improve throughput under a known validator set, which matches land registry governance better than open-mining systems 505 511 517 524 531

.

- **Delegated systems:** DPoS can improve speed and resource efficiency, but the evidence shows a real governance cost from delegate concentration and voting-attack risk 452 513 520

.

- **Measured vs theoretical:** several sources report actual experiments or simulations for throughput, latency, and energy, but many public claims are comparative and protocol-level rather than land-specific measurements, so the heatmap should label those cells as **theoretical/expected** unless the underlying study reports a measurement 510 517 521 525 531 532 533.

Alignment with RQ1–RQ4

RQ focus	What the evidence supports
RQ1: Energy efficiency	PoW is worst; PoS improves materially; DPoS and PBFT are typically lower-energy in the cited land-adjacent and permissioned contexts 462 510 512 513 515 516 .

RQ focus	What the evidence supports
RQ2: TPS / throughput	PBFT and optimized delegated/permissioned schemes usually outperform PoW and often PoS; PoW is weakest for high-frequency land workflows 506 517 519 521 523 .
RQ3: Finality	PBFT has the clearest deterministic finality story in permissioned settings; PoS and DPoS are more design-dependent; PoW is slowest to settle 505 511 520 521 525 .
RQ4: Decentralization / Sybil resistance	PoW and well-designed PoS score better on openness and Sybil resistance, while DPoS and PBFT trade decentralization for governance control and operational efficiency 452 520 524 527 528 530 .

Bottom line

For land and real estate administration, the heatmap should conclude that **no single consensus mechanism dominates all criteria**: **PoW** is the most decentralized but the least energy-efficient and least practical for routine registry workloads; **PoS** is a better public-network compromise; **DPoS** raises throughput and lowers cost but concentrates governance; and **BFT/PBFT** is the strongest operational choice for **permissioned land registries** because it delivers low latency, high TPS, and strong finality at the cost of smaller validator sets and reduced openness 503 505 509 521 524 525 526

.

Confidence level: **High** — multiple independent sources agree on the core trade-offs, and the strongest evidence comes from both review-level comparisons and measured or simulated studies, though exact scores still depend on whether the network is public or permissioned 503 505 521 525 526 531.

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Follow-up Suggestions

- How are finality and settlement delay measured in land registry blockchains?
- What validator governance model fits permissioned land registry networks?
- How do PBFT variants scale as validator counts increase?

Audit technology prevalence in blockchain-based land and real estate administration. Verify each technology against cited studies and the stated denominator. Recalculate percentages from confirmed numerators and denominators, distinguish proposed, tested, pilot, and operational systems, and clarify overlapping categories. Avoid implying commercial adoption or superiority. Flag unverifiable figures, preserve citations, and never fabricate data.

Processing steps

Prevalence of audit technologies in blockchain land administration

Prevalence of audit technologies in blockchain land administration

The evidence supports a **qualitative audit of prevalence**, but it does **not** support a defensible full percentage re-baselining for every technology because the retrieved studies use different denominators, overlapping categories, and in several cases report only proposals or pilot cases rather than counted deployments 534 535 536 537 538.

Verified prevalence signals

Technology / category	What the studies actually support	Status for prevalence audit
Blockchain for land administration overall	A 2023 systematic review examined **37 studies** on institutional success factors, and a 2025 review identified **20 relevant articles** in developing-country land administration, with **most deployments still in the prototype stage** 534 539 .	**Supported as an active research area**, not as evidence of broad operational adoption 534 539.

Technology / category	What the studies actually support	Status for prevalence audit
Land administration / land registry	This is the **dominant application category** in the real-estate literature, ahead of transactions, tokenization, and management; these categories are explicitly described as **overlapping**, especially transaction and tokenization 537 .	**Supported as the most prevalent theme**, but category overlap prevents clean percentage attribution 537.
Pilot / proof-of-concept systems	Multiple sources describe pilots or proof-of-concepts in Sweden, Georgia, Ghana, Kenya, Brazil, Ukraine, Greece, UAE, Japan, the Netherlands, and others 466 470 538 540 541 542 .	**Strongly supported as common**, and more common than operational rollouts in the retrieved evidence 470 538 539 540.
Operational / production systems	The evidence mentions systems in government or enterprise use, but often in limited, partial, or collaborative form rather than as proof of fully scaled national adoption 543 544 545 546.	**Supported only in limited cases**; the evidence does not justify claiming widespread commercial or nationwide deployment 543 544 545 546.
Public / permissionless blockchain	A Ghana-focused review says it **appears possible** to apply a permissionless public blockchain across land administration processes, but notes dependence on standardization and stakeholder consensus, plus storage, scalability, and electricity constraints 547. A Kenya pilot is described as public but with restricted permissions for sensitive data 541 .	**Supported as proposed or piloted**, not as a proven superior model 541 547.

Technology / category	What the studies actually support	Status for prevalence audit
Permissioned / private blockchain	The literature repeatedly describes private or permissioned approaches for controlled governance and pilot deployment, including Sweden-style private access and system designs with central oversight or authorized actors 545 548 549 550 551 .	**Supported as a common design choice**, especially where institutional control is required 545 548 549 550 551.
Smart contracts	Smart contracts are a recurring mechanism for automating registration, verification, transfer, and workflow control in land and real estate systems 488 552 553 554 555 .	**Supported as highly prevalent in proposals and prototypes** 488 552 553 555.
Tokenization / NFTs / fractional ownership	These appear in real-estate literature as a distinct but **overlapping** category with transactions, and are presented mainly as conceptual or emerging business models 537 556 557 558 559 .	**Supported as emerging**, not as the dominant land-administration use case 537 556 559.
IPFS / off-chain storage	Several land systems pair blockchain with IPFS or related off-chain storage to handle documents and large files 488 545 555 560 .	**Supported as a recurring complement**, not a standalone prevalence category 488 545.

What can and cannot be recalculated

- **Can be recalculated:** any percentage that is explicitly tied to a stated numerator and denominator in the source text, such as the 2025 review's **20 relevant articles** out of a search process that is described but not fully numerically specified in the excerpt 539.
- **Cannot be recalculated from the excerpt alone:** the share of each technology across the full 70-reference set, because the references are **not mutually**

exclusive, several studies cover multiple technologies at once, and the excerpt does not provide a single consistent denominator for all categories 537 550 559 561.

Unverifiable or unsupported figures

Figure in the evidence	Why it is not auditable from the provided material
"Most deployments still prototype stage"	Supported as a qualitative conclusion, but no exact count of prototypes versus operational systems is given in the excerpt 539.
"Pilot projects in Sweden, the Netherlands, Ukraine, Greece, UAE, Georgia, Japan, and other countries"	The excerpt lists examples, but does not provide a counted denominator or a deduplicated study total 540.
"Blockchain could be applied across all land administration processes"	Stated as a possibility, not as observed prevalence or measured adoption 547.
Claims of superiority or commercial adoption	The evidence repeatedly frames systems as proposals, concepts, pilots, or limited deployments, so superiority or broad commercial adoption is not verified here 470 535 539 562.

Bottom line

The audited evidence shows a **research-heavy, prototype-dominant landscape**: land administration is the most common blockchain application area in the retrieved literature, smart contracts and permissioned/private designs recur frequently, and tokenization/IPFS appear as overlapping adjuncts rather than separate dominant categories 534 537 539 550 555

. Exact prevalence percentages are **not defensible** from the provided material because the denominator is inconsistent across studies and the same paper can count toward multiple categories 537 559 561.

Confidence level: **Medium** — the evidence is strong for directional prevalence patterns and deployment stage distinctions, but insufficient for a precise percentage audit across all technologies without a unified coding scheme and denominator 534 537 539.

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Follow-up Suggestions

- Which studies reported operational blockchain land registries
- How were overlapping blockchain real estate categories coded in the review
- What denominator was used for the prototype stage prevalence

Which studies reported operational blockchain land registries

Processing steps

Operational blockchain land registries

Operational blockchain land registries

The studies that most clearly report **operational** blockchain land registries are the ones on **Georgia**, **Dubai**, and, in a more limited "implemented in a department" sense, **Jordan**; most of the other retrieved land-registry papers describe **proposals, prototypes, or pilots** rather than confirmed operational deployment 562 563 564 565 566

.

Operational evidence in the retrieved studies

Study	What it says about operational status	Confidence as "operational"
Citizen e-governance using blockchain (2021)	Says Georgia launched a pilot in 2016, then **"after a successful completion of the pilot project, the initiative took off as a full-scale project"**; it also reports that **over 1.5 million land titles have been registered on blockchain** 563 .	**Strong**

Study	What it says about operational status	Confidence as "operational"
A systematic review of the institutional success factors for blockchain-based land administration (2023)	Says **Georgia is the first-ever country to successfully adopt a blockchain land administration project** and that **in Dubai, their experience shows that blockchain for the land registry is actively working well** 564 .	**Strong**
The effect of smart contracts (blockchain technology) on Jordan's land registry and survey department for sustainable development (2025)	Reports implementation inside the Jordanian Land Registry and Survey Department, based on survey evidence from that department, but it reads more as **institutional adoption / implementation context** than a verified live registry architecture description 565 566 .	**Moderate**

Not operational, or not clearly operational

- **Sweden, the Netherlands, Ukraine, Greece, UAE, Georgia, Japan, etc.** are described as **pilot projects** or implementation efforts, not all operational rollouts 540

.

- **Honduras** is discussed as a project that failed or stalled, not an operational registry 562 564

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- **Ghana** is described as piloted or proposed, with one review explicitly saying an effective output had not yet been reported 471 564

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- **India, Turkey, Malaysia, Bangladesh, Romania, and similar studies** in the retrieved set are mostly **conceptual, prototype, or feasibility** papers rather than proof of live production operation 567 568 569 570 571 572

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Bottom line

If you mean **documented live or full-scale use**, the clearest examples in the retrieved evidence are **Georgia** and **Dubai**, with **Jordan** appearing as a departmental implementation case rather than a clearly described nationwide operational registry 563 564 565

. The rest of the land-registry literature here is dominated by **pilots, prototypes, and conceptual models**, so it should not be counted as operational without stronger evidence 540 568 570 572

.

Confidence level: **High** — the retrieved studies explicitly distinguish pilot, full-scale, and "actively working" cases for Georgia and Dubai, while most other land-registry papers in the set are framed as proposals or pilots 540 563 564.

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Follow-up Suggestions

- Which studies reported blockchain land registry pilots
- How were operational land registry cases distinguished from prototypes
- Which countries had blockchain-based land administration implementations

Audit technology prevalence and blockchain architecture patterns in land and real estate administration. Verify cited studies, denominators, percentages, architecture classifications, and jurisdiction claims. Resolve inconsistencies between counts and references. Distinguish prototypes, pilots, and operational deployments; compare on-chain, hybrid, and anchoring models. Avoid market-adoption claims, preserve citations, and never invent evidence.

Processing steps

Blockchain technology in land administration

Blockchain technology in land administration

Blockchain land and real-estate literature is **dominated by pilots, prototypes, proof-of-concepts, and hybrid designs**, while **operational deployments are rare and often reported only as partial or jurisdiction-specific implementations**; the strongest recurring architecture pattern is **hybrid/on-top-of-existing-systems**, not fully on-chain replacement. 534 539 540 545 550 573 574 575 576

Prevalence audit: what is actually supported

Technology / pattern	What the cited studies support	Audit judgment
Land administration / land registry	This is the dominant application area across the review literature, with most blockchain work in land administration, followed by property transactions and related real-estate subsectors. 534 536 578	**Supported as the main theme**, but not as evidence of widespread deployment.
Smart contracts	Frequently used to automate registration, transfer, approval, and verification workflows in proposed and prototype systems. 481 488 545 552 579 580 581 582 583	**Highly prevalent in designs and prototypes**.
Document storage / IPFS / off-chain support	Recurs in systems that keep documents or metadata off-chain while storing hashes, references, or proof anchors on-chain. 481 488 545 584 585 586 587	**Common hybrid complement**, not a standalone deployment class.
Tokenization / NFTs / fractional ownership	Appears in newer real-estate models and tokenized land/property concepts, but the literature frames these as emerging or conceptual rather than broadly adopted. 561 584 588 589	**Emerging, not dominant**.
Geospatial encoding and cadastral data management	Work on WGS84, Geohash, Mapcode, and related geospatial representations focuses on efficient encoding and storage trade-offs for land systems. 481 488 590	**Recurring technical subtopic**, not a prevalence category by itself.

Technology / pattern	What the cited studies support	Audit judgment
AI/IoT/blockchain combinations	Appears in some recent land-management and real-estate papers as an extension of blockchain-enabled governance. 591 592	**Present but secondary**.

Architecture pattern audit

Architecture type	Evidence pattern	Representative interpretation
On-chain / all-blockchain	Some papers present land records, verification steps, or transaction histories as being stored directly on-chain, sometimes with local blockchain storage and automated smart-contract flows. 481 488 593 594	**Exists in proposals and prototypes**, but the evidence does not show this as the dominant live deployment model.
Hybrid	Multiple studies explicitly describe blockchain as an add-on to existing registries, or as a complementary layer that preserves the role of the land registry as ultimate arbiter. 545 550 575 577 578 585 595	**Most consistently supported architecture pattern**.
Anchoring / off-chain + on-chain proof	Several systems store documents, images, or large files off-chain in IPFS or related storage while anchoring hashes, references, or proofs on-chain for integrity and auditability. 481 488 545 584 585 586	**Widely used for privacy, storage, and scalability trade-offs**.

Prototype, pilot, and operational status

Status	What the evidence says	Examples / jurisdictions
Prototype / proof-of-concept	A large share of the literature is explicitly conceptual or prototype-based, including proof-of-concept land-administration models and academic prototypes. 481 488 539 552 579 596 597 598 599	Bangladesh, Namibia, India, Albania, and other proposal-heavy settings. 569 586 597 600
Pilot / testbed	Pilot deployments are repeatedly described as the modal “real-world” status, especially in Sweden, Georgia, Ghana, Kenya, Ukraine, the Netherlands,	**Strongly supported as common**.

Status	What the evidence says	Examples / jurisdictions
	Greece, UAE, Japan, and Cook County. 470 471 540 541 542 573 575 601 602 603 604 605	
Operational / live use	The clearest operational claims in the retrieved evidence are **Georgia**, **Dubai**, and, more cautiously, **Jordan**; other cases are often described as pilots or initiatives still facing limited concrete development. 471 542 550 606 607 608	**Supported, but only for a small number of jurisdictions**.

Jurisdiction claims that are supported

- **Georgia** is repeatedly described as a successful or full-scale blockchain land-registry case, and is one of the strongest operational examples in the evidence. 542 607 608
- **Dubai / UAE** is also described as actively working or part of an ongoing blockchain land-registry effort. 604 607
- **Sweden** appears mainly as a pilot/proof-of-concept environment rather than confirmed nationwide operational replacement. 540 550 575 576 602
- **Ghana** is described as exploratory, pilot-based, and still limited in concrete outcomes. 471 542 547 605 609
- **India, Bangladesh, Turkey, Namibia, Albania, Serbia, Vietnam, Malaysia, and Kenya** are mainly represented by proposals, frameworks, pilots, or prototypes rather than confirmed broad operational deployments. 486 488 541 568 569 583 586 595 597 598 600 610

Inconsistencies resolved

Issue	Resolution
"Adoption" language vs actual implementation status	Many papers use adoption-style language, but the evidence often only supports **conceptual**, **prototype**, or **pilot** status; that language should not be read as operational deployment. 534 539 573 611
Counted studies vs overlapping categories	Real-estate studies overlap across land administration, transactions, tokenization, and management, so the same paper can support multiple categories; the literature review explicitly

Issue	Resolution
	notes this overlap and the need for standardized vocabulary. 536 578 611
Permissionless vs permissioned claims	Both appear, but the evidence supports **context-specific trade-offs**, not universal superiority: public chains are argued to enhance openness, while permissioned/private or hybrid systems are repeatedly used for governance, privacy, and institutional control. 541 547 550 575 583 612 613
Full on-chain vs anchoring/hybrid	The more defensible reading is that land systems commonly use **anchoring plus off-chain storage** or **hybrid integration**, because documents, privacy, scalability, and legal control make full on-chain designs less common. 481 488 545 550 575 585

Bottom line

The evidence supports a **prevalence hierarchy of concepts rather than a prevalence of mature deployments**: land administration dominates the literature, smart contracts and hybrid architectures are the most common technical pattern, and anchoring/off-chain storage is the most common practical compromise; by contrast, fully operational blockchain land registries are limited to a small number of jurisdictions, with many other cases remaining pilots or prototypes. 536 540 542 550 574 575 576

Confidence level: **High** — the retrieved studies consistently distinguish prototypes, pilots, and a few operational cases, and they repeatedly converge on hybrid/anchoring patterns rather than universal on-chain replacement.

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Follow-up Suggestions

- Which studies describe Georgia’s blockchain land registry architecture

- Which papers use hybrid blockchain with off-chain storage in land administration
- Which jurisdictions are reported as operational blockchain land registries

Which studies describe Georgia's blockchain land registry architecture

Processing steps

Georgia blockchain land registry architecture

Georgia blockchain land registry architecture

Georgia's blockchain land registry architecture is described most concretely in studies that portray it as an **add-on layer on top of an existing registry**, using a **private Exonum blockchain for timestamps/registration details, Bitcoin anchoring for hashes, and a centralized land-title database for confidential records**; other studies summarize it more broadly as a **Bitcoin-based or permissioned blockchain land-titling system** with varying emphasis on centralization versus decentralization 563 607 614 615 616.

Studies that describe the architecture

Study	What it says about Georgia's architecture
Citizen e-governance using blockchain (2021)	The clearest architectural description: Georgia's system was launched as a pilot, then became a full-scale project; it is **an add-on to the existing system**, where parties still go to the local office, the office validates and signs the agreement, the **land titles remain in a centralized database**, and the **location/registration details are timestamped on a private Exonum blockchain**, with document hashes stored on the **Bitcoin blockchain** 563 .
Blockchain-based land registry platforms: a survey on their implementation and potential challenges (2022)	Says Georgia began the project in 2016, the National Public Registry Agency acts as a **third-party enforcer**, titles can be issued digitally, and the blockchain is **"just a layer of tamper-proof security with the Bitcoin blockchain"**; it also says the process is **still centralized** and not fully automated 614 .

Study	What it says about Georgia's architecture
Algorithmic Government: Automating Public Services and Supporting Civil Servants in using Data Science Technologies (2019)	Describes a **custom blockchain system integrated into NAPR's digital records system** and **anchored to Bitcoin through distributed digital timestamping** 615 .
Blockchains: A review and research agenda for international business (2023)	Frames Georgia as a **successful pilot/implementation built on Exonum**, a **permissioned blockchain framework**, and identifies Georgia's land-titling registry as one of its successful applications 607 .
Land records on Blockchain for implementation of Land Titling in India (2020)	Presents Georgia as the first government to introduce blockchain land-registry management and says records are **immutable**, the process is **automatically added to a digital ledger**, and notary services are no longer required for transaction proof 616 .

Broader contextual descriptions

- **Georgia as a successful/first adopter** appears repeatedly in comparative and review literature, but those sources usually do **not** add new architectural detail beyond the pilot-to-full-scale story and the role of NAPR/Bitfury/Exonum/Bitcoin anchoring 534 562 564 617 618 619.
- Some sources emphasize **trust restoration, anti-corruption, and stakeholder participation** more than system design, which supports the policy rationale but not the low-level architecture 620.
- A few sources describe Georgia's system in broader terms as **permissioned / public-private / blockchain-based land registry**, but the most specific architecture remains the **hybrid, partly centralized model** in which blockchain complements rather than replaces the registry 563 607 614 615.

Bottom line

If you want the architecture description, the best-supported reading is that **Georgia did not move to a fully on-chain land registry**; instead, the evidence describes a **hybrid registry**

with **centralized title storage, a private Exonum layer for timestamping and registration data, and Bitcoin anchoring for integrity proofs**, while NAPR stayed involved as the authoritative registry actor 563 614 615

.

Confidence level: **High** — multiple independent studies converge on the same hybrid architecture, with the main variation being whether the blockchain layer is described as Exonum, Bitcoin-anchored, or more generally permissioned 563 607 614 615.

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Follow-up Suggestions

- Which studies describe Georgia's use of Exonum in land registry
- Which studies explain Bitcoin anchoring in Georgia's land registry
- Which studies compare Georgia's land registry with Dubai's blockchain registry

Which papers use hybrid blockchain with off-chain storage in land administration

Processing steps

Hybrid blockchain land administration architectures

Hybrid blockchain land administration architectures

These papers in the retrieved evidence **explicitly combine blockchain with off-chain storage** for land-administration use cases, most often by keeping documents or raw records in **IPFS / decentralized storage** while anchoring hashes or metadata on-chain 481 488 621 622 623 624 625

.

Confirmed studies

Paper	Land-administration use case	Hybrid / off-chain pattern
A Hybrid AI-Blockchain Framework with OCR, NER, and Anomaly Detection for Secure Land Record Management (2025)	Land record management	Uses **Ethereum smart contracts + IPFS** for secure, immutable storage after AI-based pre-storage validation 621 .
Blockchain-Based Data Storage Model for Land Survey (2023)	Land surveying and mapping	Uses **on-chain storage plus association retrieval of on-chain and off-chain data**, with distributed slice storage and erasure coding 622 .
A Blockchain based Secured Land Registration System using Dynamic Smart Contracts (2025)	Land registration	Combines **Hyperledger Fabric** with **decentralized off-chain storage (IPFS)** for large documents 623 .
Community Blockchain Record-Keeping Method for Agricultural Land Leases Using Design Science Research Approach (2025)	Agricultural land leases	Uses **Hyperledger Fabric + IPFS** to store lease agreements securely and immutably 624 .
Cadastral: Blockchain-Based Land Registration System (2024)	Land registration	Uses **smart contracts plus IPFS for file storage**, with records stored in the local blockchain 481 .
Cadastral: Blockchain-Based Land Registration System (2025)	Land registration	Same hybrid pattern: **Solidity smart contracts + IPFS for file storage** 488 .
Blockchain-Enabled Framework for Secure and Transparent Land	Land administration	Uses a **permissioned blockchain + off-chain document storage with cryptographic anchoring**,

Paper	Land-administration use case	Hybrid / off-chain pattern
Administration in Bangladesh (2025)		implemented with **Hyperledger Fabric and IPFS** 625 .

Closely related but not fully confirmed as off-chain-storage papers

- **Land Certificate Authenticity Using Blockchain Technology in Indonesia (2024)** is described as a **hybrid blockchain** land-certificate system, but the abstract does **not explicitly state** off-chain storage 626.
- **A Proof of Concept of the integration of blockchain with an ISO 19152:2012 based Land Administration System (2022)** is a blockchain/LADM proof of concept, but it does **not explicitly mention off-chain storage** in the retrieved abstract 552.
- **Modernizing Land Management with AI, IoT Using Blockchain, and Smart Contracts to Promote Transparency, Efficiency, and Trust (2025)** is a review of land papers that mentions **IPFS** among the surveyed technologies, but it is not itself a land-system implementation study 591.

Bottom line: the strongest matches are the **land record management**, **land survey**, **land registration**, **agricultural land lease**, and **Bangladesh land administration** papers, all of which explicitly pair blockchain with **off-chain storage or retrieval** rather than storing everything directly on-chain 481 488 621 622 623 624 625.

Confidence level: **High** — multiple retrieved abstracts directly name IPFS, off-chain storage, or on-chain/off-chain association in land-administration contexts, while the excluded items are only hybrid by implication or are review papers 481 488 621 622 623 624 625.

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Follow-up Suggestions

- Which land administration papers use IPFS for off-chain document storage
- Which studies describe on-chain and off-chain data separation in cadastral systems
- Which blockchain land registry papers use anchored hashes for integrity proofs

Which land administration papers use IPFS for off-chain document storage

Processing steps

IPFS for off-chain land administration

IPFS for off-chain land administration

Papers that explicitly use IPFS for off-chain document storage

The land-administration papers in the retrieved evidence that **explicitly** use **IPFS** or equivalent **off-chain document storage** are the following 481 488 621 623 624 625.

Paper	Land-administration context	IPFS / off-chain storage use
A Hybrid AI-Blockchain Framework with OCR, NER, and Anomaly Detection for Secure Land Record Management (2025)	Land record management	Uses **Ethereum smart contracts + IPFS** for secure immutable storage 621 .
A Blockchain based Secured Land Registration System using Dynamic Smart Contracts (2025)	Land registration	Combines **Hyperledger Fabric** with **decentralized off-chain storage (IPFS)** for large documents 623 .
Community Blockchain Record-Keeping Method for Agricultural Land Leases Using Design Science Research Approach (2025)	Agricultural land leases	Uses **Hyperledger Fabric + IPFS** to store lease agreements securely and immutably 624 .

Paper	Land-administration context	IPFS / off-chain storage use
Cadastral: Blockchain-Based Land Registration System (2024)	Land registration	Uses **smart contracts plus IPFS for file storage** 481 .
Cadastral: Blockchain-Based Land Registration System (2025)	Land registration	Uses **Solidity smart contracts + IPFS for file storage** 488 .
Blockchain-Enabled Framework for Secure and Transparent Land Administration in Bangladesh (2025)	Land administration	Uses **Hyperledger Fabric and IPFS** with **off-chain document storage and cryptographic anchoring** 625 .

Excluded from the list

- **Blockchain-Based Data Storage Model for Land Survey (2023)** is relevant because it uses on-chain/off-chain association retrieval, but the retrieved evidence does **not explicitly name IPFS** 622.
- **Land Certificate Authenticity Using Blockchain Technology in Indonesia (2024)** is described as hybrid blockchain, but the retrieved abstract does **not explicitly state off-chain document storage or IPFS** 626.
- **A Proof of Concept of the integration of blockchain with an ISO 19152:2012 based Land Administration System (2022)** also does **not explicitly mention IPFS** in the retrieved abstract 552.

Confidence level: High — the included papers were explicitly described in the retrieved evidence as using IPFS or off-chain document storage in land-administration contexts 481 488 621 623 624 625.

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Follow-up Suggestions

- Which land administration papers use Hyperledger Fabric with IPFS
- Which land administration papers store document hashes on-chain and files off-chain
- Which land administration papers use decentralized storage besides IPFS

Which studies describe on-chain and off-chain data separation in cadastral systems

Processing steps

On-chain and off-chain data separation in cadastral systems

On-chain and off-chain data separation in cadastral systems

The clearest cadastral studies describing **on-chain/off-chain data separation** are the ones that keep **metadata, hashes, or registry references on-chain while storing larger cadastral records off-chain or in a sidechain**, rather than putting all cadastral data directly on the blockchain 590 627 628 629

.

Studies that explicitly describe the split

Study	What it says about the separation
Efficient and Optimized Geospatial Data Representation in Blockchain-Based Land Administration (2024)	Discusses blockchain-based land administration systems that store geospatial cadastral data with a focus on efficient encoding and storage, including trade-offs that motivate separating data representation and storage to reduce size and improve manageability 590 .
Blockchain in real estate: Recent developments and empirical applications (2022)	Describes the **hybrid approach** of storing **hashes of original land data on-chain** while keeping the **original data off-chain**, and notes this has been piloted but also

Study	What it says about the separation
	criticized because hashing alone does not protect the underlying data 627 .
Sidechain: storage land registry data using blockchain improve performance of search records (2022)	Proposes a **sidechain mechanism** for land registry storage, explicitly framing it as a way to improve land record search performance rather than storing all registry data in one main chain 628 .
BERBLOM: Blockchain-enabled reliable land ownership management system using ERC-4907 smart contract (2026)	Describes a **hybrid decentralized storage model** that combines **IPFS for off-chain storage** with **smart contracts for on-chain operations** in land ownership management 629 .

What the separation looks like

- **On-chain layer:** usually stores **hashes, timestamps, metadata, access references, or transaction records** rather than full cadastral files 627 629

.

- **Off-chain layer:** keeps **original records, documents, or larger data objects** outside the blockchain to reduce storage load and improve practicality 627 629

.

- **Alternative split model:** some land systems use a **sidechain** to hold land registry data separately from the main chain, which is a form of separation but not the same as simple off-chain storage 628

.

Bottom line

If you want the most directly relevant cadastral references, start with **Blockchains in real estate: Recent developments and empirical applications**, **Sidechain: storage land registry data using blockchain improve performance of search records**, and **BERBLOM**; together they show the main patterns of **hash-on-chain / data-off-chain** and **main-chain / sidechain separation** in cadastral and land registry systems 627 628 629.

Confidence level: **High** — the retrieved studies explicitly describe hybrid land-registry architectures with off-chain storage, hash anchoring, or sidechains, and the separation pattern is stated directly in the evidence 627 628 629.

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Follow-up Suggestions

- Which cadastral systems store hashes on-chain and documents off-chain?
- Which land registry studies use sidechains for record storage?
- Which papers compare IPFS with other off-chain storage methods in land administration?

Critically synthesize governance and legal models for blockchain-based land and real estate administration. Evaluate state-supervised registration, precondition-based smart contracts, legal title validity, authoritative data sources, and LADM/ISO 19152 interoperability. Verify citations, distinguish conceptual proposals from implemented systems, identify jurisdictional limitations and evidence gaps, and avoid equating immutability with legal validity.

Processing steps

Governance and legal frameworks for blockchain-based land administration

Governance and legal frameworks for blockchain-based land administration

Blockchain-based land and real estate administration is best understood as a **hybrid governance problem**, not a purely technical one: the evidence supports **state-supervised registration, authoritative off-chain registries, and preconditioned smart contracts**, while also showing that **immutability alone does not create legal title validity** 569 612 630 631

. The strongest implementations preserve a government or registry authority as the legally determinative actor, with blockchain serving as a verification, audit, or workflow layer rather than a replacement for public law 563 569 614 615

.

Governance and legal models

Model	What the evidence supports	Main limitation
State-supervised registration	Georgia's architecture is repeatedly described as a **hybrid** model: a private Exonum/Bitcoin-anchored layer supports integrity, but the registry remains institutionally authoritative and not fully decentralized 563 607 614 615 . Türkiye's proposed model also keeps blockchain as a **complementary layer** rather than a replacement for centralized structures 569 .	This model is legally safer because the state remains the title authority, but it does **not** prove that blockchain by itself confers title validity 563 569 614 .
Precondition-based smart contracts	Several papers require **government database checks, inspector approval, physical survey, or other preconditions** before transfer or registration can proceed, as in the cadastral system papers and smart-registration proposals 481 488 600 . This is a strong sign that smart contracts are being used to **encode procedure**, not to replace legal judgment 481 488 600 .	These are mostly **conceptual or prototype** systems, so their legal force depends on external institutions and statutory recognition 481 488 600 .
Legal title validity	The literature repeatedly warns that blockchain's **immutability, transparency, and tamper resistance** do not automatically equal legal enforceability or valid title 612 630 632	Evidence is strongest for **policy and doctrinal caution**, weaker for jurisdiction-wide legal validation of blockchain titles 612 630 634

Model	What the evidence supports	Main limitation
	. Legal validity still depends on compliance with land law, registration rules, and dispute-resolution mechanisms 612 633 634 .	.
Authoritative data sources	The clearest systems rely on a **government database, land office, or registry authority** for verification and final status, even when blockchain is used for records or auditability 481 488 563 614 615 . Georgia's case is especially explicit that blockchain anchored hashes, while the registry authority remained central 563 614 615 .	This creates a **trust bridge** but also means the system is hybrid and institution-dependent, not fully self-executing 563 569 .
LADM / ISO 19152 interoperability	The evidence supports LADM as the **main conceptual bridge** for adapting blockchain to land administration: one proof-of-concept explicitly integrates blockchain with ISO 19152/LADM, and related land-administration work treats LADM as a de facto domain model for structuring land data and workflows 552 635 636 637 638 639 .	Most LADM-linked blockchain papers are **proofs of concept, adaptations, or profiles**, not deployed national integrations 552 637 639 .

Critical synthesis

- **State supervision is the dominant viable model** in the retrieved evidence: the state or registry office remains the source of legal authority, while blockchain improves auditability, integrity, and coordination 563 569 614 615

.

- **Smart contracts work best as conditional workflow tools**—for example, enforcing identity checks, survey confirmation, or registry approval—rather than as autonomous title-makers 481 488 600

.

- **Jurisdiction matters**: Malaysia's smart-tenancy analysis, for example, shows that enforceability depends on local stamping, electronic-signature, and court/land-registry rules 640

. Türkiye's conceptual model likewise foregrounds legal infrastructure, accessibility, and digital literacy as constraints 569

.

- **Implemented systems remain limited and hybrid**: Georgia is the clearest operational example, but even there the literature describes a registry-anchored design rather than a fully on-chain title system 563 614 615 641

.

Bottom line

The evidence supports **blockchain as a legally mediated governance layer** for land administration, not as a substitute for land law. The most defensible model is **state-supervised, LADM-aligned, and preconditioned by authoritative registry checks**, where smart contracts automate procedural steps but **legal title validity still comes from the public land regime**, not from blockchain immutability alone 552 563 569 612 614 630

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Confidence level: **High** — multiple independent sources converge on hybrid state-led designs, LADM-based interoperability, and explicit legal cautions, while the operational evidence remains limited and jurisdiction-specific 552 563 612 614 615 630 632.

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Follow-up Suggestions

- Which blockchain land administration systems are actually deployed in national registries

- How does ISO 19152 LADM model blockchain land records
- What legal preconditions make blockchain land transfers valid

Which blockchain land administration systems are actually deployed in national registries

Processing steps

Deployed blockchain land administration systems

Deployed blockchain land administration systems

Only a small number of blockchain land systems are described as actually deployed in national registry settings, and the clearest case is Georgia; Sweden is described as an official but small-scale registry use, while Dubai/UAE and Jordan are better supported as active or departmental implementations than as fully documented nationwide registry deployments. 563 564 607 614 615 642 643

National-registry deployments with the strongest support

Jurisdiction	What the evidence says	Deployment status
Georgia	Described as a blockchain land registry initiative that moved from pilot to a **full-scale project**; one source says **over 1.5 million land titles** were registered on blockchain, and others describe it as the first successful blockchain land administration case or an Exonum-based, Bitcoin-anchored hybrid registry 563 564 607 608 .	**Best-supported actual deployment**
Sweden	Described as having **official use** of blockchain for land registry **at a small scale**, and as the site of a large-scale study or pilot by the land registry authority rather than a nationwide replacement 540 602 642 .	**Pilot / limited operational use**
Dubai / UAE	Mentioned in comparative literature as a case where blockchain for land registry is **actively working well**, but the retrieved evidence does not give the same architectural or scale detail as Georgia 564	**Active implementation, scale unclear**

Jurisdiction	What the evidence says	Deployment status
	.	
Jordan	The evidence supports **institutional implementation** in the land registry and survey department, but it reads as departmental adoption rather than a clearly documented national registry deployment 565 .	**Departmental implementation**

What should not be counted as deployed national registries

- **Türkiye, Malaysia, India, Bangladesh, Namibia, Turkey proposals, and similar papers** are mainly **conceptual models, proposals, or prototypes**, not confirmed national registry deployments 539 569 570 571 578 597 644 645

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- **Andhra Pradesh, Honduras, Ghana, and other country cases** are described in the evidence as **pilots, exploratory frameworks, or limited implementations**, not as fully verified national registry rollouts 540 547 642 646 647

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- Several reviews explicitly warn that **most blockchain land deployments remain prototypes or pilots**, which is why the deployment list is so short 539 540 647 648

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Bottom line

If you mean **national registry systems that are actually in use**, the evidence most strongly supports **Georgia**; **Sweden** is a limited official use case, while **Dubai/UAE** and **Jordan** are better treated as active or departmental implementations rather than clearly documented nationwide registry deployments 563 564 565 607 608 642.

Confidence level: **High** — the retrieved sources explicitly separate full-scale use, small-scale official use, pilots, and proposals, and only Georgia is consistently described as a confirmed operational national land-registry case 540 563 564 607 608 642.

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Follow-up Suggestions

- Which national land registries use blockchain in production
- How does Georgia's blockchain land registry work
- Which blockchain land registry projects are pilots rather than deployments

Evaluate technical implementations for land and real estate administration, including Ethereum, IPFS, Hyperledger Fabric, throughput, latency, and resource utilization. Verify reported metrics and citations; distinguish prototype benchmarks from operational and national-scale performance. Compare architectural trade-offs, identify missing evaluation criteria, and avoid unsupported scalability claims or invented results.

Processing steps

Technical performance of blockchain land administration

Technical performance of blockchain land administration

The evidence shows that land and real-estate blockchain systems are usually evaluated as prototypes or test-network implementations, while only a few studies report more operational or production-like setups; the strongest technical results come from small-scale experiments, not national-scale benchmarks. 539 625 648 649 650 651 652 653 654

What the studies actually report

System / paper	Stack and architecture	Reported metrics	Audit judgment
Secured Land Registration using Ethereum Blockchain and IPFS	Ethereum, Solidity, Sepolia TestNet, IPFS via Pinata API; framed as a land-registration	The abstract emphasizes implementation and comparison with traditional processes, but does **not** provide a	**Prototype/test-net study**, not an operational deployment 649

System / paper	Stack and architecture	Reported metrics	Audit judgment
	prototype for India 649	verified throughput/latency figure in the retrieved text 649	
A Blockchain based Secured Land Registration System using Dynamic Smart Contracts	Hyperledger Fabric + IPFS, RBAC, biometric and PKI verification, PBFT consensus 623	Average ownership transfer time **420 ms vs 5600 ms**, throughput **above 175 TPS**, transaction accuracy **99.84%**; benchmarking also includes gas cost, latency, and auditability 623	**Prototype benchmark** with strong headline numbers, but not national-scale evidence 623
Blockchain-Enabled Framework for Secure and Transparent Land Administration in Bangladesh	Hyperledger Fabric, Go chaincode, Node.js middleware, IPFS; permissioned ledger with off-chain document storage and cryptographic anchoring 625	Measures **resource utilization, throughput, end-to-end latency, and IPFS cache behavior**; reports predictable operation under moderate to high loads and identifies CPU/off-chain bottlenecks 625	**Prototype implementation** with realistic load testing, not an operational registry 625
Land Registry System using Blockchain With Multiple Nodes	Ethereum with Geth and multiple nodes; distributed database plus smart contracts 655	Reports **high throughput and low latency** in a simulated node network, but no precise figures appear in the retrieved abstract 655	**Simulated prototype**, not production evidence 655
LandAdministration System using Hyperledger Fabric	Permissioned Hyperledger	No numeric throughput/latency values in the retrieved	**Conceptual/prototype-oriented** 656

System / paper	Stack and architecture	Reported metrics	Audit judgment
	Fabric for land administration 656	abstract, only a claim that blockchain improved integrity and reduced tampering risk 656	
A blockchain solution for decentralized training in ML for IoT and related non-land studies	Not land-specific, but useful for method comparison because they evaluate blockchain overhead, memory, and training time 657	Shows how experimental blockchain studies often measure **runtime overhead and memory usage** rather than deployment scale 657	**Methodologically relevant**, but not land-administration evidence 657

Technical patterns and trade-offs

Ethereum-based designs

Ethereum appears most often in **prototype land-registration and real-estate workflows**, usually with smart contracts and sometimes IPFS for documents 584 586 596 648 649

. These systems generally aim at **automation, traceability, and tamper resistance**, but their performance evidence is typically limited to small-scale test environments or simulated trading settings 584 586 596 649

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Hyperledger Fabric designs

Fabric is the dominant choice where authors want **permissioned governance, lower-latency enterprise operation, and better control over participants** 623 625 650 651 656 658

. The retrieved evidence consistently frames Fabric as suitable for **high-throughput, controlled-access land workflows**, and benchmark-style papers explicitly measure **throughput, latency, and resource consumption** 625 651 652 653 658 659

. A separate private-blockchain comparison found Fabric outperformed Ethereum on **execution time, latency, and throughput** under varying workloads, though both remained less competitive than current databases in high-load scenarios 660

.

IPFS and off-chain storage

IPFS is used to keep **documents, images, or metadata off-chain** while anchoring hashes or references on-chain 584 586 623 625 648 649 650

. This pattern reduces storage pressure and supports privacy/auditability, but the evidence also shows **off-chain interaction bottlenecks** and sensitivity to file size and access path in IPFS-based designs 625 661

. In other words, IPFS helps with storage design, but it does **not** eliminate performance costs 625 661

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What is missing from the evaluation literature

The retrieved studies often omit or under-report several criteria needed for a serious scalability claim:

- **Operational scale definition**: many papers give TPS or latency without stating whether the system was tested on a real registry, a pilot, or a lab network 625 649 655 656

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- **Workload realism**: transaction mixes, contention, and multi-agency workflows are often simplified 623 625 660

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- **Resource utilization detail**: only some studies report CPU/memory/bandwidth; when they do, bottlenecks can still appear under load 625 652 653

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- **End-to-end legal/registry latency**: many studies measure blockchain confirmation time, not the full administrative process from submission to legal effect 625 632

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- **National-scale evidence**: the retrieved land papers do **not** provide robust, audited performance results for a nationwide live registry comparable to a production public service 539 625 654

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Bottom line

For land and real-estate administration, the technical evidence supports "Fabric for permissioned enterprise prototypes" and "Ethereum for smart-contract prototypes," with IPFS mainly solving off-chain storage—not proving national-scale readiness. The strongest quantified result in the retrieved land set is the Hyperledger Fabric + IPFS land-registration prototype reporting **420 ms transfer time and >175 TPS**, but that should still be read as **benchmark evidence from a prototype**, not proof of full operational deployment or universal scalability 623 625

.

Confidence level: **High** — multiple independent land-administration studies converge on the same pattern: prototype-scale benchmarking, Fabric favored for permissioned performance, Ethereum used for smart-contract workflows, and IPFS used for off-chain storage, while operational/national-scale performance remains largely unverified 539 623 625 649 651 652 653 660.

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Follow-up Suggestions

- How do land registry papers benchmark end-to-end legal processing time
- What throughput tests do Hyperledger Fabric land prototypes report
- How is IPFS evaluated for document storage in land administration

How do land registry papers benchmark end-to-end legal processing time

Processing steps

Benchmarking land registry processing time

Benchmarking land registry processing time

Land registry papers benchmark **end-to-end legal processing time** mainly by measuring the **full workflow duration from submission to legally relevant completion**—for example, from property-document upload or ownership request to approval, title transfer, or registration completion—rather than isolating only blockchain execution time 662 663 664 665 666 667 668.

How the benchmark is usually framed

Benchmark style	What gets timed	What it captures
Workflow completion time	Start-to-finish registration, transfer, or sale-deed processing	The closest proxy for **end-to-end legal processing time** because it includes the legal workflow, not just technical execution 662 663 666 667 668
Stage-specific processing time	A named stage such as post-registration transfer, sale approval, or document verification	Shows where blockchain shortens the legal process, but does not always cover the whole registry chain 662 663 668
Comparative before/after time	Legacy process vs proposed blockchain process	Useful for demonstrating reduction in administrative delay 666 667 669
System latency / throughput	Block insertion time, transaction latency, CPU use	Technical benchmark, not a full legal-process benchmark unless the paper explicitly links it to workflow completion 664 668

What the retrieved papers actually report

- **BERBLOM** is explicit that its main performance benefit appears **after tokenization**, while **initial land registration still depends on human governmental approval**; it says post-registration operations such as peer-to-peer transfers and on-chain leasing can be completed **within seconds** 662

. That means the paper distinguishes **legal approval time** from later blockchain-enabled execution time 662.

- **LandReg** reports that land registration and transfer are completed **within a few seconds** on a test network, with a **100% functional success rate** and improved gas efficiency 663

. This is a start-to-finish prototype benchmark, but it is still a **test-network** measure rather than evidence of a national registry process 663.

- **A Blockchain based Secured Land Registration System using Dynamic Smart Contracts** reports average ownership transfer time falling from **5600 ms to 420 ms** and throughput above **175 TPS** 664

. The time metric is a workflow-performance proxy, but the paper's abstract does not show a full legal-service clock that includes every human approval step 664.

- **A secured land registration framework on Blockchain** says transaction processing time is "reasonably low," but does not provide a full end-to-end legal workflow definition in the abstract 670.

- **Blockchain-Enabled Sale Deed Registration** reports a **40% reduction in total processing time** and **35% lower transactional costs** 667

. This is one of the clearest "end-to-end" style claims because it refers to total process time, not just technical latency 667.

- **Blockchain-Based Land Record Management System for Secure Property Transactions in Indian States** reports average processing times of **1.4 days instead of 9.2 days** in the prior system 666

. That is directly in the legal-administrative time category because it compares the full transaction process before and after the proposed system 666.

- **Implementation of Land Registry System Using Ethereum Blockchain and IPFS** says benchmarking showed **low transaction processing time** and that this made the system suitable for practical use, but the abstract does not provide a full end-to-end legal-process breakdown 668.

- **Minimization of fraudulent activities in land authentication through Blockchain-based system** reports a drop from **a few months to a couple of minutes** for land ownership verification 669

. That is a strong end-to-end legal-processing claim, but it is about **ownership verification**, not necessarily the whole registry lifecycle 669.

Bottom line

The dominant pattern is **not** to benchmark “legal processing time” as a formal legal metric; instead, papers usually approximate it with **total transaction/registration/transfer turnaround time** and then show blockchain cuts that time dramatically—sometimes from days or months to seconds or minutes 662 663 666 667 669

. The most defensible end-to-end benchmarks are the papers that compare **whole-process duration before and after the proposed system**, while pure blockchain metrics like throughput or latency only become legal-process evidence when the authors explicitly tie them to registration or transfer completion 664 668.

Confidence level: **High** — several retrieved abstracts explicitly report total processing time, turnaround time, or workflow completion time in land-registration contexts, while others only measure technical latency and are therefore weaker proxies 662 663 666 667 668.

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Follow-up Suggestions

- Which land registry papers define end-to-end processing time explicitly?
- How do papers separate legal approval time from blockchain confirmation time?
- What metrics do land registry benchmarks use besides throughput and latency?

Critically refine this four-layer hybrid blockchain model for land and real estate administration. Strengthen technical coherence, legal governance, off-chain storage, smart-contract validation, consensus, and cryptographic anchoring. Verify citations and correct overclaims about security, GDPR, performance, and legal validity. Distinguish

proposed design from validated implementation, align with LADM/ISO 19152, and identify testable propositions and research gaps. Never invent evidence.

Processing steps

Hybrid blockchain land administration architecture

Hybrid blockchain land administration architecture

A stronger four-layer model should treat **land administration as a state-supervised, LADM-aligned workflow with blockchain used for integrity and auditability, not as a substitute for legal title or registry authority**. The most defensible design is: **(1) authoritative legal/data layer, (2) off-chain document layer, (3) on-chain validation and anchoring layer, and (4) consensus/governance layer**—with explicit validation boundaries, privacy controls, and testable claims rather than blanket security or GDPR guarantees 547 575 671 672 673 674 675.

Refined four-layer model

Layer	What it should do	What to tighten
1. Legal-authoritative layer	Keep the land registry or cadastre as the **legal source of truth**, with blockchain acting as a complementary system rather than the title-maker 547 569 575	Do **not** claim immutability creates legal validity; legality still depends on institutional and statutory recognition 547 676 677
2. Off-chain document layer	Store large or sensitive records off-chain, using IPFS or equivalent, while keeping only hashes, CIDs, timestamps, and access metadata on-chain 674 675 678 679	Avoid saying off-chain storage is automatically private or GDPR-compliant; GDPR-sensitive designs still need encryption, access control, and careful treatment of persistence/erasure 677 680 681 682
3. Smart-contract validation layer	Encode procedural checks such as identity, approval, transaction type, and workflow constraints before state changes are accepted 575 683 684	Frame smart contracts as **precondition enforcers**, not autonomous legal adjudicators; failures and misconfigurations can still cause reverts or validation gaps 681 682

Layer	What it should do	What to tighten
4. Consensus + anchoring layer	Use permissioned consensus where governance needs controlled participation; anchor document fingerprints or state transitions on-chain for tamper evidence 575 674 685 686	Avoid overstating throughput or security; performance depends on workload, storage, and consensus choice, and public-chain latency/gas costs remain material 547 682 687

Corrections to overclaims

- **Security:** The evidence supports *tamper evidence, auditability, and controlled access*, not absolute security. Several sources explicitly note remaining risks from off-chain access control, smart-contract failure, and implementation overhead 675 681 682.
- **GDPR:** You should not claim "GDPR compliance" by default. The retrieved evidence shows that compliance depends on design choices such as keeping personal data off-chain, encrypting off-chain content, and limiting what is written on-chain; even then, data-localization and persistence issues remain 677 680 681 688.
- **Performance:** Replace generic "scalable" claims with measurable propositions. The literature shows that blockchain systems often improve workflow control, but real scalability depends on consensus, gas cost, throughput, and off-chain load handling 674 682 685 687.
- **Legal validity:** Do not equate hashed records or immutable logs with valid title. The literature repeatedly separates technical integrity from legal recognition and stresses the need for institutional control 547 575 676 677.

Alignment with LADM / ISO 19152

LADM is the right conceptual anchor because it is explicitly presented as a core land-administration model with legal, spatial, versioning, and metadata support, and it has been used as a basis for country profiles and implementation methodologies 671 672 673. The blockchain model should therefore map: **rights/restrictions/responsibilities, spatial units, parties, source documents, and version history** to LADM-compatible structures rather than inventing a parallel registry ontology 671 673 689 690.

Testable propositions and research gaps

Testable propositions

- **P1:** A permissioned, hybrid design with off-chain documents and on-chain hashes reduces registry storage burden without losing traceability 674 679 685

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- **P2:** LADM-aligned data structures improve interoperability between blockchain components and existing land-registry workflows 671 673 689

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- **P3:** Smart contracts that enforce preconditions reduce procedural errors, but they still require registry approval for legal validity 575 683 684

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- **P4:** Permissioned consensus and off-chain storage outperform public-chain designs on latency and operational practicality in land-administration settings 547 682 685 687

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Research gaps

- **Validated implementation gap:** Most land-administration work remains conceptual, proof-of-concept, or pilot-based rather than nationally validated 575 675 676 691

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- **Legal integration gap:** More work is needed on legal recognition, institutional workflows, and jurisdiction-specific constraints, especially where digital registry entries must coexist with existing law 569 676 677.
- **Privacy and compliance gap:** Off-chain storage is not enough; papers still under-specify GDPR-grade governance, deletion/retention handling, and access revocation 680 681 682.
- **Evaluation gap:** Many studies report architecture claims without complete benchmarking of throughput, latency, resource use, or failure modes under realistic registry loads 674 682 685.

Bottom line: the refined model should be presented as a **hybrid legal-technical architecture for workflow integrity and auditability**, not as a fully decentralized title system; its strongest version is LADM-aligned, permissioned, off-chain for documents, on-

chain for hashes and state transitions, and explicitly bounded by registry authority and jurisdiction-specific law 547 575 671 675 676.

Confidence level: **High** — the retrieved evidence consistently supports hybrid, registry-anchored, LADM-aligned designs, while also explicitly warning against overclaiming legal validity, privacy compliance, and scalability 547 575 671 676 677.

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Follow-up Suggestions

- How can a land registry map LADM classes to blockchain data structures?
- What GDPR controls are needed for off-chain land records on IPFS?
- How should a permissioned consensus be benchmarked for registry workloads?

Strengthen the threats-to-validity and system-architecture discussion for this systematic review of hybrid blockchain in real estate administration. Address search and selection bias, heterogeneous metrics, jurisdictional differences, limited deployment evidence, and generalizability. Clarify RCUBE and smart-contract workflows, distinguish proposed from validated architecture, temper unsupported superiority claims, and identify testable future research. Preserve citations; never invent evidence.

Processing steps

Hybrid blockchain in real estate administration

Hybrid blockchain in real estate administration

The review should frame hybrid blockchain in real estate as an early-stage, hybrid, institution-dependent design space: most evidence is conceptual or prototype-level, empirical deployments are sparse, performance metrics are heterogeneous, and legal validity still depends on jurisdiction-specific land law rather than blockchain immutability. 563 614 615 676 677 692 693 694

Strengthened threats-to-validity discussion

Threat	What the evidence supports	How to state it in the review
Search and selection bias	Systematic reviews in adjacent blockchain domains explicitly warn that keyword choice, database coverage, and duplicate screening can distort the final sample, even when PRISMA is used 695 696 697 .	Say the review may undercount papers indexed under adjacent terms such as **real estate**, **land administration**, **cadastre**, **tokenization**, or **smart contracts**, and that inclusion decisions may favor English-language, indexed, or architecture-focused studies 695 696 697.
Heterogeneous metrics	Land-registry papers benchmark different things: workflow time, latency, throughput, storage cost, fraud detection, or qualitative trust gains, often in test networks or simulations rather than comparable deployments 623 625 663 664 666 667 668 669 .	State that cross-paper comparisons are only indicative because the studies do not use a shared benchmark for "performance," and technical metrics are not directly equivalent to legal-processing outcomes 623 625 663 664 666 667 668 669.
Jurisdictional differences	The literature repeatedly shows that legal force depends on local registration, e-signature, stamping, and title rules; Georgia is the clearest hybrid deployment, while many other country cases remain proposals or pilots 563 608 614 615 640 642 643 .	Explicitly separate **technical portability** from **legal portability** and note that a design validated in one jurisdiction cannot be assumed to transfer to another without statutory alignment 563 608 614 615 640 642 643.
Limited deployment evidence	Most blockchain real-estate applications are still "hybrid, smaller-scale settings," with blockchain acting as an add-on layer; conceptual benefits remain	Avoid language that implies widespread adoption; instead say the evidence base is dominated by prototypes, proofs of concept, and case studies, with few operational

Threat	What the evidence supports	How to state it in the review
	largely unconfirmed empirically 692 693 694 698 699 .	national-registry implementations 692 693 694 698 699.
Generalizability.	The most defensible implementations rely on state-supervised or registry-anchored models, not fully decentralized title systems, and LADM-aligned approaches are mostly conceptual or proof-of-concept 552 563 575 671 673	Conclude that generalization is limited because the architecture is institution-dependent, workflow-dependent, and often designed to fit a specific registry, not a universal legal model 552 563 575 671 673.

System-architecture discussion to sharpen

Distinguish proposed from validated architecture

The review should explicitly separate **proposed architecture** from **validated implementation**. The retrieved evidence shows that many papers propose hybrid designs with smart contracts, off-chain storage, and permissioned consensus, but only a small subset reports actual institutional use or realistic testing 563 607 614 625 692 694 700 701

. Georgia is the clearest operational national-registry case, while most other systems are prototypes or jurisdiction-specific pilots 563 607 614 615 642 643

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Clarify smart-contract workflows

A stronger description is that smart contracts **encode preconditions and workflow checks**, not legal judgment. The evidence supports patterns such as identity verification, database checks, multi-party approval, and staged execution before a record changes state 481 488 575 623 662 663 664

. Some systems also keep sensitive documents off-chain and use blockchain for hashes, timestamps, and audit trails 481 488 623 625 627 629

. That means the workflow is best described as **registry-governed automation**, not autonomous title creation 563 575 614 615

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Clarify RCUBE

RCUBE is not named in the retrieved evidence, so it should not be described as an established architecture unless the review has an internal definition elsewhere. The supported architectural language in the evidence is **hybrid**, **permissioned**, **state-supervised**, **off-chain document storage**, and **on-chain anchoring/validation** 563 614 625 671 673 692.

Temper superiority claims

The review should replace claims like "blockchain is superior" with narrower statements: blockchain can improve **verifiability, traceability, and procedural integrity** in specific workflows, but its advantages are most visible in hybrid settings and are not uniformly demonstrated across legal systems or deployment scales 623 625 666 667 669 692 694 698. Reviews in adjacent fields likewise caution against overstating generalized benefits without institutional fit and governance support 632 702.

Testable future research

The evidence supports several concrete, testable propositions:

1. **Hybrid architectures outperform full-chain designs on storage and operational practicality** when documents are kept off-chain and hashes or metadata are anchored on-chain 623 625 629 701 703 704

.

2. **Permissioned consensus is more suitable than public-chain designs** for land administration workflows that require controlled participation and institutional oversight 623 625 704

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3. **LADM/ISO 19152 alignment improves interoperability** between blockchain components and existing land-administration structures, but this remains mostly a proof-of-concept claim that needs broader validation 552 671 672 673

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4. **Smart contracts reduce procedural delay only when paired with authoritative registry validation**, not when treated as standalone legal instruments 575 623 662 664 666 667

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5. **Jurisdiction-specific legal design is a necessary moderator** of adoption and performance, because legal enforceability and institutional fit vary across countries 569 632 640 677

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6. **Comparability needs a common benchmark suite** that separates technical metrics from legal-processing outcomes, since current studies mix latency, throughput, resource use, turnaround time, and qualitative trust claims 623 625 663 664 666 667 668 669

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Bottom line

The review is strongest if it presents hybrid blockchain in real estate administration as a **jurisdiction-bound, registry-anchored governance tool** whose technical promise is real but unevenly validated; the safest synthesis is that blockchain mainly adds **auditability, workflow control, and off-chain/on-chain integrity**, while legal validity, scalability, and generalizability still depend on institutional recognition, local law, and better comparative evidence 563 575 614 625 632 671 673 692 702.

Confidence level: **High** — multiple independent sources converge on hybrid, state-supervised, and prototype-heavy land/real-estate blockchain architectures, with consistent warnings about limited deployment evidence, legal dependence, and heterogeneous evaluation methods 563 614 625 671 673 692 694 698.

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Follow-up Suggestions

- How can a benchmark suite compare hybrid land-registry papers consistently

- How should a systematic review code jurisdiction-specific legal validity claims
- What evidence qualifies as a validated blockchain deployment in real estate administration

What technical, regulatory, and interoperability constraints hinder the implementation and widespread adoption of blockchain-based systems in land records management?

Processing steps

Blockchain implementation in land administration

Blockchain implementation in land administration

Blockchain-based land records management is hindered mainly by **(1) technical constraints**, **(2) regulatory/legal constraints**, and **(3) interoperability and institutional-fit constraints**; the evidence consistently shows that the biggest adoption barrier is not immutability itself, but whether the system can integrate with existing land administration workflows, laws, and data infrastructures 563 570 705 706 707 708

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Main constraints

Constraint area	What blocks implementation and adoption	Evidence
Technical	**Scalability, throughput, storage burden, transaction costs, and processing speed** repeatedly appear as adoption limits; several studies also note complexity of implementation, dependency on PKI/Internet, and the need for stronger consensus and security mechanisms 570 644 705 709 710 .	Land-record studies explicitly cite PKI and Internet dependencies, privacy and security issues, and implementation difficulty 644 705 711. Other reviews identify scalability, storage cost, throughput, and processing speed as recurring bottlenecks 570 709 710.
Regulatory / legal	**Legal enforceability, privacy rules, and the need to align with existing land law** are major barriers; immutability can conflict with legitimate record correction or	Bangladesh-focused work says blockchain must coexist with traditional legal systems and will require regulatory reform 706 715. Malaysia-focused work highlights the

Constraint area	What blocks implementation and adoption	Evidence
	ownership changes, so legal frameworks must be adapted before broad deployment 632 706 712 713 714 .	legal enforceability of smart contracts and tokenized assets as unresolved 713.
Interoperability.	**Lack of standardization and poor compatibility with existing land systems** obstruct data exchange across agencies, blockchain networks, and legacy infrastructure; this is especially important in public-sector land administration, where multiple departments share records 707 716 717 718 719	Public-sector and blockchain-governance sources stress that systems often cannot easily communicate with existing infrastructures without common standards 707 716. Land-administration literature similarly notes that interdepartmental data harmonisation is essential to reduce mismatch and overlap 717 718.

What the evidence says in practice

- **Implementation is technically complex** because land registries must handle large documents, legacy workflows, and secure identity verification, not just blockchain transactions 570 644 720

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- **Privacy and security are double-edged**: blockchain can improve auditability, but transparency can expose sensitive land information unless the system uses access control, off-chain storage, or privacy-preserving techniques 712 713 714

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- **Adoption depends on institutional readiness**, including staff capacity, governance, and user training; one land-administration study emphasizes that compatibility, training, and cross-department coordination are necessary for smooth uptake 718

.

- **The literature does not support treating blockchain as a drop-in replacement** for land systems; instead, it repeatedly frames adoption as phased, hybrid, and dependent on legal and technical alignment 563 706 708 715

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Bottom line

The strongest pattern is that **blockchain-based land records fail to scale when technical design outpaces legal and institutional readiness**: systems need interoperable data standards, a legally valid correction/ownership-change framework, and integration with existing registry processes, or they remain prototypes rather than deployable public infrastructure 563 614 632 707 708 713

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Confidence level: **High** — multiple independent land-administration, public-sector, and governance sources converge on the same three constraint classes, with the clearest evidence coming from studies that explicitly name scalability, legal/regulatory alignment, and interoperability as adoption barriers 705 706 707 708 709 717 718.

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Follow-up Suggestions

- How do LADM and ISO 19152 map to blockchain land registry data models?
- What legal reforms are needed for blockchain land record correction and rectification?
- How should a benchmark suite measure blockchain land registry interoperability?

How can a hybrid architecture—integrating an Ethereum Proof-of-Stake (PoS) network, off-chain storage (via IPFS and MongoDB), and Python-based validation logic—be engineered to resolve these bottlenecks while preserving data security and operational efficiency?

Processing steps

Hybrid blockchain architecture for land records

Hybrid blockchain architecture for land records

A robust way to engineer that hybrid stack is to **treat Ethereum PoS as the integrity-and-governance layer, IPFS as the immutable content layer, MongoDB as the query/index layer, and Python as the validation-and-orchestration layer**—with only hashes, CIDs, metadata, and access references written on-chain, while full documents stay off-chain. That design aligns with the repeatedly observed pattern of **on-chain anchoring plus off-chain payload storage** and avoids the storage, latency, and gas bottlenecks that appear when large files are pushed directly to the chain 621 661 721 722 723 724

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Recommended architecture

Layer	What it should do	Why this resolves bottlenecks
Ethereum PoS smart-contract layer	Store document hashes/CIDs, ownership state, timestamps, access-control decisions, and audit events; avoid storing raw records 621 722 723 724	Keeps the chain small, lowers gas use, and preserves tamper evidence without making the ledger the primary file store 721 723 725
IPFS off-chain document layer	Store encrypted PDFs/images/scans and other large payloads; pin critical records; return CIDs for on-chain anchoring 661 725 726 727 728 729	Reduces on-chain storage pressure and supports scalable retrieval of large documents 721 723 730
MongoDB metadata/index layer	Store normalized metadata for fast search, filtering, workflow status, user/session state, and document pointers to IPFS CIDs 731	Acts as a fast operational index for application queries without overloading the blockchain or IPFS with lookup-heavy requests 648 731

Layer	What it should do	Why this resolves bottlenecks
Python validation/orchestration layer	Run OCR/NER/anomaly checks, schema validation, signature verification, deduplication, and business-rule checks before writing to IPFS or blockchain 621 732 733 734	Prevents invalid or fraudulent documents from becoming permanent ledger entries and reduces downstream rework 621

How the flow should work

1. **Ingest** the document through a Python service.
2. **Validate first**: use Python to check document type, completeness, identity fields, duplicate hashes, and authenticity signals before any on-chain write; this is the key lesson from the land-record AI validation paper, which argues that pre-storage validation prevents invalid records from becoming permanent 621

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3. **Encrypt and store** the full file in IPFS; keep only the resulting CID/hash on-chain and optionally in MongoDB for fast lookup 661 722 725 727

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4. **Anchor the transaction** on Ethereum PoS with a smart contract that records the CID, document hash, owner/agent identity, and workflow state 621 723 724

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5. **Index metadata in MongoDB** so the UI can search by parcel ID, applicant name, status, date, or office without querying the chain for every request 648 731

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6. **Retrieve and verify** by pulling the CID from MongoDB or the chain, fetching the file from IPFS, and recomputing the hash to confirm integrity against the blockchain anchor 725 735 736

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Design choices that matter

- **Use Ethereum PoS for finality and governance, not bulk storage.** The evidence consistently shows that the best-performing hybrid systems keep the blockchain for

verification, access control, and workflow state, while large payloads are moved off-chain 721 723 724

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- **Encrypt before IPFS.** Several systems explicitly warn that off-chain storage alone does not guarantee privacy; encryption and access control remain necessary 737 738 739

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- **Pin critical files.** IPFS designs in the evidence rely on pinning or replicated nodes to preserve availability and persistence 729 740 741

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- **Keep MongoDB as a cache/index, not the source of truth.** The strongest patterns in the evidence place integrity on the blockchain and document persistence in IPFS, while MongoDB-like storage is best used for flexible application-side storage and retrieval support 731 735

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- **Make Python the policy gatekeeper.** A Python layer is well suited to pre-storage checks, smart-contract orchestration, IPFS interaction, and validation pipelines; prior implementations use Python for off-chain cryptographic operations and blockchain/IPFS coordination 733 734 742

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Security and efficiency safeguards

- **Integrity:** store only hashes/CIDs on-chain and recompute hashes on retrieval 725 735

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- **Confidentiality:** encrypt documents before IPFS and restrict access by smart contract 738 739

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- **Efficiency:** batch writes, deduplicate uploads, and avoid duplicate hash entries to reduce gas and storage costs 723

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- **Resilience:** pin IPFS content and replicate critical records across nodes to avoid single-point availability failures 729 740

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- **Workflow control:** let Python reject malformed or suspicious records before anchoring, which directly addresses the problem of permanent invalid ledger entries 621

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Bottom line

The most defensible engineering pattern is **"validate in Python, persist the file in encrypted IPFS, index it in MongoDB, and anchor only the proof and workflow state on Ethereum PoS."** That arrangement is the clearest way to reduce blockchain bloat, lower latency and gas cost, preserve auditability, and keep legal workflows from recording unverified data as immutable truth 621 723 724 743

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Confidence level: **High** — the evidence strongly supports this hybrid off-chain/on-chain split and pre-storage validation pattern, although the exact Ethereum PoS + MongoDB + Python combination is an inference synthesized from multiple implementations rather than a single directly studied land-record architecture 621 723 724 731 733.

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Follow-up Suggestions

- How should smart contracts handle land title correction workflows?
- What privacy controls are suitable for off-chain land record storage?

- How can land registry interoperability be modeled across legacy databases and blockchain?

Which governance models and international data standards (notably LADM/ISO 19152) are required to ensure legal validity, regulatory compliance, and the seamless integration of distributed ledgers with legacy centralized cadastres?

Processing steps

Governance and standards for blockchain land registries

Governance and standards for blockchain land registries

To ensure legal validity, regulatory compliance, and integration with legacy cadastres, the evidence points to a state-supervised, legally anchored, standards-based hybrid model: blockchain should sit behind or alongside the cadastral authority, while LADM/ISO 19152 provides the core conceptual data model, national LADM country profiles adapt it to local law, and semantic/technical mediation standards handle interoperability with existing systems. 671 677 744 745 746 747

What governance model is required

Governance element	What the evidence supports	Why it matters
State-supervised / registry-anchored governance	Blockchain adoption in land administration needs **clear legal and institutional support** and **well-defined supervisory structures**; the public registry remains the legally determinative actor rather than the ledger itself 746 747 .	This is what makes entries legally meaningful instead of merely technically immutable 746 747.
Hybrid governance with off-chain authority	The literature frames blockchain as a tool for transparency and compliance, but not as a replacement for public law; on-chain records must be tied to registry procedures, data governance, and regulatory oversight 748 749 750 751 .	This avoids the false equation of immutability with legal validity 749 751.
Jurisdiction-specific	Successful adoption depends on local legal and institutional settings; Georgia is cited as a case	A model that works in one jurisdiction cannot

Governance element	What the evidence supports	Why it matters
institutional design	where flexible regulatory policy supported implementation, while most countries still lack adoption strategies 746 752 .	be assumed to transfer unchanged to another 746 752.

Which standards are needed

Standard / framework	Role in integration	Evidence
LADM / ISO 19152	The core international standard for land administration; it provides a **conceptual data model** with parties, rights/restrictions/responsibilities, spatial units, surveying/representation, versioning/history, source documents, and metadata 671 753 .	It is repeatedly described as the main foundation for modern cadastre and land administration design 671 753.
National LADM country profiles	Needed to adapt the standard to local legal and systematic requirements; multiple studies recommend or develop profiles for China, Turkey, Brazil, Malaysia, Mongolia, and others 690 744 745 754 755 756 .	National profiles bridge the gap between the international model and domestic cadastre law 690 744 756.
Semantic alignment tools / code lists	LADM and LandInfra require mediating platforms such as **CaLAThe** to align code lists and terminology, creating a shared universe of discourse for interoperability 757 .	This is key when multiple systems and agencies must exchange cadastral data consistently 757.
LADM edition II / implementation aspects	Edition II explicitly adds **implementation aspects** and expands land registration, 3D profiles, valuation, and spatial planning support 758 .	This signals that implementation and interoperability are part of the standard's

Standard / framework	Role in integration	Evidence
		evolution, not an afterthought 758.
INTERLIS / formal constraints	For legal 3D spaces and integrated models, the literature highlights the need for **formal semantics, explicit constraints, and implementable code lists**, with INTERLIS proposed as a communication standard for LADM-based country profiles 689 .	This is especially important where legal and physical boundaries must be unambiguous 689.
BIM / IFC and 3D cadastral exchange	For 3D cadastres, open BIM standards and IFC can supply geometric/legal space data, while LADM provides the land-administration framework 759 760 .	This supports data exchange between legacy cadastral systems and newer 3D/urban models 759 760.

What “seamless integration” actually requires

The evidence does **not** support a simple plug-in blockchain layer. Instead, integration needs:

- **Proper data integration and linkage between existing registers**, after analysing the contents of legacy datasets 744 745 747

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- **Interoperable data dissemination practices**, since cadastral systems differ by jurisdiction and legacy format 677 761

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- **Shared terminology, code lists, and semantics**, especially where multiple standards or agencies intersect 689 757

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- **Versioning, history, and source-document linkage**, which LADM is designed to support and which help preserve legal traceability 671 753

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- **Legal and institutional adaptation**, including data protection, liability, and administrative coordination requirements 677 746 762

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Bottom line

The most defensible model is **state-supervised blockchain governance embedded within an LADM-based cadastral architecture**: LADM/ISO 19152 supplies the legal-spatial data backbone, national profiles adapt it to domestic law, and semantic/implementation standards such as CaLAThe, INTERLIS, and IFC help legacy cadastres interoperate with distributed ledgers 671 689 744 745 757 760

. The evidence is strongest for **hybrid, jurisdiction-specific integration**, not for blockchain as a standalone source of legal validity 746 747 749 752

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Confidence level: **High** — multiple LADM and blockchain governance sources converge on the same requirements: legal supervision, national profile adaptation, semantic interoperability, and explicit institutional compliance rather than reliance on immutability alone 671 677 689 744 746 747 757.

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Follow-up Suggestions

- How should a national LADM country profile be developed for blockchain integration
- What role do INTERLIS code lists play in cadastral interoperability
- How can legacy cadastre datasets be mapped to LADM classes

): How does the existing literature evaluate the scalability and practical viability of hybrid blockchain architectures under realistic registry workloads, specifically regarding transaction throughput, latency, and consensus overhead?

Processing steps

Scalability and viability of hybrid blockchain architectures

Scalability and viability of hybrid blockchain architectures

Hybrid blockchain architectures are **usually evaluated as prototype or simulated systems rather than as full registry-scale deployments**, and the literature consistently shows that their practical viability depends on whether the design can keep **throughput acceptable, latency low, and consensus overhead bounded under realistic transaction loads** 623 625 660 763 764 765 766

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What the literature actually measures

Evaluation focus	What is reported	What it implies for registry viability
Throughput	Hybrid and permissioned designs report higher TPS when they use sharding, rollups, sidechains, or permissioned consensus; e.g., a hybrid land-registration prototype reported **>175 TPS** 623 , while a hybrid land-transfer system reported **34%–45% throughput gains** over comparators 766 .	Throughput gains are plausible, but they are mostly shown in **controlled prototypes**, not in nationwide registry backlogs 623 766.
Latency	Land and registry systems often report major reductions in confirmation or transfer time: **420 ms vs 5600 ms** in one dynamic-smart-contract system 623 , **18% to 50% lower record-confirmation latency** in LandChain 766	The best results come from architectures that **move bulk work off-chain** and keep the chain for anchoring and final confirmation 767 768.

Evaluation focus	What is reported	What it implies for registry viability
	, and sub-second / sub-50 ms figures in non-land hybrid systems that use batching or off-chain storage 767 768 .	
Consensus overhead	Land-record papers explicitly target consensus overhead: one study reduced **message overhead by 30%–40%** with a modified round-robin consensus 763 , while trusted-node consensus cut message exchange by **58.94% vs PoW** and **26.44% vs a load-balanced method** 769 .	Consensus choice is a first-order determinant of viability; **permissioned or hybrid consensus** is favored where registry throughput matters 763 769 770.
Realistic workload testing	Some studies use transaction-volume or concurrency sweeps, including simulated workloads and concurrent requests 764 765 771, but many land papers still stop at conceptual claims or small-scale prototypes 640 643 692 693.	Evidence for **realistic registry workload robustness is still thin**; most results are workload tests, not production registry traces 640 643 692 693.

Main pattern across the evidence

The literature's strongest practical lesson is that **hybrid architectures are viable only when they externalize expensive work**: documents, metadata, or validation steps are pushed off-chain, while the blockchain handles hashes, state transitions, and auditability 767 772 773 774

. That pattern consistently improves throughput and latency relative to fully on-chain designs, but the gains are typically demonstrated in **benchmarks, simulators, or small pilots**, not in national registry operations 600 623 625 691.

Consensus trade-off

The consensus evidence is consistent: **PoW-style or fully public-chain consensus is usually too expensive for registry workloads**, while **permissioned or hybrid schemes** reduce communication overhead and improve commit times 578 763 769 770

. The practical trade-off is that lower overhead comes from stronger governance assumptions—trusted nodes, validator sets, or registry-controlled participants—rather than from decentralization alone 578 763 769

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Bottom line

Under realistic registry workloads, the literature supports hybrid blockchain as **operationally promising but not universally proven**: it can deliver meaningful throughput and latency improvements when paired with **off-chain storage, batching, permissioned consensus, and preconditioned workflows**, yet the strongest evidence still comes from **prototype benchmarks and simulated loads**, not from sustained national-scale registry stress tests 623 625 763 764 766 767

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Confidence level: **High** — multiple independent land-registry and broader hybrid-blockchain studies converge on the same performance pattern, but the evidence base remains dominated by prototypes and workload simulations rather than long-run operational deployments 623 625 763 764 766.